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(54) **ENZYME VARIANTS WITH IMPROVED
ESTER SYNTHASE PROPERTIES**

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C12N 9/10 (2006.01)
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C12P 7/6436 (2022.01)

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CPC **C12N 9/18** (2013.01); **C12N 9/1007** (2013.01); **C12P 7/6436** (2013.01); **C12Y 201/01015** (2013.01)

(58) **Field of Classification Search**
None

See application file for complete search history.

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(57) **ABSTRACT**

The disclosure relates to enzyme variants with improved ester synthase properties for the production of fatty acid esters. Further contemplated are recombinant host cells that express such variants, cell cultures comprising the recombinant host cells and fatty acid ester compositions produced by such recombinant host cells.

15 Claims, 7 Drawing Sheets

Specification includes a Sequence Listing.

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FIG. 1

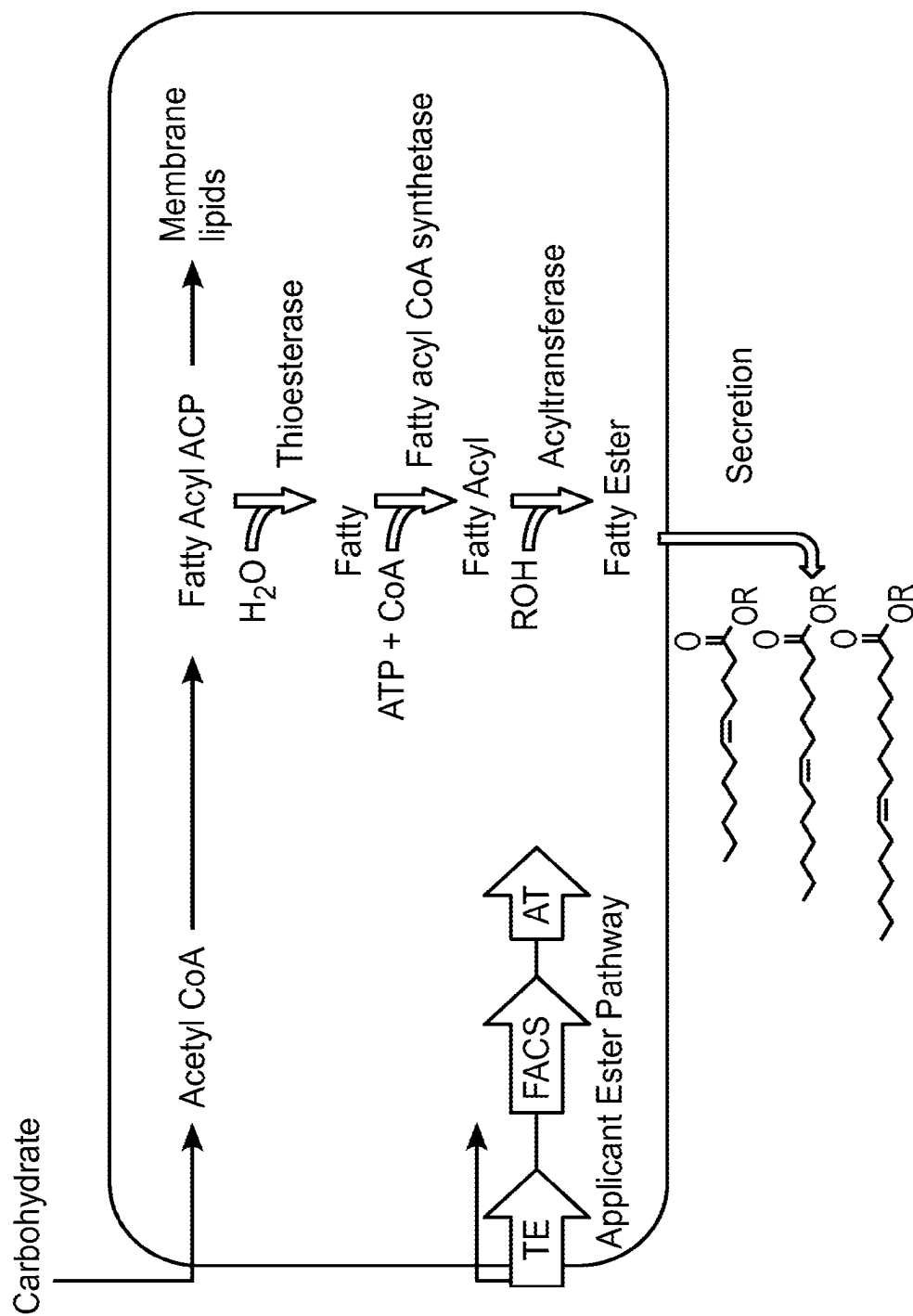


FIG. 2

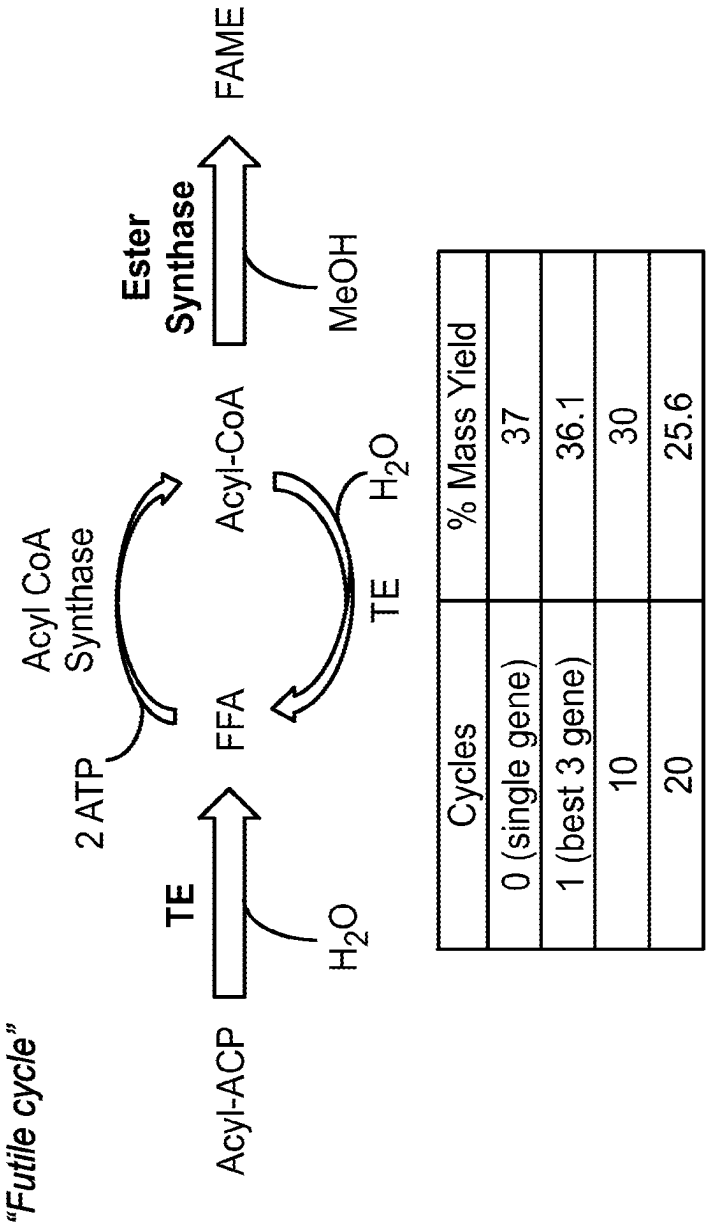
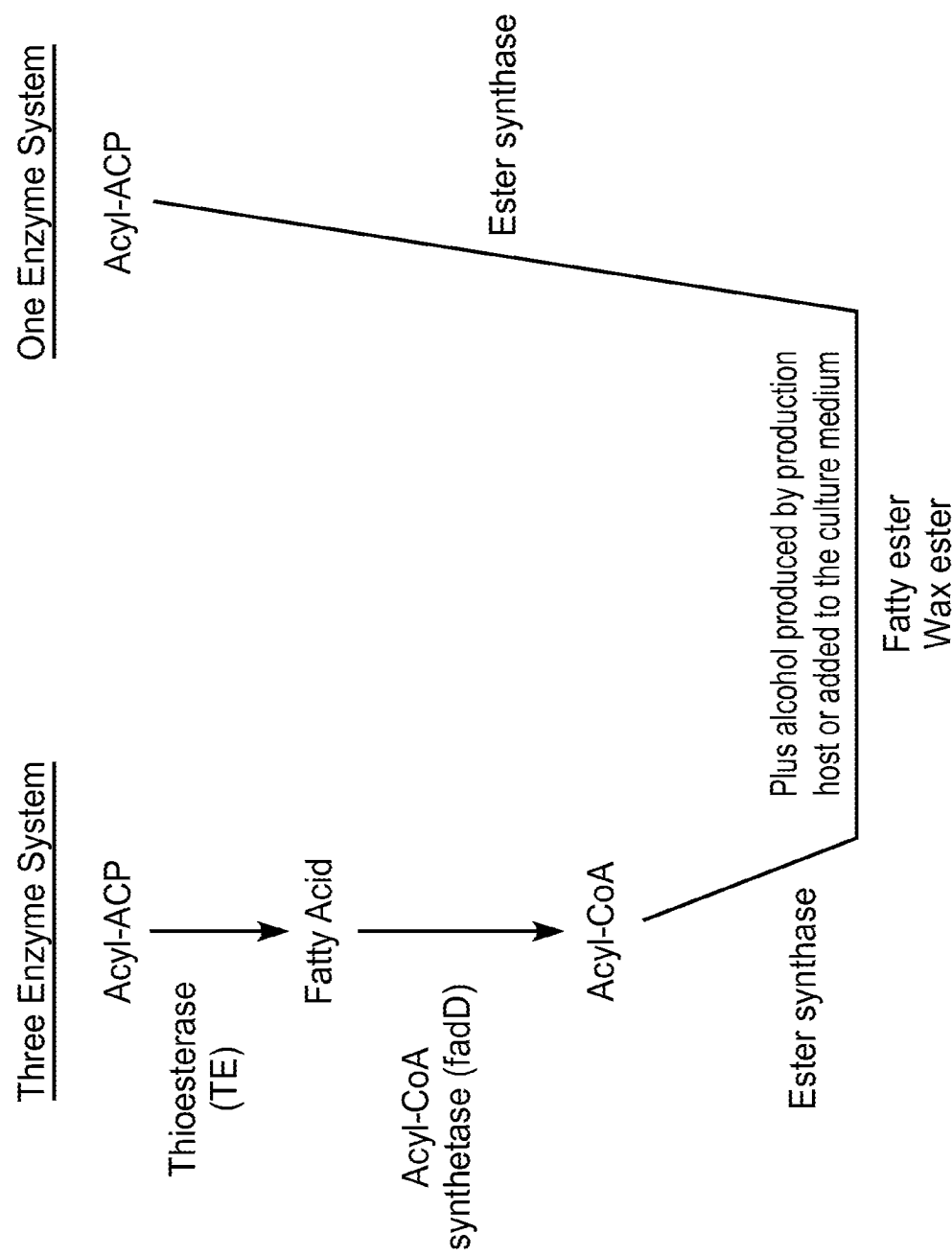


FIG. 3



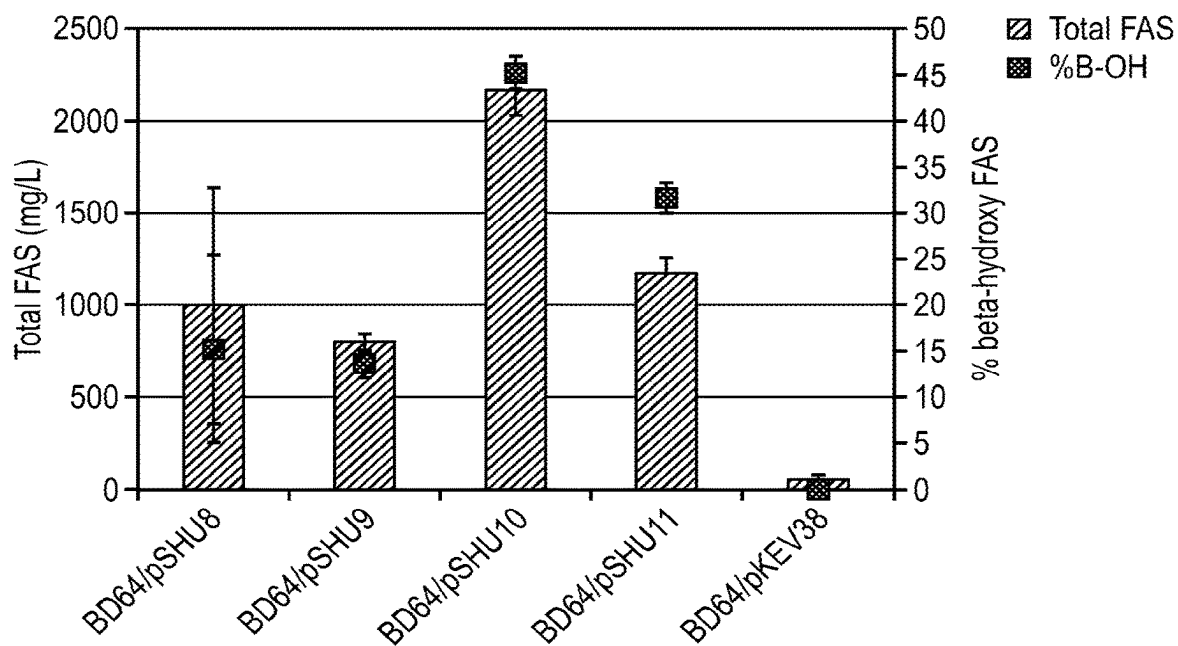


FIG. 4

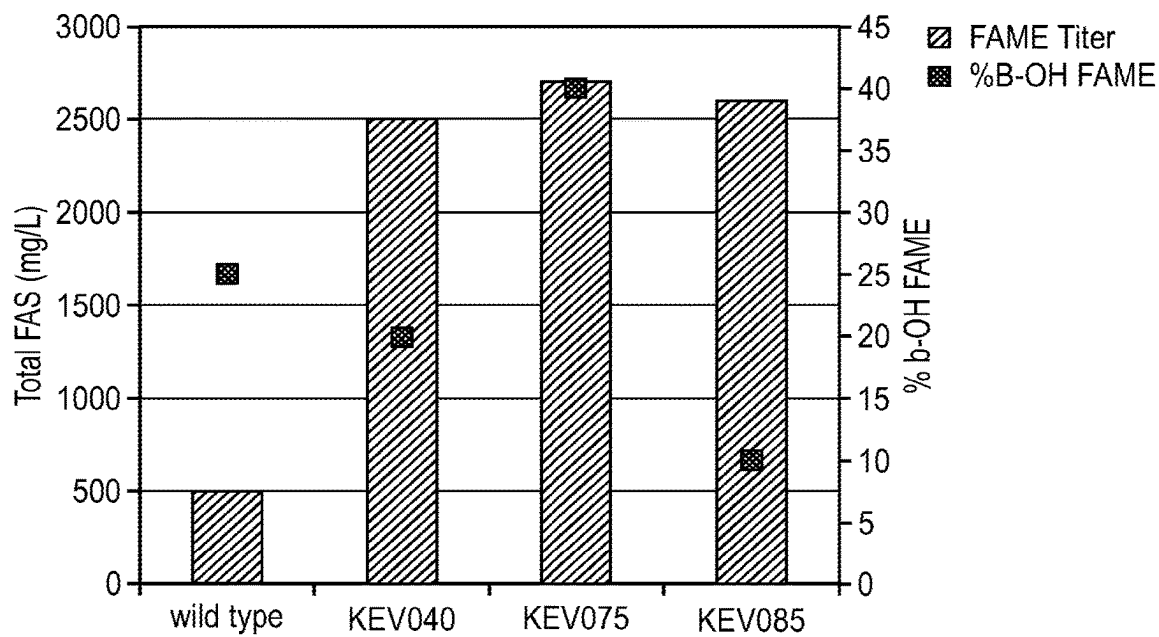


FIG. 5

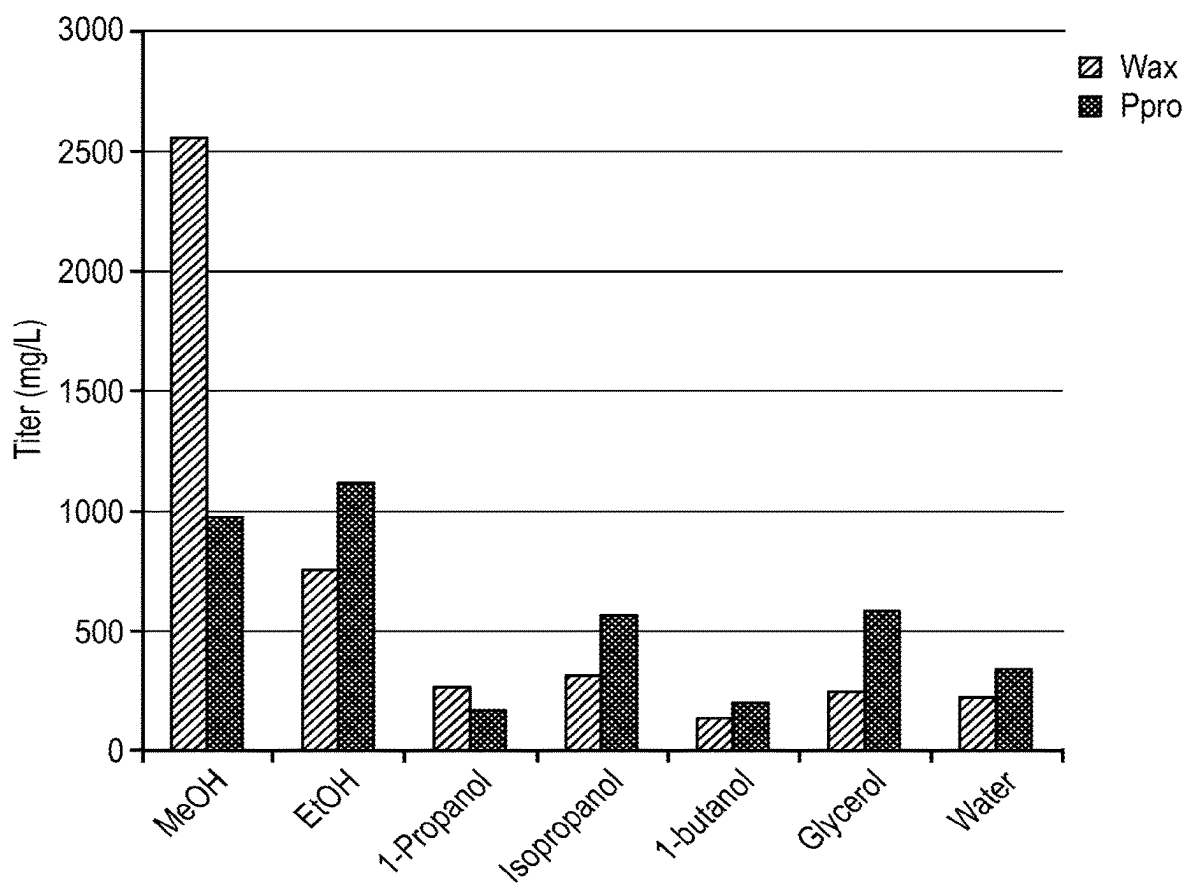


FIG. 6

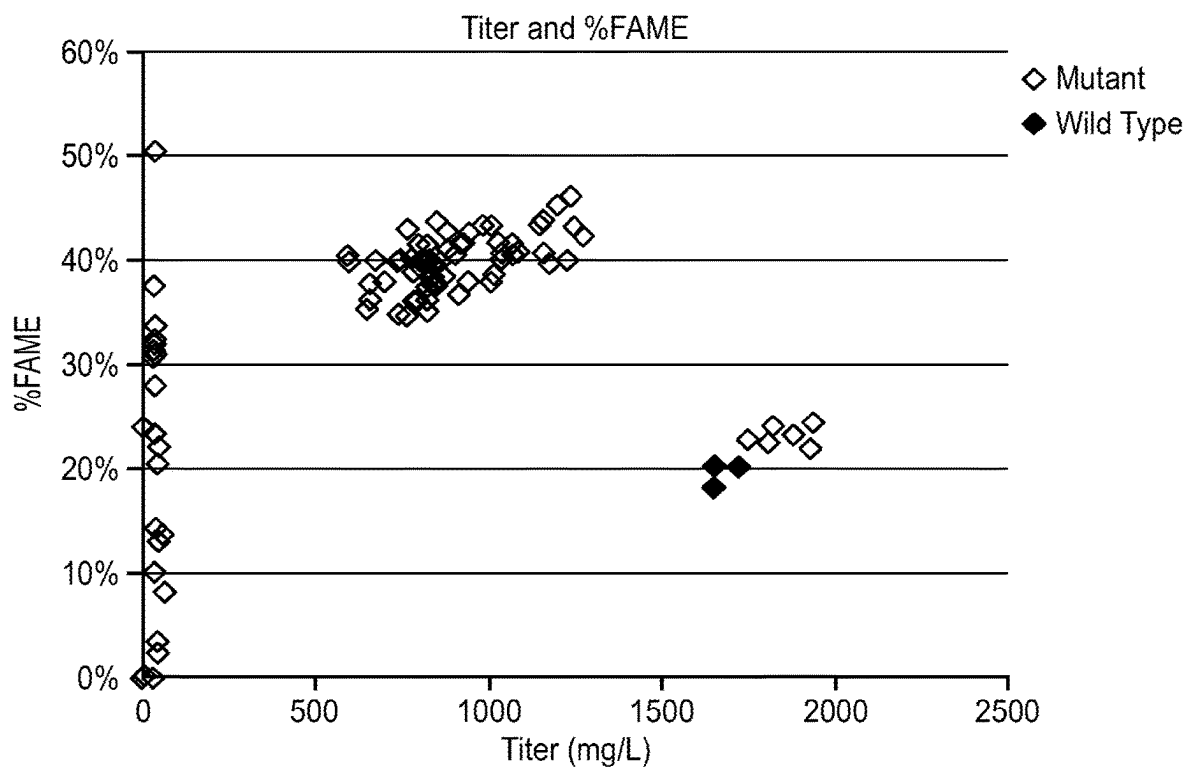


FIG. 7A

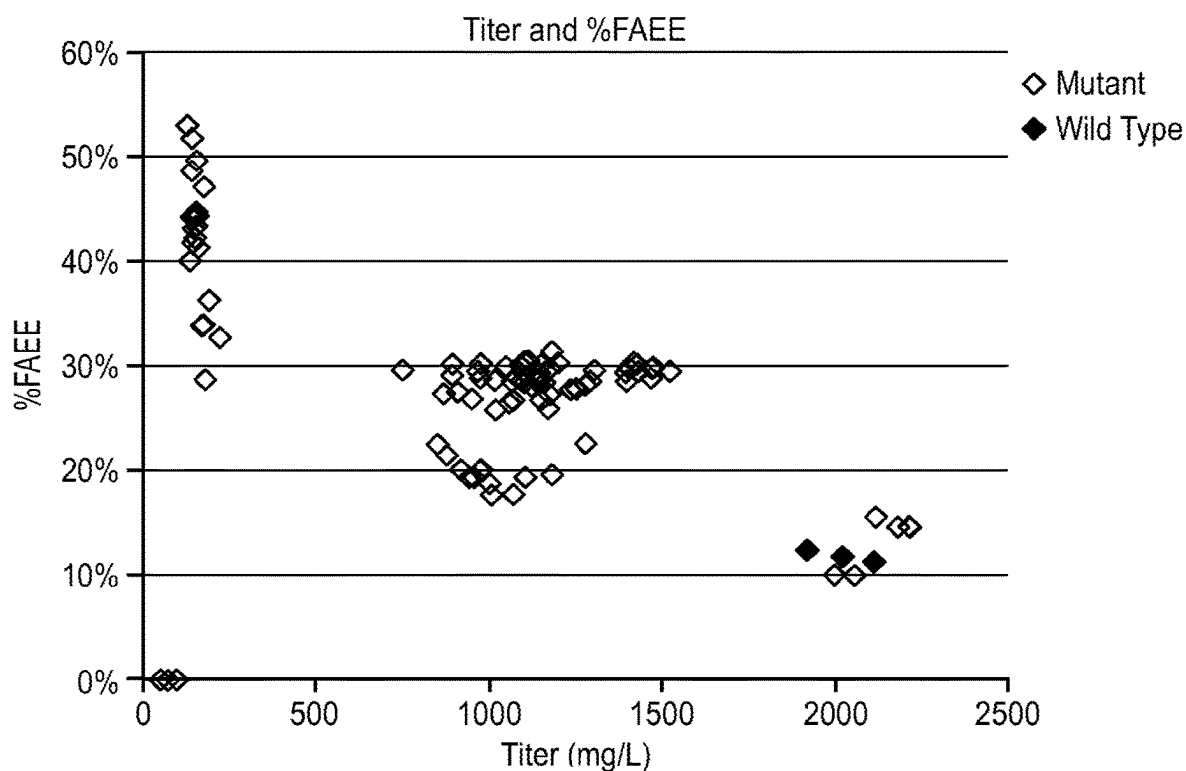
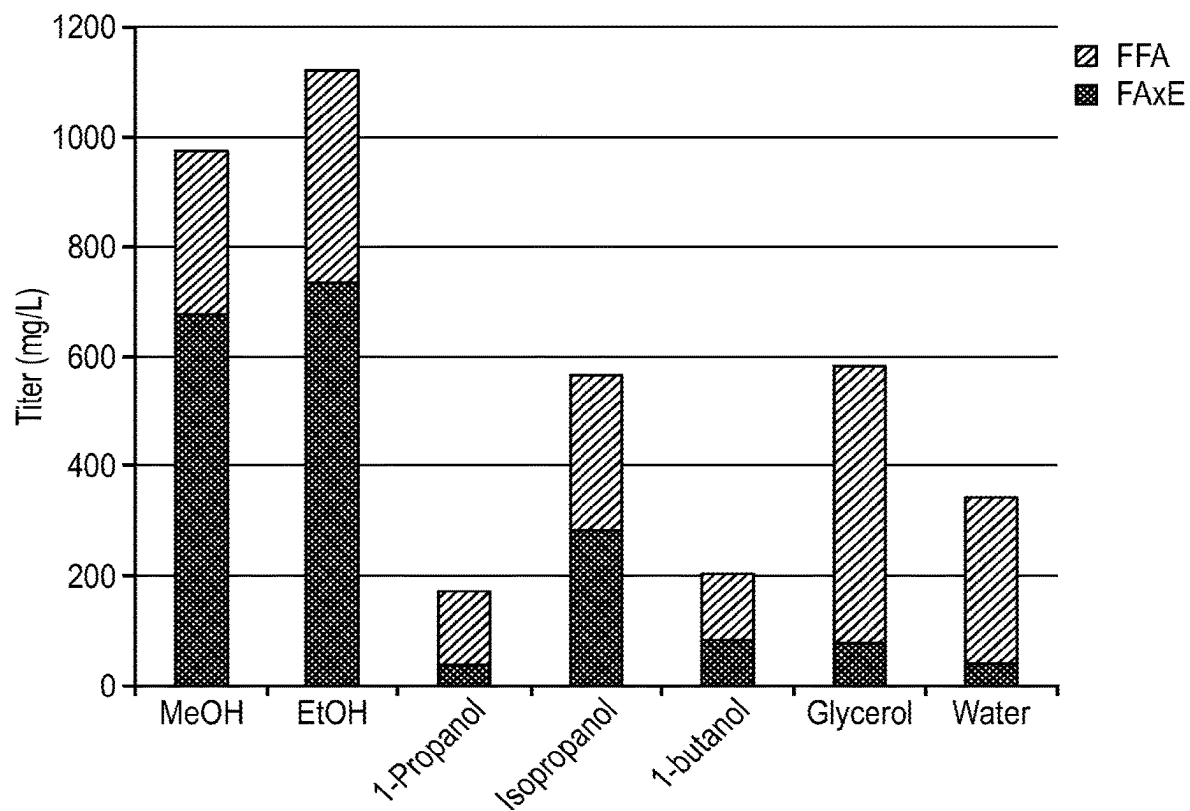


FIG. 7B

FIG. 8



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ENZYME VARIANTS WITH IMPROVED ESTER SYNTHASE PROPERTIES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/116,817 filed 9 Dec. 2020, which is a continuation of U.S. patent application Ser. No. 15/873,977, now U.S. Pat. No. 10,894,953, filed 18 Jan. 2018, which is a continuation of U.S. patent application Ser. No. 14/424,387, now U.S. Pat. No. 9,879,239, filed 26 Feb. 2015, which is a 371 of International Application No. PCT/US2013/032564 filed 15 Mar. 2013, which claims the benefit of U.S. Provisional Application No. 61/708,424 filed 1 Oct. 2012, and U.S. Provisional Application No. 61/701,191 filed 14 Sep. 2012, the entire disclosures of which are hereby incorporated by reference.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

This application contains references to amino acid sequences and/or nucleic acid sequences which have been submitted concurrently herewith as the sequence listing .xml file entitled "18561-000043-US-COD_13_May_2024-ST26.xml", file size 114 KiloBytes (KB), created on 13 May 2024. The aforementioned sequence listing is hereby incorporated by reference in its entirety.

FIELD

The disclosure relates to enzyme variants with improved ester synthase properties for the production of fatty esters.

BACKGROUND

Petroleum is a limited, natural resource found in the earth in liquid, gaseous, or solid forms. However, petroleum products are developed at considerable costs, both financial and environmental. In its natural form, crude petroleum extracted from the Earth has few commercial uses. It is a mixture of hydrocarbons, e.g., paraffins (or alkanes), olefins (or alkenes), alkynes, naphthenes (or cycloalkanes), aliphatic compounds, aromatic compounds, etc. of varying length and complexity. In addition, crude petroleum contains other organic compounds (e.g., organic compounds containing nitrogen, oxygen, sulfur, etc.) and impurities (e.g., sulfur, salt, acid, metals, etc.). Due to its high energy density and its easy transportability, most petroleum is refined into fuels, such as transportation fuels (e.g., gasoline, diesel, aviation fuel, etc.), heating oil, liquefied petroleum gas, etc.

Petrochemicals can be used to make specialty chemicals, such as plastics, resins, fibers, elastomers, pharmaceuticals, lubricants, or gels. Examples of specialty chemicals which can be produced from petrochemical raw materials include fatty acids, hydrocarbons (e.g., long chain hydrocarbons, branched chain hydrocarbons, saturated hydrocarbons, unsaturated hydrocarbons, etc.), fatty alcohols, fatty esters, fatty aldehydes, ketones, lubricants, etc. Specialty chemicals have many commercial uses. Fatty acids are used commercially as surfactants. Surfactants can be found in detergents and soaps. Fatty acids can also be used as additives in fuels, lubricating oils, paints, lacquers, candles, shortenings, cosmetics, and emulsifiers. In addition, fatty acids are used as accelerator activators in rubber products. Fatty acids can

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also be used as a feedstock to produce methyl esters, amides, amines, acid chlorides, anhydrides, ketene dimers, and peroxy acids and esters.

Esters have many commercial uses. For example, biodiesel, an alternative fuel, is comprised of esters (e.g., fatty acid methyl ester, fatty acid ethyl esters, etc.). Some low molecular weight esters are volatile with a pleasant odor which makes them useful as fragrances or flavoring agents. In addition, esters are used as solvents for lacquers, paints, and varnishes. Furthermore, some naturally occurring substances, such as waxes, fats, and oils are comprised of esters. Esters are also used as softening agents in resins and plastics, plasticizers, flame retardants, and additives in gasoline and oil. In addition, esters can be used in the manufacture of polymers, films, textiles, dyes, and pharmaceuticals.

There is a need for alternative routes to create both fuels and products currently derived from petroleum. As such, microbial systems offer the potential for the biological production of numerous types of biofuels and chemicals. Renewable fuels and chemicals can be derived from genetically engineered organisms (such as bacteria, yeast and algae). Naturally occurring biosynthetic pathways can be genetically altered to enable engineered organisms to synthesize renewable fuel and chemical products. In addition, microbes can be tailored, or metabolically engineered, to utilize various carbon sources as feedstock for the production of fuel and chemical products. For example, FAME (fatty acid methyl ester) can be produced by adding methanol to a culture of a recombinant host cell expressing a wax ester synthase, for example, a wax ester synthase derived from *Marinobacter hydrocarbonoclasticus* ("*M. hydrocarbonoclasticus*"). However, the expression of the wild type wax ester synthase from *M. hydrocarbonoclasticus* does not produce significant amounts of FAME, thus development of an improved wax ester synthase is highly desirable. In addition, the wild-type wax ester synthase from *M. hydrocarbonoclasticus* produces significant amounts of beta hydroxy (" β -OH") esters, which are not typically found in fuel or chemicals derived from plant sources. See, e.g., Patent Cooperation Treaty Application No. PCT/US12/31682, which is incorporated by reference herein. Thus, it would be desirable to engineer an ester synthase to produce higher yields of fatty ester; and an ester synthase that produces higher yields of fatty ester without producing significant amounts of β -OH ester, when expressed in a recombinant host cell. Finally, it would be desirable to engineer other enzymes that have ester synthase activity to further improve the yield of fatty ester production.

Notwithstanding the advances in the field, there remains a need for improvements in genetically modified enzymes, recombinant host cells, methods and systems in order to achieve robust and cost-effective production of fuels and chemicals of interest by fermentation of recombinant host cells. The present disclosure addresses this need by providing a number of ester synthase enzyme variants that increase the yield of fatty esters. Some of the ester synthase enzyme variants of the disclosure also alter (i.e., increase or reduce) the β -OH ester content of the fatty ester composition produced by fermentation of recombinant host cells that express the ester synthase enzyme variants. The present disclosure further addresses this need by providing a number of thioesterase variants that have been engineered to have greater ester synthase activity, thereby further enhancing fatty ester production.

SUMMARY

One aspect of the disclosure provides a variant ester synthase polypeptide having SEQ ID NO: 2, wherein the

variant ester synthase polypeptide is genetically engineered to have at least one mutation at an amino acid position including, but not limited to, amino acids 4, 5, 7, 15, 24, 30, 33, 39, 40, 41, 43, 44, 58, 73, 76, 77, 78, 80, 98, 99, 101, 102, 111, 122, 131, 146, 147, 149, 150, 155, 157, 162, 164, 166, 170, 171, 172, 173, 177, 179, 182, 184, 185, 186, 187, 188, 190, 192, 193, 195, 197, 201, 202, 203, 206, 207, 212, 216, 219, 234, 239, 242, 243, 244, 246, 255, 257, 258, 259, 260, 262, 263, 264, 265, 266, 267, 285, 288, 289, 293, 294, 301, 302, 303, 304, 306, 307, 309, 310, 311, 313, 314, 315, 316, 317, 319, 320, 323, 328, 334, 348, 349, 351, 352, 353, 356, 357, 360, 366, 375, 381, 393, 394, 395, 409, 411, 413, 420, 424, 442, 443, 447, 454, 455, 457, 458, 461, 466, 468, and 472, wherein the variant ester synthase polypeptide has improved fatty acid methyl ester activity compared to a corresponding wild type polypeptide. In a related aspect, the variant polypeptide is a *Marinobacter hydrocarbonoclasticus* (WS377) ester synthase polypeptide. In another related aspect, the expression of the variant ester synthase polypeptide in a recombinant host cell results in a higher titer of fatty ester compositions compared to a recombinant host cell expressing the corresponding wild type polypeptide. The titer is at least about 5 percent or greater. The fatty ester compositions include, but are not limited to, fatty acid methyl esters (FAME), fatty acid ethyl esters (FAEE), fatty acid propyl esters, fatty acid isopropyl esters, fatty acid butyl esters, monoglycerides, fatty acid isobutyl esters, fatty acid 2-butyl esters, and fatty acid tert-butyl esters. The variant ester synthase has improved ester synthase activity, particularly improved fatty acid methyl ester activity that results in improved properties including, but not limited to, increased beta hydroxy esters, decreased beta hydroxyl esters, increased chain lengths of fatty acid esters, and decreased chain lengths of fatty acid esters.

Another aspect of the disclosure provides a variant ester synthase polypeptide having SEQ ID NO: 2, wherein the variant ester synthase polypeptide is genetically engineered to have at least one mutation at an amino acid position including, but not limited to, amino acids 7, 24, 30, 41, 99, 111, 122, 146, 147, 149, 150, 171, 172, 173, 179, 187, 192, 212, 234, 239, 244, 246, 258, 264, 266, 267, 285, 301, 302, 303, 304, 306, 307, 309, 310, 311, 313, 314, 315, 316, 320, 348, 349, 352, 356, 357, 360, 375, 381, 393, 409, 411, 424, 443, 455, 457, 458, 461 and 472. The expression of the variant ester synthase polypeptide in a recombinant host cell may result in the production of an altered percentage of beta hydroxy esters as compared to a recombinant host cell expressing the corresponding wild type ester synthase polypeptide, wherein the altered percentage of beta hydroxy esters is an increased or decreased percentage of beta hydroxy esters.

Another aspect of the disclosure provides a variant ester synthase polypeptide having SEQ ID NO: 2, wherein the variant ester synthase polypeptide is genetically engineered to have at least one mutation at an amino acid position including, but not limited to, amino acids 5, 7, 15, 24, 33, 44, 58, 73, 78, 101, 111, 162, 171, 179, 184, 187, 188, 192, 201, 207, 212, 234, 243, 244, 255, 257, 267, 307, 310, 317, 320, 348, 349, 353, 357, 366, 381, 394, 409, 411, 413, 442, 443, and 461.

Another aspect of the disclosure provides a variant ester synthase polypeptide having SEQ ID NO: 2, wherein the variant ester synthase polypeptide is genetically engineered to have at least one mutation at an amino acid position including, but not limited to, amino acids G4R, T5P, T5S, D7N, S15G, T24M, T24W, L30H, G33D, G33S, L39A, L39M, L39S, R40S, D41A, D41G, D41H, D41Y, V43K,

V43S, T44A, T44F, T44K, Y58C, Y58L, A73Q, V76L, D77A, K78F, K78W, I80V, R98D, E99Q, G101L, I102R, P111D, P111G, P111S, H122S, R131M, I146K, I146L, I146R, S147A, V149L, R150P, V155G, T157S, R162E, N164D, N164R, P166S, T170R, T170S, V171E, V171F, V171H, V171R, V171W, R172S, R172W, P173W, R177V, A179S, A179V, D182G, E184F, E184G, E184L, E184R, E184S, A185L, A185M, S186T, V187G, V187R, P188R, A190P, A190R, A190W, S192A, S192L, S192V, Q193R, Q193S, M195G, A197T, A197V, Q201A, Q201V, Q201W, A202L, D203R, P206F, R207A, G212A, G212L, V216I, V219L, V234C, H239G, T242K, T242R, A243R, Q244G, R246A, R246G, R246L, R246Q, R246V, R246W, D255E, L257I, K258R, N259E, N259Q, L260V, H262Q, A263V, S264D, S264V, S264W, G265N, G266A, G266S, S267G, A285L, A285R, A285V, N288D, N289G, T293A, P294G, V301A, N302G, I303G, I303R, I303W, R304W, A306G, D307F, D307G, D307L, D307N, D307R, D307V, E309A, E309G, E309S, G310H, G310R, G310V, T311S, T313S, Q314G, I315F, S316G, F317W, I319G, A320C, A323G, D328F, Q334K, Q334S, Q348A, Q348R, K349A, K349C, K349H, K349Q, P351G, K352I, K352N, S353K, S353T, T356G, T356W, Q357V, M360Q, M360R, M360S, M360W, Y366G, Y366W, G375A, G375V, G375S, V381F, E393G, E393R, E393W, G394E, T395E, V409L, L411A, A413T, I420V, S424G, S424Q, S442E, S442G, M443G, A447C, A447I, A447L, L454V, D455E, L457Y, E458W, I461G, I461L, I461V, K466N, A468G, K472T and K472. In a particular aspect of the disclosure the variant ester synthase polypeptide includes mutations D7N, A179V and V381F; D7N, A179V, Q348R and V381F; D7N, A179V, V187R, G212A, Q357V, V381F and M443G; T5S, S15G, P111S, V171R, P188R, F317W, S353T, V409L and S442G; T5S, S15G, K78F, P111S, V171R, P188R, S192V, A243R, F317W, K349H, S353T, V409L and S442G; T5S, S15G, V76L, P111S, V171R, P188R, K258R, S316G, F317W, S353T, M360R, V409L and S442G; T5S, S15G, P111S, V171R, P188R, Q244G, S267G, G310V, F317W, A320C, S353T, Y366W, V409L and S442G; S15G, P111S, V155G, P166S, V171R, P188R, F317W, Q348A, S353T, V381F, V409L and S442G; S15G, L39S, D77A, P111S, V171R, P188R, T313S, F317W, Q348A, S353T, V381F, V409L, I420V and S442G; T5S, S15G, T24W, T44F, P111S, I146L, V171R, P188R, D307N, F317W, S353T, V409L and S442G; T5S, S15G, K78F, P111S, V171R, P188R, A243R, D255E, L257I, N259Q, L260V, H262Q, G265N, A285V, N288D, N289G, F317W, Q348R, S353T, V381F, V409L and S442G; T5S, S15G, K78F, P111S, V171R, P188R, S192V, R207A, A243R, D255E, L257I, N259Q, L260V, H262Q, G265N, A285V, N288D, N289G, F317W, H349K, S353T, V381F, V409L and S442G; T5P, S15G, G33D, T44K, Y58L, P111S, R162E, V171R, P188R, R207A, V234C, Q244G, D255E, S267G, D307N, G310V, F317W, A320C, S353T, Y366W, G394E, V409L, S442G and I461V; T5P, S15G, G33D, T44K, Y58C, P111S, V171R, P188R, R207A, V234C, Q244G, L257I, S267G, D307N, G310V, F317W, A320C, Q334K, S353T, Y366W, G394E, V409L, S442G and I461V; or mutations T5P, S15G, Y58L, P111S, R162E, T170S, V171R, A179S, P188R, R207A, V234C, Q244G, L257I, S267G, D307N, G310V, F317W, A320C, S353K, Y366W, G394E, V409L, S442G and I461V.

Another aspect of the disclosure provides a recombinant host cell genetically engineered to express a variant ester

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synthase polypeptide. In a related aspect, the disclosure provides a cultured recombinant host cell having a polynucleotide sequence encoding a variant ester synthase polypeptide, wherein the recombinant host cell when expressing the variant ester synthase polypeptide produces a fatty acid ester composition with higher titer, higher yield and/or higher productivity of fatty acid esters compared to a host cell expressing the corresponding wild type polypeptide, when cultured in medium containing a carbon source under conditions effective to express the variant ester synthase polypeptide. The titer is at least about 5 percent or greater. In a related aspect, the recombinant host cell when expressing the variant ester synthase polypeptide produces a fatty ester composition with an altered percentage of beta hydroxy esters as compared to a host cell that expresses the wild type ester synthase polypeptide. Herein, the altered percentage of beta hydroxy esters is an increased or decreased percentage of beta hydroxy esters.

Still another aspect of the invention provides a cell culture including the recombinant host cell and a fatty ester composition. The fatty ester composition may include one or more of a C6, C8, C10, C12, C13, C14, C15, C16, C17 or C18 fatty ester. In addition, the fatty ester composition may include unsaturated fatty esters and/or saturated fatty esters and/or branched chain fatty esters. The fatty ester composition may further include a fatty ester having a double bond at position 7 in the carbon chain between C7 and C8 from the reduced end of the fatty ester.

The disclosure further contemplates a variant *Marinobacter hydrocarbonoclasticus* (WS377) ester synthase polypeptide having SEQ ID NO: 2, wherein the variant WS377 polypeptide has improved fatty acid methyl ester activity compared to the wild type WS377 polypeptide, and wherein the variant WS377 has at least one mutation at an amino acid position including, but not limited to, amino acids 4, 5, 7, 15, 24, 30, 33, 39, 40, 41, 43, 44, 58, 73, 76, 77, 78, 80, 98, 99, 101, 102, 111, 122, 131, 146, 147, 149, 150, 155, 157, 162, 164, 166, 170, 171, 172, 173, 177, 179, 182, 184, 185, 186, 187, 188, 190, 192, 193, 195, 197, 201, 202, 203, 206, 207, 212, 216, 219, 234, 239, 242, 243, 244, 246, 255, 257, 258, 259, 260, 262, 263, 264, 265, 266, 267, 285, 288, 289, 293, 294, 301, 302, 303, 304, 306, 307, 309, 310, 311, 313, 314, 315, 316, 317, 319, 320, 323, 328, 334, 348, 349, 351, 352, 353, 356, 357, 360, 366, 375, 381, 393, 394, 395, 409, 411, 413, 420, 424, 442, 443, 447, 454, 455, 457, 458, 461, 466, 468, and 472.

Another aspect of the disclosure provides a variant thioesterase polypeptide comprising an amino acid sequence having at least about 90% sequence identity to SEQ ID NO: 51, wherein the variant ester synthase polypeptide is genetically engineered to have at least one mutation at an amino acid position including, but not limited to, amino acids 32, 33, 34, 38, 45, 46, 48, 49, 51, 52, 76, 81, 82, 84, 88, 112, 115, 116, 118, 119, 148, 149, 156, 159 and 164, wherein the variant thioesterase polypeptide has greater ester synthase activity compared to the corresponding wild type polypeptide. In a related aspect of the disclosure, the variant polypeptide is a *Photobacterium profundum* (Ppro) thioesterase polypeptide. The expression of the variant thioesterase polypeptide in a recombinant host cell results in a higher titer of fatty ester compositions compared to a recombinant host cell expressing the corresponding wild type polypeptide, wherein said titer is at least about 5 percent or greater.

Another aspect of the disclosure provides a variant thioesterase polypeptide having at least one mutation including, but not limited to, amino acids K32M, Q33A, Q33R, Q34R, I38Y, I45R, S46K, S46V, S46W, D48E, D48F, D48L,

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D48M, T49D, T49L, T49V, T49W, G51R, N52C, N52F, N52K, N52L, N52R, N76I, N76L, N76V, G81C, F82G, F82I, F82R, Q84A, Q84L, Q84V, R88H, V112L, N115F, N115W, N115Y, Y116H, Y116N, Y116P, Y116R, Y116S, Y116T, K118Q, R119C, I148Y, L149F, L149M, L149T, N156K, N156R, N156Y, L159K, L159M and D164T. A related aspect of the disclosure provides a variant thioesterase polypeptide having mutations S46V and Y116H; S46V, D48M, G51R, N76V and Y116H; S46V, D48M, T49L, G51R, N76V, Y116H and I148Y; S46V, D48M, T49L, G51R, N76V, N115F, Y116H and I148Y; Q33R, Q34R, S46V, D48M, T49L, G51R, N52R, N76V, V112L, N115W, Y116H and I148Y; Q33R, S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115W, Y116H, I148Y and L149M; or mutations S46V, D48M, T49L, G51R, N52R, N76V, Q84A, V112L, N115W, Y116H, I148Y and L149M.

The disclosure further encompasses a recombinant host cell genetically engineered to express a variant thioesterase polypeptide that has ester synthase activity. A cultured recombinant host cell including a polynucleotide sequence encoding a variant thioesterase polypeptide is further contemplated by the disclosure. The recombinant host cell, when expressing the variant thioesterase polypeptide, produces a fatty acid ester composition with higher titer, higher yield and/or higher productivity of fatty acid esters compared to a host cell expressing the corresponding wild type thioesterase polypeptide, when cultured in medium containing a carbon source under conditions effective to express the variant thioesterase polypeptide. The titer is at least about 5 percent or greater. The cell culture further includes the recombinant host cell and a fatty ester composition. The fatty ester composition may include one or more of a C6, C8, C10, C12, C13, C14, C15, C16, C17 or C18 fatty ester. In addition, the fatty ester composition may include unsaturated fatty esters and/or saturated fatty esters and/or branched chain fatty esters. The fatty ester composition may further include a fatty ester having a double bond at position 7 in the carbon chain between C7 and C8 from the reduced end of the fatty ester.

Another aspect of the disclosure provides a variant thioesterase polypeptide with ester synthase activity, wherein the variant thioesterase polypeptide includes amino acids 73-81 of SEQ ID NO: 51 as follows LG[AG][VIL]D[AG]LRG.

Yet, another aspect of the disclosure provides a variant *Photobacterium profundum* (Ppro) thioesterase polypeptide having at least about 90% sequence identity to a corresponding wild type Ppro polypeptide sequence having SEQ ID NO: 51, wherein the variant Ppro polypeptide has greater ester synthase activity compared to the wild type Ppro polypeptide, and where the variant Ppro polypeptide has at least one mutation including, but not limited to, amino acids 32, 33, 34, 38, 45, 46, 48, 49, 51, 52, 76, 81, 82, 84, 88, 112, 115, 116, 118, 119, 148, 149, 156, 159 and 164.

The disclosure still encompasses a recombinant host cell including (a) a polynucleotide encoding a variant polypeptide with increased ester synthase activity, (b) an activated fatty acid, and (c) an alcohol, wherein the recombinant host cell produces a fatty ester composition under conditions effective to express an ester synthase activity of the variant polypeptide. The activated fatty acid is an Acyl-ACP or an Acyl-CoA. The recombinant host cells may further utilize a carbon source. The fatty ester composition is produced at a titer of at least about 5 percent or higher.

Another aspect of the disclosure provides a polynucleotide encoding a variant ester synthase polypeptide, wherein the polynucleotide has at least about 85% sequence identity

to SEQ ID NO: 1, wherein the polynucleotide further includes SEQ ID NO: 27 prior to ATG start.

Yet, another aspect of the disclosure provides a variant ester synthase polypeptide having an amino acid sequence that has at least about 95% sequence identity to SEQ ID NO: 29; or 95% sequence identity to SEQ ID NO: 33; or 95% sequence identity to SEQ ID NO: 35; or 95% sequence identity to SEQ ID NO: 37; or 95% sequence identity to SEQ ID NO: 39; or 95% sequence identity to SEQ ID NO: 41; or 95% sequence identity to SEQ ID NO: 43. Herein, the expression of the variant ester synthase polypeptide in a recombinant host cell results in a production of a higher titer, higher yield, or increased productivity of fatty acid esters as compared to a recombinant host cell expressing a wild type *Marinobacter hydrocarbonoclasticus* (WS377) ester synthase polypeptide. The titer is at least about 5% or higher.

BRIEF DESCRIPTION OF THE FIGURES

The present disclosure is best understood when read in conjunction with the accompanying figures, which serve to illustrate the preferred embodiments. It is understood, however, that the disclosure is not limited to the specific embodiments disclosed in the figures.

FIG. 1 is a schematic depiction of an exemplary three gene biosynthetic pathway that may be incorporated into a recombinant host cells for production of fatty acid esters of various chain lengths. The abbreviations used include TE for thioesterase, FACS for fatty acyl CoA synthetase, and AT for acyltransferase.

FIG. 2 is a schematic depiction of a proposed futile cycle catalyzed by thioesterase.

FIG. 3 is a schematic depiction of two exemplary biosynthetic pathways for production of fatty esters starting with acyl-ACP, where the production of fatty esters is accomplished by a one enzyme system or a three enzyme system.

FIG. 4 shows the results of a GC-FID screen from a plate fermentation of combination library hits. The combination library was built using Group 0 primers and pKEV38 as a template, transformed into BD64.

FIG. 5 shows FAME titers and % β -OH FAME produced by mutant strains KEV040, KEV075 and KEV085 compared to a strain which contains wild-type '377 ester synthase.

FIG. 6 shows the titer of total fatty acid species produced by KEV040 (Wax) and becos.128 (Ppro) when methanol, ethanol, 1-propanol, isopropanol, 1-butanol or glycerol was fed to the strains.

FIGS. 7A and 7B show the titer and % FAME (FIG. 7A) or FAEE (FIG. 7B) produced when a Ppro combination library built was fed methanol to evaluate the ability of Ppro variants to use methanol to make FAME (FIG. 7A) or ethanol to make FAEE (FIG. 7B).

FIG. 8 shows the composition of fatty acid species produced by becos.128 when cultured in the presence of (fed) methanol, ethanol, 1-propanol, isopropanol, 1-butanol or glycerol. FFA refers to free fatty acid species and FAXE refers to the ester species.

DETAILED DESCRIPTION

General Overview

The disclosure relates to variant enzymes with improved ester synthase properties for the production of fatty esters. Herein, the disclosure relates to enzymes such as variant

ester synthase enzymes having improved ester synthase properties and variant thioesterase enzymes designed to have ester synthase activity. Both types of enzymes are engineered to improve the production of fatty esters when used alone or in combination with other proteins. In order to illustrate this, the applicants have engineered an ester synthase from *Marinobacter hydrocarbonoclasticus* and a thioesterase from *Photobacterium profundum* to produce fatty esters in vivo without the need to overexpress a fatty acyl-CoA synthetase or a different thioesterase. So far, it has been thought that in order to produce higher yields of fatty esters, at least an acyl-CoA synthetase or a thioesterase enzyme has to be simultaneously overexpressed with an ester synthase because any fatty ester production requires the co-expression of thioesterase, acyl-CoA synthetase, and ester synthase. Moreover, native *Marinobacter* ester synthases, and native thioesterases in general are not known to utilize short chain alcohols. However, the applicants have engineered variant enzymes with ester synthase activity such that the enzymes can utilize short chain alcohols, including methanol and ethanol, allowing the enzymes to produce fatty esters. For example, the variant ester synthase and variant thioesterase with ester synthase activity can produce fatty acid methyl ester(s) (FAME) and/or fatty acid ethyl ester(s) (FAEE) at high quantities. As such, the applicants have engineered variant enzymes that each lead to a higher production of esters in vivo.

Herein, the applicants employ a convenient one-gene-system (see FIG. 3) for the production of fatty esters. A one-gene-system typically employs a gene that encodes a variant enzyme with improved ester synthase activity in combination with an activated fatty acid, such as an Acyl-ACP or Acyl-CoA, (i.e., as a substrate) and alcohol to produce fatty esters in a recombinant host cell. There are certain advantages of using a one-gene-system (i.e., rather than a multi-gene-system) for the production of fatty esters. For example, one such advantage is the simplicity of the system itself which allows it to be faster implemented and more readily monitored. Another advantage is that a one-gene-system is more energy efficient for the cell because it likely bypasses or avoids the so-called futile cycle (i.e., a substrate futile cycle, wherein two metabolic pathways run simultaneously in opposite directions and have no overall effect), thereby increasing the yield of ester production in vivo.

More specifically, fatty acid ester production in *Escherichia coli* typically relies on the co-expression of a thioesterase ('tesA), an acyl-CoA synthetase (fadD) and a wax or ester synthase. In order to maximize ester production the three genes are carefully balanced in order to minimize production of free fatty acids ("FFA"). The thioesterase can act on both acyl-ACP (product of fatty acid biosynthesis) and acyl-CoA (product of acyl-CoA synthetase) generating free fatty acids (substrate of acyl-CoA synthetase). The present disclosure is directed to a modified ester synthase which has improved properties relative to the wild type enzyme, cloned from *Marinobacter hydrocarbonoclasticus* DSM 8798 GenBank: EF219377 (referred to herein as "'377"). The modified '377 has been shown to act on both acyl-ACP and acyl-CoA to generate fatty acid esters such as fatty acid methyl ester ("FAME").

In addition, the percentage of FAME of the total acyl species (FAME+FFA) generated varies depending on the thioesterase being used. The 'tesA of *Photobacterium profundum* ("P. profundum" or "Ppro") naturally produces a small percentage of FAME. The *Photobacterium profundum* 'tesA was engineered to produce a higher percentage of

FAME. Thus, the present disclosure is further directed to a modified thioesterase ("tesA") from *Photobacterium profundum* (referred to herein as "Ppro") that was genetically engineered to produce a higher percentage of FAME without the need to overexpress any other gene in a host cell such as *E. coli*. While not wanting to be bound by theory, an ester synthase that acts directly on acyl-ACP (a product of fatty acid biosynthesis) could eliminate the futile cycle that theoretically exists in the current biodiesel pathway, thus reserving resources from the cell, which could be directed towards fatty acid biosynthesis, which should also improve the production.

Definitions

As used in this specification and the appended claims, the singular forms "a," "an" and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "a recombinant host cell" includes two or more such recombinant host cells, reference to "a fatty ester" includes one or more fatty esters, or mixtures of esters, reference to "a nucleic acid coding sequence" includes one or more nucleic acid coding sequences, reference to "an enzyme" includes one or more enzymes, and the like.

Sequence Accession numbers throughout this description were obtained from databases provided by the NCBI (National Center for Biotechnology Information) maintained by the National Institutes of Health, U.S.A. (which are identified herein as "NCBI Accession Numbers" or alternatively as "GenBank Accession Numbers"), and from the UniProt Knowledgebase (UniProtKB) and Swiss-Prot databases provided by the Swiss Institute of Bioinformatics (which are identified herein as "UniProtKB Accession Numbers").

EC numbers are established by the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (IUBMB), description of which is available on the IUBMB Enzyme Nomenclature website on the World Wide Web. EC numbers classify enzymes according to the reaction catalyzed.

As used herein, the term "nucleotide" refers to a monomeric unit of a polynucleotide that consists of a heterocyclic base, a sugar, and one or more phosphate groups. The naturally occurring bases (guanine, (G), adenine, (A), cytosine, (C), thymine, (T), and uracil (U)) are typically derivatives of purine or pyrimidine, though it should be understood that naturally and non-naturally occurring base analogs are also included. The naturally occurring sugar is the pentose (five-carbon sugar) deoxyribose (which forms DNA) or ribose (which forms RNA), though it should be understood that naturally and non-naturally occurring sugar analogs are also included. Nucleic acids are typically linked via phosphate bonds to form nucleic acids or polynucleotides, though many other linkages are known in the art (e.g., phosphorothioates, boranophosphates, and the like).

The term "polynucleotide" refers to a polymer of ribonucleotides (RNA) or deoxyribonucleotides (DNA), which can be single-stranded or double-stranded and which can contain non-natural or altered nucleotides. The terms "polynucleotide," "nucleic acid sequence," and "nucleotide sequence" are used interchangeably herein to refer to a polymeric form of nucleotides of any length, either RNA or DNA. These terms refer to the primary structure of the molecule, and thus include double- and single-stranded DNA, and double- and single-stranded RNA. The terms include, as equivalents, analogs of either RNA or DNA made from nucleotide analogs and modified polynucleotides such

as, though not limited to methylated and/or capped polynucleotides. The polynucleotide can be in any form, including but not limited to, plasmid, viral, chromosomal, EST, cDNA, mRNA, and rRNA.

As used herein, the terms "polypeptide" and "protein" are used interchangeably to refer to a polymer of amino acid residues. The term "recombinant polypeptide" refers to a polypeptide that is produced by recombinant techniques, wherein generally DNA or RNA encoding the expressed protein is inserted into a suitable expression vector that is in turn used to transform a host cell to produce the polypeptide.

As used herein, the terms "homolog," and "homologous" refer to a polynucleotide or a polypeptide comprising a sequence that is at least about 50% identical to the corresponding polynucleotide or polypeptide sequence. Preferably homologous polynucleotides or polypeptides have polynucleotide sequences or amino acid sequences that have at least about 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least about 99% homology to the corresponding amino acid sequence or polynucleotide sequence. As used herein the terms sequence "homology" and sequence "identity" are used interchangeably.

One of ordinary skill in the art is well aware of methods to determine homology between two or more sequences. Briefly, calculations of "homology" between two sequences can be performed as follows. The sequences are aligned for optimal comparison purposes (e.g., gaps can be introduced in one or both of a first and a second amino acid or nucleic acid sequence for optimal alignment and non-homologous sequences can be disregarded for comparison purposes). In one preferred embodiment, the length of a first sequence that is aligned for comparison purposes is at least about 30%, preferably at least about 40%, more preferably at least about 50%, even more preferably at least about 60%, and even more preferably at least about 70%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 98%, or about 100% of the length of a second sequence. The amino acid residues or nucleotides at corresponding amino acid positions or nucleotide positions of the first and second sequences are then compared. When a position in the first sequence is occupied by the same amino acid residue or nucleotide as the corresponding position in the second sequence, then the molecules are identical at that position. The percent homology between the two sequences is a function of the number of identical positions shared by the sequences, taking into account the number of gaps and the length of each gap, that need to be introduced for optimal alignment of the two sequences. The comparison of sequences and determination of percent homology between two sequences can be accomplished using a mathematical algorithm, such as BLAST (Altschul et al., J. Mol. Biol., 215 (3): 403-410 (1990)). The percent homology between two amino acid sequences also can be determined using the Needleman and Wunsch algorithm that has been incorporated into the GAP program in the GCG software package, using either a Blossum 62 matrix or a PAM250 matrix, and a gap weight of 16, 14, 12, 10, 8, 6, or 4 and a length weight of 1, 2, 3, 4, 5, or 6 (Needleman and Wunsch, J. Mol. Biol., 48:444-453 (1970)). The percent homology between two nucleotide sequences also can be determined using the GAP program in the GCG software package, using a NWSgapdna.CMP matrix and a gap weight of 40, 50, 60, 70, or 80 and a length weight of 1, 2, 3, 4, 5, or 6. One of ordinary skill in the art can perform initial homology calculations and adjust the algorithm parameters accordingly. A preferred set of parameters (and the one that should be used if a practi-

tioner is uncertain about which parameters should be applied to determine if a molecule is within a homology limitation of the claims) are a Blossum 62 scoring matrix with a gap penalty of 12, a gap extend penalty of 4, and a frameshift gap penalty of 5. Additional methods of sequence alignment are known in the biotechnology arts (see, e.g., Rosenberg, *BMC Bioinformatics*, 6:278 (2005); Altschul, et al., *FEBS J.*, 272 (20): 5101-5109 (2005)).

The term “hybridizes under low stringency, medium stringency, high stringency, or very high stringency conditions” describes conditions for hybridization and washing. Guidance for performing hybridization reactions can be found in *Current Protocols in Molecular Biology*, John Wiley & Sons, N.Y. (1989), 6.3.1-6.3.6. Aqueous and non-aqueous methods are described in that reference and either method can be used. Specific hybridization conditions referred to herein are as follows: 1) low stringency hybridization conditions—6× sodium chloride/sodium citrate (SSC) at about 45° C., followed by two washes in 0.2×SSC, 0.1% SDS at least at 50° C. (the temperature of the washes can be increased to 55° C. for low stringency conditions); 2) medium stringency hybridization conditions—6×SSC at about 45° C., followed by one or more washes in 0.2×SSC, 0.1% SDS at 60° C.; 3) high stringency hybridization conditions—6×SSC at about 45° C., followed by one or more washes in 0.2×SSC, 0.1% SDS at 65° C.; and 4) very high stringency hybridization conditions—0.5M sodium phosphate, 7% SDS at 65° C., followed by one or more washes at 0.2×SSC, 1% SDS at 65° C. Very high stringency conditions (4) are the preferred conditions unless otherwise specified.

An “endogenous” polypeptide refers to a polypeptide encoded by the genome of the parental microbial cell (also termed “host cell”) from which the recombinant cell is engineered (or “derived”). An “exogenous” polypeptide refers to a polypeptide which is not encoded by the genome of the parental microbial cell. A variant (i.e., mutant) polypeptide is an example of an exogenous polypeptide.

The term “heterologous” as used herein typically refers to a nucleotide, polypeptide or protein sequence, not naturally present in a given organism. For example, a polynucleotide sequence endogenous to a plant can be introduced into a host cell by recombinant methods, and the plant polynucleotide is then heterologous to that recombinant host cell. The term “heterologous” may also be used with reference to a nucleotide, polypeptide, or protein sequence which is present in a recombinant host cell in a non-native state. For example, a “heterologous” nucleotide, polypeptide or protein sequence may be modified relative to the wild type sequence naturally present in the corresponding wild type host cell, e.g., a modification in the level of expression or in the sequence of a nucleotide, polypeptide or protein.

As used herein, the term “fragment” of a polypeptide refers to a shorter portion of a full-length polypeptide or protein ranging in size from four amino acid residues to the entire amino acid sequence minus one amino acid residue. In certain embodiments of the disclosure, a fragment refers to the entire amino acid sequence of a domain of a polypeptide or protein (e.g., a substrate binding domain or a catalytic domain).

The term “mutagenesis” refers to a process by which the genetic information of an organism is changed in a stable manner. Mutagenesis of a protein coding nucleic acid sequence produces a mutant protein. Mutagenesis also refers to changes in non-coding nucleic acid sequences that result in modified protein activity.

A “mutation”, as used herein, refers to a permanent change in a nucleic acid position of a gene or in an amino

acid position of a polypeptide or protein. Mutations include substitutions, additions, insertions, and deletions. For example, a mutation in an amino acid position can be a substitution of one type of amino acid with another type of amino acid (e.g., a serine(S) may be substituted with an alanine (A); a lysine (L) may be substituted with a T (Threonine); etc.). As such, a polypeptide or a protein can have one or more mutations wherein one amino acid is substituted with another amino acid. For example, an ester synthase polypeptide or protein can have one or more mutations in its amino acid sequence.

An “ester synthase variant” is an ester synthase polypeptide or protein that has one or more mutations in its amino acid sequence. In one example, the amino acid sequence ranges from 0 (i.e., the initial methionine (M) based on the ATG start site) to 473. Such an ester synthase variant can have one or more mutation in the amino acid position 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, and/or 473. Notably, amino acid positions 139 and 140 are part of a conserved motif, and thus, are important for the activity of the enzyme. In one embodiment, the mutations include G4R, T5P, T5S, D7N, S15G, T24M, T24N, T24W, Q26T, L30H, G33D, G33N, G33S, L39A, L39M, L39S, R40S, D41A, D41G, D41H, D41Y, V43K, V43S, T44A, T44F, T44K, E48A, E48Q, A49D, A49T, Y58C, Y58L, I70L, I70V, A73G, A73Q, A73S, A73T, V76L, D77A, K78F, K78W, D79H, D79K, I80V, R98D, E99Q, G101L, I102R, P111D, P111G, P111S, H122S, V123I, V123M, R131M, I146K, I146L, I146R, S147A, V149L, R150P, V155G, T157S, T158E, T158K, R162E, R162K, C163I, C163R, N164D, N164R, P166L, P166S, T170R, T170S, V171E, V171F,

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V171H, V171R, V171W, R172S, R172W, P173W, H174E, Q175R, Q175S, R176T, R177V, A179K, A179S, A179V, D182G, K183S, E184F, E184G, E184L, E184R, E184S, A185L, A185M, S186T, V187G, V187R, P188R, A189G, A190P, A190R, A190W, V191I, V191L, S192A, S192L, S192V, Q193R, Q193S, M195G, D196E, A197T, A197V, Q201A, Q201V, Q201W, A202L, D203R, P206F, R207A, G212A, G212L, G212M, G212S, V216I, V219L, V234C, V236A, V236K, L237I, H239G, T242K, T242R, A243G, A243R, Q244G, R246A, R246G, R246L, R246Q, R246V, R246W, D255E, L257I, L257M, K258R, N259A, N259E, N259Q, L260M, L260V, H262Q, H262R, A263V, S264D, S264V, S264W, G265N, G266A, G266S, S267G, A285L, A285R, A285V, N288D, N289E, N289G, T293A, T293I, P294G, V301A, N302G, I303G, I303R, I303W, R304W, A306G, A306S, D307F, D307G, D307L, D307N, D307R, D307V, E309A, E309G, E309S, G310H, G310R, G310V, T311S, T313S, Q314G, I315F, S316G, F317W, I319G, A320C, A323G, D328F, N331I, N331K, N331T, Q334K, Q334S, Q335C, Q335N, Q335S, T338A, T338E, T338H, Q348A, Q348R, K349A, K349C, K349H, K349Q, P351G, K352I, K352N, S353K, S353T, T356G, T356W, Q357V, M360R, M360Q, M360S, M360W, Y366G, Y366W, G375A, G375V, G375S, V381F, E393G, E393R, E393W, G394E, G394R, T395E, R402K, V409L, L411A, A413T, I420V, S424G, S424Q, S442E, S442G, M443G, A447C, A447I, A447L, L454V, D455E, L457Y, E458W, I461G, I461L, I461V, K466N, A468G, K472T and K472.

The terms “thioesterase variant with ester synthase activity” or “thioesterase variant” are used interchangeably herein and refer to a thioesterase polypeptide or protein that has ester synthase activity and has one or more mutations in its amino acid sequence. In one example, the amino acid sequence ranges from 0 (i.e., the initial methionine (M) based on the ATG start site) to 182. Such a thioesterase variant can have one or more mutation(s) in the amino acid position 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 158, 159, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181 and/or 182. However, amino acid positions 10, 157 and 160 are part of the catalytic triad, and thus, are important for the activity of the enzyme. In one embodiment, the mutations include R18C, K32M, Q33A, Q33R, Q34R, E37D, I38Y, I45R, I45T, S46K, S46V, S46W, D48E, D48F, D48L, D48M, T49D, T49L, T49V, T49W, G51R, N52C, N52F, N52K, N52L, N52R, K65M, N76I, N76L, N76V, G81C, F82G, F82I, F82R, Q84A, Q84L, Q84V, R88H, D102V, V112L, N115F, N115W, N115Y, Y116H, Y116N, Y116P, Y116R, Y116S, Y116T, K118Q, R119C, I148Y, L149F, L149M, L149T, N156K, N156R, N156Y, L159K, L159M and D164T.

As used herein, the term “gene” refers to nucleic acid sequences encoding either an RNA product or a protein product, as well as operably-linked nucleic acid sequences affecting the expression of the RNA or protein (e.g., such sequences include but are not limited to promoter or enhancer sequences) or operably-linked nucleic acid

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sequences encoding sequences that affect the expression of the RNA or protein (e.g., such sequences include but are not limited to ribosome binding sites or translational control sequences).

Expression control sequences are known in the art and include, for example, promoters, enhancers, polyadenylation signals, transcription terminators, internal ribosome entry sites (IRES), and the like, that provide for the expression of the polynucleotide sequence in a host cell. Expression control sequences interact specifically with cellular proteins involved in transcription (Maniatis et al., *Science*, 236:1237-1245 (1987)). Exemplary expression control sequences are described in, for example, Goeddel, *Gene Expression Technology: Methods in Enzymology*, Vol. 185, Academic Press, San Diego, Calif. (1990). In the methods of the disclosure, an expression control sequence is operably linked to a polynucleotide sequence. By “operably linked” is meant that a polynucleotide sequence and an expression control sequence(s) are connected in such a way as to permit gene expression when the appropriate molecules (e.g., transcriptional activator proteins) are bound to the expression control sequence(s). Operably linked promoters are located upstream of the selected polynucleotide sequence in terms of the direction of transcription and translation. Operably linked enhancers can be located upstream, within, or downstream of the selected polynucleotide.

As used herein, the term “vector” refers to a nucleic acid molecule capable of transporting another nucleic acid, i.e., a polynucleotide sequence, to which it has been linked. One type of useful vector is an episome (i.e., a nucleic acid capable of extra-chromosomal replication). Useful vectors are those capable of autonomous replication and/or expression of nucleic acids to which they are linked. Vectors capable of directing the expression of genes to which they are operatively linked are referred to herein as “expression vectors.” In general, expression vectors of utility in recombinant DNA techniques are often in the form of “plasmids,” which refer generally to circular double stranded DNA loops that, in their vector form, are not bound to the chromosome. The terms “plasmid” and “vector” are used interchangeably herein, in as much as a plasmid is the most commonly used form of vector. However, also included are such other forms of expression vectors that serve equivalent functions and that become known in the art subsequently hereto. In some embodiments, a recombinant vector further includes a promoter operably linked to the polynucleotide sequence. In some embodiments, the promoter is a developmentally-regulated, an organelle-specific, a tissue-specific, an inducible, a constitutive, or a cell-specific promoter. The recombinant vector typically comprises at least one sequence selected from an expression control sequence operatively coupled to the polynucleotide sequence; a selection marker operatively coupled to the polynucleotide sequence; a marker sequence operatively coupled to the polynucleotide sequence; a purification moiety operatively coupled to the polynucleotide sequence; a secretion sequence operatively coupled to the polynucleotide sequence; and a targeting sequence operatively coupled to the polynucleotide sequence. In certain embodiments, the nucleotide sequence is stably incorporated into the genomic DNA of the host cell, and the expression of the nucleotide sequence is under the control of a regulated promoter region. The expression vectors described herein include a polynucleotide sequence described herein in a form suitable for expression of the polynucleotide sequence in a host cell. It will be appreciated by those skilled in the art that the design of the expression vector can depend on such factors as the choice of the host

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cell to be transformed, the level of expression of polypeptide desired, etc. The expression vectors described herein can be introduced into host cells to produce polypeptides, including fusion polypeptides, encoded by the polynucleotide sequences as described herein.

As used herein, a recombinant or engineered "host cell" is a host cell, e.g., a microorganism used to produce one or more of fatty acid derivatives include, for example, acyl-CoA, fatty acids, fatty aldehydes, short and long chain alcohols, hydrocarbons, fatty alcohols, fatty esters (e.g., waxes, fatty acid esters, fatty esters, and/or fatty fatty esters), terminal olefins, internal olefins, and ketones. In some embodiments, the recombinant host cell comprises one or more polynucleotides, each polynucleotide encoding a polypeptide having fatty acid biosynthetic enzyme activity, wherein the recombinant host cell produces a fatty ester composition when cultured in the presence of a carbon source under conditions effective to express the polynucleotides.

As used herein "acyl-ACP" refers to an acyl thioester formed between the carbonyl carbon of alkyl chain and the sulphydryl group of the phosphopantetheinyl moiety of an acyl carrier protein (ACP). The phosphopantetheinyl moiety is post-translationally attached to a conserved serine residue on the ACP by the action of holo-acyl carrier protein synthase (ACPS), a phosphopantetheinyl transferase. In some embodiments an acyl-ACP is an intermediate in the synthesis of fully saturated acyl-ACPs. In other embodiments an acyl-ACP is an intermediate in the synthesis of unsaturated acyl-ACPs. In some embodiments, the carbon chain will have about 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, or 26 carbons. Each of these acyl-ACPs are substrates for enzymes that convert them to fatty acid derivatives.

As used herein, the term "fatty acid derivative" means a "fatty acid" or a "fatty acid derivative", which may be referred to as a "fatty acid or derivative thereof". The term "fatty acid" means a carboxylic acid having the formula RCOOH . R represents an aliphatic group, preferably an alkyl group. R can comprise between about 4 and about 22 carbon atoms. Fatty acids can have a branched chain or straight chain and may be saturated, monounsaturated, or polyunsaturated. A "fatty acid derivative" is a product made in part from the fatty acid biosynthetic pathway of the production host organism. "Fatty acid derivatives" include products made in part from acyl-ACP or acyl-ACP derivatives. Exemplary fatty acid derivatives include, acyl-CoA, fatty acids, fatty aldehydes, short and long chain alcohols, fatty alcohols, hydrocarbons, esters (e.g., waxes, fatty acid esters, fatty esters), terminal olefins, internal olefins, and ketones.

A "fatty acid derivative composition" as referred to herein is produced by a recombinant host cell and typically includes a mixture of fatty acid derivatives. In some cases, the mixture includes more than one type of fatty acid derivative product (e.g., fatty acids, fatty esters, fatty alcohols, fatty aldehydes, fatty ketones, hydrocarbons, etc.). In other cases, a fatty acid derivative composition may include, for example, a mixture of fatty esters (or another fatty acid derivative) with different chain lengths, saturation and/or branching characteristics. In still other cases, the fatty acid derivative composition may comprise both a mixture of more than one type of fatty acid derivative product and fatty acid derivatives with different chain lengths, saturation and/or branching characteristics. In still other cases, a fatty acid derivative composition may include, for example, a mixture of fatty esters and beta hydroxy esters.

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As used herein, the term "fatty acid biosynthetic pathway" means a biosynthetic pathway that produces fatty acid derivatives. The fatty acid biosynthetic pathway may include additional enzymes to produce fatty acids derivatives having desired characteristics.

As used herein, "fatty ester" means an ester having the formula RCOOR' . A fatty ester as referred to herein can be any ester made from a fatty acid, for example a fatty acid ester. In some embodiments, the R group is at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, or at least 19, carbons in length. Alternatively, or in addition, the R group is 20 or less, 19 or less, 18 or less, 17 or less, 16 or less, 15 or less, 14 or less, 13 or less, 12 or less, 11 or less, 10 or less, 9 or less, 8 or less, 7 or less, or 6 or less carbons in length. Thus, the R group can have an R group bounded by any two of the above endpoints. For example, the R group can be 6-16 carbons in length, 10-14 carbons in length, or 12-18 carbons in length. In some embodiments, the fatty ester composition comprises one or more of a C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, and a C26 fatty ester. In other embodiments, the fatty ester composition includes one or more of a C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, and a C18 fatty ester. In still other embodiments, the fatty ester composition includes C12, C14, C16 and C18 fatty esters; C12, C14 and C16 fatty esters; C14, C16 and C18 fatty esters; or C12 and C14 fatty esters.

The R group of a fatty acid derivative, for example a fatty ester, can be a straight chain or a branched chain. Branched chains may have more than one point of branching and may include cyclic branches. In some embodiments, the branched fatty acid, branched fatty aldehyde, or branched fatty ester is a C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, or a C26 branched fatty acid, branched fatty aldehyde, or branched fatty ester. In particular embodiments, the branched fatty acid, branched fatty aldehyde, or branched fatty ester is a C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, or C18 branched fatty acid, or branched fatty ester. A fatty ester of the present disclosure may be referred to as containing an A side and a B side. As used herein, an "A side" of an ester refers to the carbon chain attached to the carboxylate oxygen of the ester. As used herein, a "B side" of an ester refers to the carbon chain comprising the parent carboxylate of the ester. When the fatty ester is derived from the fatty acid biosynthetic pathway, the A side is typically contributed by an alcohol, and the B side is contributed by a fatty acid.

Any alcohol can be used to form the A side of the fatty esters. For example, the alcohol can be derived from the fatty acid biosynthetic pathway, such as those describe herein. Alternatively, the alcohol can be produced through non-fatty acid biosynthetic pathways. Moreover, the alcohol can be provided exogenously. For example, the alcohol can be supplied in the fermentation broth in cases where the fatty ester is produced by an organism. Alternatively, a carboxylic acid, such as a fatty acid or acetic acid, can be supplied exogenously in instances where the fatty ester is produced by an organism that can also produce alcohol.

The carbon chains comprising the A side or B side of the ester can be of any length. In one embodiment, the A side of the ester is at least about 1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 16, or 18 carbons in length. When the fatty ester is a fatty acid methyl ester, the A side of the ester is 1 carbon in length. When the fatty ester is a fatty acid ethyl ester, the A side of

the ester is 2 carbons in length. The B side of the ester can be at least about 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, or 26 carbons in length. The A side and/or the B side can be straight or branched chain. The branched chains can have one or more points of branching. In addition, the branched chains can include cyclic branches. Furthermore, the A side and/or B side can be saturated or unsaturated. If unsaturated, the A side and/or B side can have one or more points of unsaturation. In addition, the alcohol group of a fatty ester produced in accordance with the present disclosure need not be in the primary (C1) position. In one embodiment, the fatty ester is produced biosynthetically. In this embodiment, first the fatty acid is "activated." Non-limiting examples of "activated" fatty acids are acyl-CoA, acyl ACP, and acyl phosphate. Acyl-CoA can be a direct product of fatty acid biosynthesis or degradation. In addition, acyl-CoA can be synthesized from a free fatty acid, a CoA, and an adenosine nucleotide triphosphate (ATP). An example of an enzyme which produces acyl-CoA is acyl-CoA synthase.

In certain embodiments, the branched fatty acid derivative is an iso-fatty acid derivative, for example an iso-fatty ester, or an anteiso-fatty acid derivative, e.g., an anteiso-fatty ester. In exemplary embodiments, the branched fatty acid derivative is selected from iso-C7:0, iso-C8:0, iso-C9:0, iso-C10:0, iso-C11:0, iso-C12:0, iso-C13:0, iso-C14:0, iso-C15:0, iso-C16:0, iso-C17:0, iso-C18:0, iso-C19:0, anteiso-C7:0, anteiso-C8:0, anteiso-C9:0, anteiso-C10:0, anteiso-C11:0, anteiso-C12:0, anteiso-C13:0, anteiso-C14:0, anteiso-C15:0, anteiso-C16:0, anteiso-C17:0, anteiso-C18:0, and an anteiso-C19:0 branched fatty ester.

The R group of a branched or unbranched fatty acid derivative can be saturated or unsaturated. If unsaturated, the R group can have one or more than one point of unsaturation. In some embodiments, the unsaturated fatty acid derivative is a monounsaturated fatty acid derivative. In certain embodiments, the unsaturated fatty acid derivative is a C6:1, C7:1, C8:1, C9:1, C10:1, C11:1, C12:1, C13:1, C14:1, C15:1, C16:1, C17:1, C18:1, C19:1, C20:1, C21:1, C22:1, C23:1, C24:1, C25:1, or a C26:1 unsaturated fatty acid derivative. In certain embodiments, the unsaturated fatty acid derivative, is a C10:1, C12:1, C14:1, C16:1, or C18:1 unsaturated fatty acid derivative. In other embodiments, the unsaturated fatty acid derivative is unsaturated at the omega-7 position. In certain embodiments, the unsaturated fatty acid derivative comprises a cis double bond.

As used herein, the term "clone" typically refers to a cell or group of cells descended from and essentially genetically identical to a single common ancestor, for example, the bacteria of a cloned bacterial colony arose from a single bacterial cell.

As used herein, the term "culture" typically refers to a liquid media comprising viable cells. In one embodiment, a culture comprises cells reproducing in a predetermined culture media under controlled conditions, for example, a culture of recombinant host cells grown in liquid media comprising a selected carbon source and nitrogen. "Culturing" or "cultivation" refers to growing a population of recombinant host cells under suitable conditions in a liquid or solid medium. In particular embodiments, culturing refers to the fermentative bioconversion of a substrate to an end-product. Culturing media are well known and individual components of such culture media are available from commercial sources, e.g., Difco™ media and BBL™ media. In one non-limiting example, the aqueous nutrient medium is a "rich medium" including complex sources of nitrogen, salts, and carbon, such as YP medium, comprising 10 g/L of peptone and 10 g/L yeast extract of such a medium.

As used herein, the term "under conditions effective to express said heterologous nucleotide sequence(s)" means any conditions that allow a host cell to produce a desired fatty acid derivative. Suitable conditions include, for example, fermentation conditions.

As used herein, "modified" or an "altered level of" activity of a protein, for example an enzyme, in a recombinant host cell refers to a difference in one or more characteristics in the activity determined relative to the parent or native host cell. Typically, differences in activity are determined between a recombinant host cell, having modified activity, and the corresponding wild-type host cell (e.g., comparison of a culture of a recombinant host cell relative to the corresponding wild-type host cell). Modified activities can be the result of, for example, modified amounts of protein expressed by a recombinant host cell (e.g., as the result of increased or decreased number of copies of DNA sequences encoding the protein, increased or decreased number of mRNA transcripts encoding the protein, and/or increased or decreased amounts of protein translation of the protein from mRNA); changes in the structure of the protein (e.g., changes to the primary structure, such as, changes to the protein's coding sequence that result in changes in substrate specificity, changes in observed kinetic parameters); and changes in protein stability (e.g., increased or decreased degradation of the protein). In some embodiments, the polypeptide is a mutant or a variant of any of the polypeptides described herein, e.g., WS377 ("377"), Ppro. In certain instances, the coding sequences for the polypeptides described herein are codon optimized for expression in a particular host cell. For example, for expression in *E. coli*, one or more codons can be optimized as described in, e.g., Grosjean et al., Gene 18:199-209 (1982).

The term "regulatory sequences" as used herein typically refers to a sequence of bases in DNA, operably-linked to DNA sequences encoding a protein that ultimately controls the expression of the protein. Examples of regulatory sequences include, but are not limited to, RNA promoter sequences, transcription factor binding sequences, transcription termination sequences, modulators of transcription (such as enhancer elements), nucleotide sequences that affect RNA stability, and translational regulatory sequences (such as, ribosome binding sites (e.g., Shine-Dalgarno sequences in prokaryotes or Kozak sequences in eukaryotes), initiation codons, termination codons). As used herein, the phrase "the expression of said nucleotide sequence is modified relative to the wild type nucleotide sequence," means an increase or decrease in the level of expression and/or activity of an endogenous nucleotide sequence or the expression and/or activity of a heterologous or non-native polypeptide-encoding nucleotide sequence. The terms "altered level of expression" and "modified level of expression" are used interchangeably and mean that a polynucleotide, polypeptide, or hydrocarbon is present in a different concentration in an engineered host cell as compared to its concentration in a corresponding wild-type cell under the same conditions. As used herein, the term "express" with respect to a polynucleotide is to cause it to function. A polynucleotide which encodes a polypeptide (or protein) will, when expressed, be transcribed and translated to produce that polypeptide (or protein). As used herein, the term "overexpress" means to express or cause to be expressed a polynucleotide or polypeptide in a cell at a greater concentration than is normally expressed in a corresponding wild-type cell under the same conditions.

As used herein, the term "titer" refers to the quantity of fatty acid derivative produced per unit volume of host cell

culture. In any aspect of the compositions and methods described herein, a fatty acid derivative is produced at a titer of about 25 mg/L, about 50 mg/L, about 75 mg/L, about 100 mg/L, about 125 mg/L, about 150 mg/L, about 175 mg/L, about 200 mg/L, about 225 mg/L, about 250 mg/L, about 275 mg/L, about 300 mg/L, about 325 mg/L, about 350 mg/L, about 375 mg/L, about 400 mg/L, about 425 mg/L, about 450 mg/L, about 475 mg/L, about 500 mg/L, about 525 mg/L, about 550 mg/L, about 575 mg/L, about 600 mg/L, about 625 mg/L, about 650 mg/L, about 675 mg/L, about 700 mg/L, about 725 mg/L, about 750 mg/L, about 775 mg/L, about 800 mg/L, about 825 mg/L, about 850 mg/L, about 875 mg/L, about 900 mg/L, about 925 mg/L, about 950 mg/L, about 975 mg/L, about 1000 mg/L, about 1050 mg/L, about 1075 mg/L, about 1100 mg/L, about 1125 mg/L, about 1150 mg/L, about 1175 mg/L, about 1200 mg/L, about 1225 mg/L, about 1250 mg/L, about 1275 mg/L, about 1300 mg/L, about 1325 mg/L, about 1350 mg/L, about 1375 mg/L, about 1400 mg/L, about 1425 mg/L, about 1450 mg/L, about 1475 mg/L, about 1500 mg/L, about 1525 mg/L, about 1550 mg/L, about 1575 mg/L, about 1600 mg/L, about 1625 mg/L, about 1650 mg/L, about 1675 mg/L, about 1700 mg/L, about 1725 mg/L, about 1750 mg/L, about 1775 mg/L, about 1800 mg/L, about 1825 mg/L, about 1850 mg/L, about 1875 mg/L, about 1900 mg/L, about 1925 mg/L, about 1950 mg/L, about 1975 mg/L, about 2000 mg/L (2 g/L), 3 g/L, 5 g/L, 10 g/L, 20 g/L, 30 g/L, 40 g/L, 50 g/L, 60 g/L, 70 g/L, 80 g/L, 90 g/L, 100 g/L or a range bounded by any two of the foregoing values. In other embodiments, a fatty acid derivative is produced at a titer of more than 100 g/L, more than 200 g/L, or more than 300 g/L. One preferred titer of fatty acid derivative produced by a recombinant host cell according to the methods of the disclosure is from 5 g/L to 200 g/L, 10 g/L to 150 g/L, 20 g/L to 120 g/L and 30 g/L to 100 g/L. The titer may refer to a particular fatty acid derivative or a combination of fatty acid derivatives produced by a given recombinant host cell culture. For example, the expression of a variant polypeptide with ester synthase activity in a recombinant host cell such as *E. coli* results in the production of a higher titer as compared to a recombinant host cell expressing the corresponding wild type polypeptide. In one embodiment, the higher titer ranges from at least about 5 g/L to about 200 g/L.

As used herein, the “yield of fatty acid derivative produced by a host cell” refers to the efficiency by which an input carbon source is converted to product (i.e., a fatty ester) in a host cell. Host cells engineered to produce fatty acid derivatives according to the methods of the disclosure have a yield of at least about 3%, at least about 4%, at least about 5%, at least about 6%, at least about 7%, at least about 8%, at least about 9%, at least about 10%, at least about 11%, at least about 12%, at least about 13%, at least about 14%, at least about 15%, at least about 16%, at least about 17%, at least about 18%, at least about 19%, at least about 20%, at least about 21%, at least about 22%, at least about 23%, at least about 24%, at least about 25%, at least about 26%, at least about 27%, at least about 28%, at least about 29%, or at least about 30% or a range bounded by any two of the foregoing values. In other embodiments, a fatty acid derivative or derivatives is produced at a yield of more than about 30%, about 40%, about 50%, about 60%, about 70%, about 80%, about 90% or more. Alternatively, or in addition, the yield is about 30% or less, about 27% or less, about 25% or less, or about 22% or less. Thus, the yield can be bounded by any two of the above endpoints. For example, the yield of a fatty acid derivative or derivatives produced by the

recombinant host cell according to the methods of the disclosure can be about 5% to about 15%, about 10% to about 25%, about 10% to about 22%, about 15% to about 27%, about 18% to about 22%, about 20% to about 28%, or about 20% to about 30%. The yield may refer to a particular fatty acid derivative or a combination of fatty acid derivatives produced by a given recombinant host cell culture. For example, the expression of a variant polypeptide with ester synthase activity in a recombinant host cell such as *E. coli* results in the production of a higher yield of fatty acid esters as compared to a recombinant host cell expressing the corresponding wild type polypeptide. In one embodiment, the higher yield ranges from about 10% to about 100% of theoretical yield.

As used herein, the term “productivity” refers to the quantity of a fatty acid derivative or derivatives produced per unit volume of host cell culture per unit time. In any aspect of the compositions and methods described herein, the productivity of a fatty acid derivative or derivatives produced by a recombinant host cell is at least 100 mg/L/hour, at least 200 mg/L/hour, at least 300 mg/L/hour, at least 400 mg/L/hour, at least 500 mg/L/hour, at least 600 mg/L/hour, at least 700 mg/L/hour, at least 800 mg/L/hour, at least 900 mg/L/hour, at least 1000 mg/L/hour, at least 1100 mg/L/hour, at least 1200 mg/L/hour, at least 1300 mg/L/hour, at least 1400 mg/L/hour, at least 1500 mg/L/hour, at least 1600 mg/L/hour, at least 1700 mg/L/hour, at least 1800 mg/L/hour, at least 1900 mg/L/hour, at least 2000 mg/L/hour, at least 2100 mg/L/hour, at least 2200 mg/L/hour, at least 2300 mg/L/hour, at least 2400 mg/L/hour, 2500 mg/L/hour, or as high as 10 g/L/hr (dependent upon cell mass). For example, the productivity of a fatty acid derivative or derivatives produced by a recombinant host cell according to the methods of the may be from 500 mg/L/hour to 2500 mg/L/hour, or from 700 mg/L/hour to 2000 mg/L/hour. The productivity may refer to a particular fatty acid derivative or a combination of fatty acid derivatives produced by a given recombinant host cell culture. For example, the expression of a variant polypeptide with ester synthase activity in a recombinant host cell such as *E. coli* results in the production of an increased productivity of fatty acid esters as compared to a recombinant host cell expressing the corresponding wild type polypeptide. In one embodiment, the higher productivity ranges from about 0.3 g/L/h to about 3 g/L/h.

As used herein, the term “total fatty species” and “total fatty acid product” may be used interchangeably herein with reference to the amount of fatty esters and fatty acids, as evaluated by GC-FID. The same terms may be used to mean fatty alcohols, fatty aldehydes and free fatty acids when referring to a fatty alcohol analysis.

As used herein, the term “glucose utilization rate” means the amount of glucose used by the culture per unit time, reported as grams/liter/hour (g/L/hr).

As used herein, the term “greater ester synthase activity as compared to the wild type polypeptide” is used with reference to an ester synthase or thioesterase with ester synthase activity with a higher titer, higher yield and/or higher productivity than the corresponding wild type enzyme. In one preferred embodiment, the titer, yield or productivity is at least twice that of the wild type polypeptide. In other preferred embodiments, the titer, yield or productivity is at least 3 times, 4 times, 5 times, 6 times, 7 times, 8 times, 9 times or 10 times that of the wild type polypeptide. In one particular embodiment, the titer is at least about 5% or greater. In some cases, “greater ester synthase activity as compared to the wild type polypeptide”, also means that in

addition to the higher titer, yield and/or productivity than the corresponding wild type ester synthase enzyme, the ester synthase produces a low or lower percentage (e.g., 1-5%) of beta hydroxy esters (see Examples, *infra*). In other cases, “greater ester synthase activity as compared to the wild type polypeptide”, also means that in addition to the higher titer, yield and/or productivity than the corresponding wild type ester synthase enzyme, the ester synthase produces a higher percentage of beta hydroxy esters (see Examples, *infra*).

The term “improved fatty acid methyl ester activity” means that the variant polypeptide or enzyme can produce fatty acid methyl esters, fatty acid ethyl esters, fatty acid propyl esters, fatty acid isopropyl esters, fatty acid butyl esters, monoglycerides, fatty acid isobutyl esters, fatty acid 2-butyl esters, and fatty acid tert-butyl esters, and the like. A variant polypeptide or enzyme with improved fatty acid methyl ester activity has improved properties including, but not limited to, increased beta hydroxy esters, decreased beta hydroxyl esters, increased chain lengths of fatty acid esters, and decreased chain lengths of fatty acid esters. A variant ester synthase polypeptide with improved fatty acid methyl ester activity produces a titer that is at least about 5% or greater when the variant polypeptide is expressed in a recombinant microorganism.

As used herein, the term “carbon source” refers to a substrate or compound suitable to be used as a source of carbon for prokaryotic or simple eukaryotic cell growth. Carbon sources can be in various forms, including, but not limited to, polymers, carbohydrates, acids, alcohols, aldehydes, ketones, amino acids, peptides, and gases (e.g., CO and CO₂). Exemplary carbon sources include, but are not limited to, monosaccharides, such as glucose, fructose, mannose, galactose, xylose, and arabinose; oligosaccharides, such as fructo-oligosaccharide and galacto-oligosaccharide; polysaccharides such as starch, cellulose, pectin, and xylan; disaccharides, such as sucrose, maltose, cellobiose, and turanose; cellulosic material and variants such as hemicelluloses, methyl cellulose and sodium carboxymethyl cellulose; saturated or unsaturated fatty acids, succinate, lactate, and acetate; alcohols, such as ethanol, methanol, and glycerol, or mixtures thereof. The carbon source can also be a product of photosynthesis, such as glucose. In certain preferred embodiments, the carbon source is biomass. In other preferred embodiments, the carbon source is glucose. In other preferred embodiments the carbon source is sucrose. In other embodiments the carbon source is glycerol.

As used herein, the term “biomass” refers to any biological material from which a carbon source is derived. In some embodiments, a biomass is processed into a carbon source, which is suitable for bioconversion. In other embodiments, the biomass does not require further processing into a carbon source. The carbon source can be converted into a composition comprising fatty esters. Fatty esters find utility in a number of products including, but not limited to, surfactants, polymers, films, textiles, dyes, pharmaceuticals, fragrances and flavoring agents, lacquers, paints, varnishes, softening agents in resins and plastics, plasticizers, flame retardants, and additives in gasoline and oil.

An exemplary source of biomass is plant matter or vegetation, such as corn, sugar cane, or switchgrass. Another exemplary source of biomass is metabolic waste products, such as animal matter (e.g., cow manure). Further exemplary sources of biomass include algae and other marine plants. Biomass also includes waste products from industry, agriculture, forestry, and households, including, but not limited to, glycerol, fermentation waste, ensilage, straw, lumber, sewage, garbage, cellulosic urban waste, and food leftovers

(e.g., soaps, oils and fatty acids). The term “biomass” also can refer to sources of carbon, such as carbohydrates (e.g., monosaccharides, disaccharides, or polysaccharides).

As used herein, the term “isolated,” with respect to products (such as fatty acids and derivatives thereof) refers to products that are separated from cellular components, cell culture media, or chemical or synthetic precursors. The fatty acids and derivatives thereof produced by the methods described herein can be relatively immiscible in the fermentation broth, as well as in the cytoplasm. Therefore, the fatty acids and derivatives thereof can collect in an organic phase either intracellularly or extracellularly.

As used herein, the terms “purify,” “purified,” or “purification” mean the removal or isolation of a molecule from its environment by, for example, isolation or separation. “Substantially purified” molecules are at least about 60% free (e.g., at least about 70% free, at least about 75% free, at least about 85% free, at least about 90% free, at least about 95% free, at least about 97% free, at least about 99% free) from other components with which they are associated. As used herein, these terms also refer to the removal of contaminants from a sample. For example, the removal of contaminants can result in an increase in the percentage of fatty acid derivatives in a sample. For example, when a fatty acid derivative is produced in a recombinant host cell, the fatty acid derivative can be purified by the removal of host cell proteins. After purification, the percentage of fatty acid derivative in the sample is increased. The terms “purify,” “purified,” and “purification” are relative terms which do not require absolute purity. Thus, for example, when a fatty acid derivative is produced in recombinant host cells, a purified fatty acid derivative is a fatty acid derivative that is substantially separated from other cellular components (e.g., nucleic acids, polypeptides, lipids, carbohydrates, or other hydrocarbons). As used herein, the term “attenuate” means to weaken, reduce, or diminish. For example, a polypeptide can be attenuated by modifying the polypeptide to reduce its activity (e.g., by modifying a nucleotide sequence that encodes the polypeptide).

Ester Synthase Variants

The present disclosure relates to, among other things, variant ester synthase enzymes, polypeptide sequences of such variants and functional fragments thereof with improved properties; polynucleotides encoding variant ester synthase polypeptide sequences; recombinant microorganisms including a nucleic acid encoding an improved ester synthase polypeptide; microorganisms capable of expressing the improved ester synthase polypeptide; cultures of such microorganisms; processes for producing fatty acid esters; fatty acid ester compositions and other compositions derived therefrom using the improved ester synthase polypeptides; and the resultant compositions.

Particularly, ester synthase polypeptides with improved properties and microorganisms expressing these polypeptides are provided herein. Wild-type ester synthase polypeptides from *Marinobacter hydrocarbonoclasticus* strain DSM8798, have been described in Holtzapfel et al. (see *J Bacteriol.* (2007) 189 (10): 3804-3812) and U.S. Pat. No. 7,897,369. SEQ ID NO: 2 is the amino acid sequence of the wild-type ester synthase, i.e., ES9/DSM8798 from *Marinobacter hydrocarbonoclasticus* (*M. hydrocarbonoclasticus*; GenBankAccession No. ABO21021), which was used as a template to generate the improved ester synthase enzymes in order to illustrate the disclosure (see Examples, *infra*). A preferred ester synthase variant has at least about 90% sequence identity to the amino acid sequence of the wild type *M. hydrocarbonoclasticus* ester synthase enzyme (re-

ferred to herein as “WS377” or “377”) of SEQ ID NO: 2. Furthermore, SEQ ID NO: 1 is the polynucleotide sequence encoding SEQ ID NO: 2. With respect to the ester synthase polypeptide sequences described herein “M” (ATG) is considered to be amino acid “0”. The first amino acid after the ATG is designated amino acid “1”.

In one aspect, the disclosure provides improved ester synthase polypeptides with enhanced ester synthase activity and nucleotide sequences that encode them. In another aspect, it can be seen that substitutions introduced at numerous different amino acid positions (also referred to herein as “residues”) within the wild-type 377 polypeptide of SEQ ID NO: 2 yield improved 377 polypeptides, capable of catalyzing increased production of fatty esters relative to wild-type 377. Depending upon the position mutated, single amino acid changes at specified positions give rise to increases or decreases in fatty ester production. Single amino acid changes at specified positions may also give rise to increases or decreases in beta hydroxy (β -OH) ester production. In one embodiment, a single amino acid change results in an increase in fatty ester production and a decrease in β -OH ester production. In another embodiment, a single amino acid change results in an increase in fatty ester production and no change in β -OH ester production. In still another embodiment, a single amino acid change results in an increase in fatty ester production and an increase in β -OH ester production. A single amino acid change may also result in a decrease in fatty ester production and an increase, decrease or no change in β -OH ester production. In other embodiments, single or multiple amino acid changes result in an increase in fatty ester production. In still other embodiments, single or multiple amino acid changes result in an increase or decrease in β -OH production.

Thus, combinations of two or more amino acid changes at specified positions may give rise to increases or decreases in fatty ester and/or free fatty acid production. The effect of each individual amino acid change on fatty ester and free fatty acid production may or may not be additive to the effect of other individual amino acid changes on fatty ester and free fatty acid production. In some preferred embodiments, a combination of two or more amino acid changes at specified positions results in an increase in fatty ester production and a decrease in free fatty acid production. Accordingly, multiple amino acid changes at specified positions give rise to increases or decreases in fatty ester production. As such, multiple amino acid changes at specified positions may also give rise to decreases in β -OH ester production. In some embodiments, multiple amino acid changes result in an increase in fatty ester production and a decrease in β -OH ester production. In other embodiments, multiple amino acid changes result in an increase in fatty ester production and no change in β -OH ester production. In still other embodiments, multiple amino acid changes result in an increase in fatty ester production and an increase in β -OH ester production. Multiple amino acid changes may also result in a decrease in fatty ester production and an increase, decrease or no change in β -OH ester production.

A full saturation library was prepared using WS377 ester synthase as a template, as described in Example 1 (*infra*). Over 200 beneficial mutations were identified based on increased titer of fatty ester and/or a decrease in the percentage of beta hydroxy ester produced. A combination library was prepared based on “hits” from the saturation library, as detailed in Examples 2 and 5. In one aspect, the disclosure relates to improved ester synthase polypeptides with at least about 70% sequence identity to SEQ ID NO: 2. In some embodiments, an improved ester synthase polypep-

tide shows at least about 75% (e.g., at least 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99%) sequence identity to the wild-type 377 sequence of SEQ ID NO: 2 and also includes one or more substitutions which results in useful characteristics and/or properties as described herein. In one aspect of the disclosure, the improved ester synthase polypeptide has about 100% sequence identity to SEQ ID NO: 4. In another aspect of the disclosure, the improved ester synthase polypeptide has about 100% sequence identity to any one of the following SEQ ID NOS, including, but not limited to, SEQ ID NO: 6, SEQ ID NO: 12, SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, SEQ ID NO: 20, SEQ ID NO: 22, SEQ ID NO: 24, SEQ ID NO: 26, SEQ ID NO: 29, SEQ ID NO: 33, SEQ ID NO: 35, SEQ ID NO: 37, SEQ ID NO: 39, SEQ ID NO: 41 and SEQ ID NO: 43.

In a related aspect, an improved ester synthase polypeptide is encoded by a nucleotide sequence having 100% sequence identity to SEQ ID NO: 3 or SEQ ID NO: 7. In another related aspect, an improved ester synthase polypeptide is encoded by a nucleotide sequence having about 100% sequence identity to any one of the following SEQ ID NOS including, but not limited to, SEQ ID NO: 5, SEQ ID NO: 9, SEQ ID NO: 11, SEQ ID NO: 13, SEQ ID NO: 15, SEQ ID NO: 17, SEQ ID NO: 19, SEQ ID NO: 21, SEQ ID NO: 23, SEQ ID NO: 25, SEQ ID NO: 28, SEQ ID NO: 32, SEQ ID NO: 34, SEQ ID NO: 36, SEQ ID NO: 38, SEQ ID NO: 40 and SEQ ID NO: 42.

In another aspect, the disclosure relates to improved ester synthase polypeptides with at least about 70% sequence identity to SEQ ID NO: 4. In some embodiments, an improved ester synthase polypeptide has at least about 75% (e.g., at least 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99%) sequence identity to the ester synthase sequence of SEQ ID NO: 4 and also includes one or more substitutions which results in improved characteristics and/or properties as described herein. In another aspect, the disclosure relates to improved ester synthase polypeptides with at least about 70% sequence identity to any one of the following SEQ ID NOS including, but not limited to, SEQ ID NO: 6, SEQ ID NO: 12, SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, SEQ ID NO: 20, SEQ ID NO: 22, SEQ ID NO: 24, SEQ ID NO: 26, SEQ ID NO: 29, SEQ ID NO: 33, SEQ ID NO: 35, SEQ ID NO: 37, SEQ ID NO: 39, SEQ ID NO: 41 and SEQ ID NO: 43. In some embodiments, an improved ester synthase polypeptide has at least about 75% (e.g., at least 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99%) sequence identity to the ester synthase sequence of any one of the following SEQ ID NOS including, but not limited to, SEQ ID NO: 6, SEQ ID NO: 12, SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, SEQ ID NO: 20, SEQ ID NO: 22, SEQ ID NO: 24, SEQ ID NO: 26, SEQ ID NO: 29, SEQ ID NO: 33, SEQ ID NO: 35, SEQ ID NO: 37, SEQ ID NO: 39, SEQ ID NO: 41 and SEQ ID NO: 43, and also includes one or more substitutions which results in improved characteristics and/or properties as described herein.

In another aspect, the disclosure relates to improved ester synthase polypeptides that comprise an amino acid sequence encoded by a nucleic acid sequence that has at least about 75% (e.g., at least 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or and at least 99%)

sequence identity to the ester synthase sequence of SEQ ID NO: 5. In some embodiments the nucleic acid sequence encodes an ester synthase variant with one or more substitutions which results in improved characteristics and/or properties as described herein. In other embodiments, the improved or variant ester synthase nucleic acid sequence is derived from a species other than *Marinobacter hydrocarbonoclasticus*. In another aspect, the disclosure relates to improved ester synthase polypeptides that comprise an amino acid sequence encoded by a nucleic acid sequence that has at least about 75% (e.g., at least 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or and at least 99%) sequence identity to the ester synthase sequence of any one of the following SEQ ID NOS including, but not limited to, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 11, SEQ ID NO: 13, SEQ ID NO: 15, SEQ ID NO: 17, SEQ ID NO: 19, SEQ ID NO: 21, SEQ ID NO: 23, SEQ ID NO: 25, SEQ ID NO: 28, SEQ ID NO: 32, SEQ ID NO: 34, SEQ ID NO: 36, SEQ ID NO: 38, SEQ ID NO: 40 and SEQ ID NO: 42. In some embodiments the nucleic acid sequence encodes an ester synthase variant with one or more substitutions which results in improved characteristics and/or properties as described herein. In other embodiments, the improved or variant ester synthase nucleic acid sequence is derived from a species other than *Marinobacter hydrocarbonoclasticus*.

In another aspect, the disclosure relates to ester synthase polypeptides that comprise an amino acid sequence encoded by a nucleic acid that hybridizes under stringent conditions over substantially the entire length of a nucleic acid corresponding to any one of the following SEQ ID NOS including, but not limited to, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 11, SEQ ID NO: 13, SEQ ID NO: 15, SEQ ID NO: 17, SEQ ID NO: 19, SEQ ID NO: 21, SEQ ID NO: 23, SEQ ID NO: 25, SEQ ID NO: 28, SEQ ID NO: 32, SEQ ID NO: 34, SEQ ID NO: 36, SEQ ID NO: 38, SEQ ID NO: 40 and SEQ ID NO: 42. In some embodiments the nucleic acid sequence encodes an improved or variant ester synthase nucleic acid sequence derived from a species other than *Marinobacter hydrocarbonoclasticus*. In a related aspect, the disclosure provides ester synthase encoded by a nucleotide sequence having at least about 70% (e.g., at least 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99%) sequence identity to SEQ ID NO: 1 and comprises one or more of the substitutions disclosed herein.

Improved Properties of Ester Synthase Variants

The wild type ester synthase from *Marinobacter hydrocarbonoclasticus* DSM 8798 ("377") was engineered to produce a high percentage of fatty ester without the need to overexpress any other gene using expression in *E. coli* as an illustrative model. In addition, when expressed in a recombinant host cell such as *E. coli*, variants of the wild type ester synthase enzyme that result in a higher titer, yield or productivity also produce greater than about 20%, 25%, 30%, 35%, 40%, 45%, 50% or more β -OH esters. In one aspect, the present disclosure provides an improved variant of an ester synthase that has a higher titer, yield or productivity than the wild type ester synthase, and also produces a lower percentage of β -OH esters than the wild type enzyme. In a preferred embodiment, the variant ester synthase enzymes of the disclosure produce less than about 20%, 18%, 16%, 14%, 12%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2% or less than 1% β -OH esters. As such, the ester synthase variants of the disclosure exhibit improved characteristics

and/or properties as described herein, for example, an increase in titer, yield and/or productivity of fatty acid esters; and/or a decrease in titer, yield or productivity of beta hydroxyl esters. Thus, the engineering of an ester synthase to have a higher titer, yield and/or productivity essentially prevents *E. coli* from using resources in a futile cycle (see FIG. 2), while allowing for a high level production of fatty acid esters, e.g., for use as biodiesel or in the manufacture of surfactants.

Photobacterium profundum Variants

The present disclosure relates to, among other things, variant thioesterase enzymes with increased ester synthase activity; polypeptide sequences of such variants and functional fragments thereof with improved properties; polynucleotides encoding variant thioesterase polypeptide sequences with increased ester synthase activity; recombinant microorganisms comprising a nucleic acid encoding an improved thioesterase polypeptide with increased ester synthase activity; microorganisms capable of expressing the improved thioesterase polypeptide sequences with increased ester synthase activity; cultures of such microorganisms; processes for producing fatty acid esters; fatty acid ester compositions and other compositions derived therefrom using the improved thioesterase polypeptide sequences with increased ester synthase activity; and the resultant compositions.

Particularly, thioesterase polypeptides with improved properties such as an increased ester synthase activity and microorganisms expressing these polypeptides are provided herein. Wild-type *Photobacterium profundum* (Ppro) polypeptides have been described in Vezzi et al. (see Science (2005) 307:1459-1461). SEQ ID NO: 51 is the wild-type amino acid sequence for thioesterase from *Photobacterium profundum*. SEQ ID NO: 50 is the polynucleotide sequence encoding SEQ ID NO: 51. The leader sequence that targets the thioesterase into the periplasm was removed from SEQ ID NO: 50. Improved thioesterase polypeptides, for example, variant Ppro polypeptides, have enhanced ester synthase activity. With respect to the Ppro polypeptide sequences described herein "M" (ATG) is considered to be amino acid "O". The first amino acid after the ATG is designated amino acid "1".

In one aspect, the disclosure provides improved Ppro polypeptides with enhanced ester synthase activity and nucleotide sequences that encode them. Improved Ppro polypeptides are examples of variant thioesterase polypeptides with increased or enhanced ester synthase activity. In another aspect, it can be seen that substitutions introduced at numerous different amino acid positions (also referred to herein as "residue" positions) within the wild-type Ppro of SEQ ID NO: 51 yield improved Ppro polypeptides capable of catalyzing increased production of fatty esters relative to wild-type Ppro. Depending upon the position mutated, single amino acid changes at specified positions gives rise to increases or decreases in fatty ester production. Single amino acid changes at specified positions may also give rise to increases or decreases in free fatty acid production. In some preferred embodiments, a single amino acid change results in an increase in fatty ester production and a decrease in free fatty acid production. In other embodiments, a single amino acid change results in an increase in fatty ester production and no change in free fatty acid production. In still other embodiments, a single amino acid change results in an increase in fatty ester production and an increase in free fatty acid production. A single amino acid change may also result in a decrease in fatty ester production and an increase, decrease or no change in free fatty acid production.

Combinations of two or more amino acid changes at specified positions may also give rise to increases or decreases in fatty ester and/or free fatty acid production. The effect of each individual amino acid change on fatty ester and free fatty acid production may or may not be additive to the effect of other individual amino acid changes on fatty ester and free fatty acid production. In some preferred embodiments, a combination of two or more amino acid changes at specified positions results in an increase in fatty ester production and a decrease in free fatty acid production.

In one aspect, the disclosure relates to Ppro polypeptides that have at least about 70% sequence identity with SEQ ID NO: 51. In some embodiments, a Ppro polypeptide shows at least about 75% (e.g., at least 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or and at least 99%) sequence identity to the wild-type Ppro sequence of SEQ ID NO: 51 and also includes one or more substitutions which results in improved characteristics and/or properties as described herein. In a related aspect the Ppro polypeptide has a 100% sequence identity with SEQ ID NO: 51. In other related aspects the Ppro polypeptides have 100% sequence identity with SEQ ID NO: 53, SEQ ID NO: 55, SEQ ID NO: 58, SEQ ID NO: 61, SEQ ID NO: 63, SEQ ID NO: 65, or SEQ ID NO: 31.

In one aspect, the disclosure relates to Ppro polypeptides that have at least about 70% sequence identity with SEQ ID NO: 61. In some embodiments, a Ppro polypeptide has at least about 75% (e.g., at least 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or and at least 99%) sequence identity to the wild-type Ppro sequence of SEQ ID NO: 61 and also includes one or more substitutions which results in improved characteristics and/or properties as described herein.

In another aspect the disclosure relates to Ppro polypeptides that have at least about 70% sequence identity with SEQ ID NO: 31. In some embodiments, a Ppro polypeptide has at least about 75% (e.g., at least 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or and at least 99%) sequence identity to the wild-type Ppro sequence of SEQ ID NO: 31 and also includes one or more substitutions which results in improved characteristics and/or properties as described herein.

In another aspect, the disclosure relates to Ppro polypeptides that comprise an amino acid sequence encoded by a nucleic acid sequence that has at least about 75% (e.g., at least 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or and at least 99%) sequence identity to the wild-type Ppro sequence of SEQ ID NO: 50, SEQ ID NO: 52, SEQ ID NO: 54, SEQ ID NO: 56, SEQ ID NO: 57, SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 62, SEQ ID NO: 64 or SEQ ID NO: 30. In some embodiments the nucleic acid sequence encodes a Ppro variant with one or more substitutions which results in improved characteristics and/or properties as described herein. In other embodiments, the Ppro nucleic acid sequence is derived from a species other than *Photobacterium profundum*.

In another aspect, the disclosure relates to Ppro polypeptides that comprise an amino acid sequence encoded by a nucleic acid that hybridizes under stringent conditions over substantially the entire length of a nucleic acid corresponding to SEQ ID NO: 50, SEQ ID NO: 52, SEQ ID NO: 54, SEQ ID NO: 56, SEQ ID NO: 57, SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 62, SEQ ID NO: 64 or SEQ ID NO:

30. In some embodiments the nucleic acid sequence encodes a Ppro variant derived from a species other than *Photobacterium profundum*. In a related aspect, the disclosure provides Ppro encoded by a nucleotide sequence having at least about 70% (e.g., at least 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99%) sequence identity to SEQ ID NO: 50 and comprises substitutions disclosed herein (see, e.g., Table 13).

Improved Properties of Ppro Variants

In order to illustrate the disclosure, the *tesA* from *P. profundum* was engineered to produce a higher percentage of FAME without the need to overexpress any other gene in *E. coli*. An ester synthase that acts directly on acyl-ACP (product of fatty acid biosynthesis) could be directed towards increased ester production, as well as allow direct pull of the pathway from the fatty acid biosynthesis, which should also improve the production. In addition, one of the libraries was tested for production of FAEE, and it was observed that engineered *tesA* from *P. profundum* for FAME, also produced higher levels of FAEE than the wild type strain. These Ppro variants exhibit improved characteristics and/or properties as described herein, for example an increase in titer, yield and/or productivity of fatty acid esters; and/or a decrease in titer, yield and/or productivity of free fatty acids. As an example, the *tesA* of *P. profundum* (Ppro) was engineered to have ester synthase activity. In other words, the *tesA* of Ppro was engineered to act as an ester synthase. When screened in plates, the wild type *tesA* Ppro gene produced about 15-18% FAME of the total acyl products (FAME+free fatty acids ("FFA")). Mutants were identified that resulted in an increase of FAME production of 42-44% when tested on plates. When these same mutants were tested in shake flasks, they produced 62-65% FAME when compared to the wild type *tesA* Ppro which only produced 30% FAME (see Example 7, *infra*).

Methods of Making Ester Synthase or Ppro Variants

In practicing the methods of the present disclosure, mutagenesis is used to prepare groups of recombinant host cells for screening. Typically, the recombinant host cells comprise one or more polynucleotide sequences that include an open reading frame for an ester synthase or Ppro polypeptide, such as a variant of ester synthase or a variant of thioesterase (e.g., Ppro) together with operably-linked regulatory sequences. Numerous examples of variant ester synthase polypeptides and variant thioesterase polypeptides useful in the practice of the methods of the present disclosure are described herein. Examples of regulatory sequences useful in the practice of the methods of the present disclosure are also described herein. Mutagenesis of such polynucleotide sequences can be performed using genetic engineering techniques, such as site directed mutagenesis, random chemical mutagenesis, exonuclease III deletion procedures, or standard cloning techniques. Alternatively, mutations in polynucleotide sequences can be created using chemical synthesis or modification procedures.

Mutagenesis methods are well known in the art and include, for example, the following. In error prone PCR (see, e.g., Leung et al., Technique 1:11-15, 1989; and Caldwell et al., PCR Methods Applic. 2:28-33, 1992), PCR is performed under conditions where the copying fidelity of the DNA polymerase is low, such that a high rate of point mutations is obtained along the entire length of the PCR product. Briefly, in such procedures, polynucleotides to be mutagenized are mixed with PCR primers, reaction buffer, MgCl₂, MnCl₂, Taq polymerase, and an appropriate concentration of

dNTPs for achieving a high rate of point mutation along the entire length of the PCR product. For example, the reaction can be performed using 20 fmole of nucleic acid to be mutagenized, 30 pmole of each PCR primer, a reaction buffer comprising 50 mM KCl, 10 mM Tris HCl (pH 8.3), and 0.01% gelatin, 7 mM MgCl₂, 0.5 mM MnCl₂, 5 units of Taq polymerase, 0.2 mM dGTP, 0.2 mM dATP, 1 mM dCTP, and 1 mM dTTP. PCR can be performed for 30 cycles of 94° C. for 1 min., 45° C. for 1 min., and 72° C. for 1 min. It will be appreciated that these parameters can be varied as appropriate. The mutagenized polynucleotides are then cloned into an appropriate vector and the activities of the affected polypeptides encoded by the mutagenized polynucleotides are evaluated. Mutagenesis can also be performed using oligonucleotide directed mutagenesis (see, e.g., Reidhaar-Olson et al., *Science* 241:53-57, 1988) to generate site-specific mutations in any cloned DNA of interest. Briefly, in such procedures a plurality of double stranded oligonucleotides bearing one or more mutations to be introduced into the cloned DNA are synthesized and assembled into the cloned DNA to be mutagenized. Clones containing the mutagenized DNA are recovered, and the activities of affected polypeptides are assessed. Another mutagenesis method for generating polynucleotide sequence variants is assembly PCR. Assembly PCR involves the assembly of a PCR product from a mixture of small DNA fragments. A large number of different PCR reactions occur in parallel in the same vial, with the products of one reaction priming the products of another reaction. Assembly PCR is described in, for example, U.S. Pat. No. 5,965,408. Still another mutagenesis method of generating polynucleotide sequence variants is sexual PCR mutagenesis (Stemmer, PNAS, USA 91:10747-10751, 1994). In sexual PCR mutagenesis, forced homologous recombination occurs between DNA molecules of different, but highly related, DNA sequence in vitro as a result of random fragmentation of the DNA molecule based on sequence homology. This is followed by fixation of the crossover by primer extension in a PCR reaction.

Ester synthase (e.g., '377) or thioesterase (e.g., Ppro) sequence variants can also be created by in vivo mutagenesis. In some embodiments, random mutations in a nucleic acid sequence are generated by propagating the polynucleotide sequence in a bacterial strain, such as an *E. coli* strain, which carries mutations in one or more of the DNA repair pathways. Such "mutator" strains have a higher random mutation rate than that of a wild-type strain. Propagating a DNA sequence in one of these strains will eventually generate random mutations within the DNA. Mutator strains suitable for use for in vivo mutagenesis are described in, for example, PCT International Publication No. WO 91/16427.

Ester synthase (e.g., '377) or thioesterase (e.g., Ppro) sequence variants can also be generated using cassette mutagenesis. In cassette mutagenesis, a small region of a double stranded DNA molecule is replaced with a synthetic oligonucleotide "cassette" that differs from the starting polynucleotide sequence. The oligonucleotide often contains completely and/or partially randomized versions of the starting polynucleotide sequence. There are many applications of cassette mutagenesis; for example, preparing mutant proteins by cassette mutagenesis (see, e.g., Richards, J. H., *Nature* 323, 187 (1986); Ecker, D. J., et al., *J. Biol. Chem.* 262:3524-3527 (1987)); codon cassette mutagenesis to insert or replace individual codons (see, e.g., Kegler-Ebo, D. M., et al., *Nucleic Acids Res.* 22 (9): 1593-1599 (1994)); preparing variant polynucleotide sequences by randomization of non-coding polynucleotide sequences comprising regulatory sequences (e.g., ribosome binding sites, see, e.g.,

Barrick, D., et al., *Nucleic Acids Res.* 22 (7): 1287-1295 (1994); Wilson, B. S., et al., *Biotechniques* 17:944-953 (1994)).

Recursive ensemble mutagenesis (see, e.g., Arkin et al., PNAS, USA 89:7811-7815, 1992) can also be used to generate polynucleotide sequence variants. Recursive ensemble mutagenesis is an algorithm for protein engineering (i.e., protein mutagenesis) developed to produce diverse populations of phenotypically related mutants whose members differ in amino acid sequence. This method uses a feedback mechanism to control successive rounds of combinatorial cassette mutagenesis. Exponential ensemble mutagenesis (see, e.g., Delegrave et al., *Biotech. Res.* 11:1548-1552, 1993) can also be used to generate polynucleotide sequence variants. Exponential ensemble mutagenesis is a process for generating combinatorial libraries with a high percentage of unique and functional mutants, wherein small groups of residues are randomized in parallel to identify, at each altered position, amino acids which lead to functional proteins. Random and site-directed mutagenesis can also be used (see, e.g., Arnold, *Curr. Opin. Biotech.* 4:450-455, 1993).

Further, standard methods of in vivo mutagenesis can be used. For example, host cells, comprising one or more polynucleotide sequences that include an open reading frame for an ester synthase (e.g., '377) or thioesterase (e.g., Ppro) polypeptide, as well as operably-linked regulatory sequences, can be subject to mutagenesis via exposure to radiation (e.g., UV light or X-rays) or exposure to chemicals (e.g., ethylating agents, alkylating agents, or nucleic acid analogs). In some host cell types, for example, bacteria, yeast, and plants, transposable elements can also be used for in vivo mutagenesis.

The mutagenesis of one or more polynucleotide sequences that encode an ester synthase variant such as a '377 variant generally results in expression of an ester synthase polypeptide product that demonstrates a modified and improved biological function. Similarly, the mutagenesis of one or more polynucleotide sequences that encode a Ppro variant generally results in expression of a Ppro polypeptide product that demonstrates a modified and improved biological function such as enhanced ester synthase activity. For example, when preparing a group of recombinant microorganisms by mutagenesis of one or more polynucleotide sequences including the open reading frame encoding a '377 polypeptide and operably-linked regulatory sequences, the protein expressed from the resulting mutagenized polynucleotide sequences may maintain the '377 ester synthase biological function, however, an improved yield of fatty esters, a decreased yield of beta hydroxy esters and/or an improved compositions comprising a modified mixture of fatty ester products (in terms of chain length, saturation, and the like) is observed upon culture of the recombinant microorganism under conditions effective to express the mutant '377 polynucleotide. Similarly, when preparing a group of recombinant microorganisms by mutagenesis of one or more polynucleotide sequences including the open reading frame encoding a Ppro polypeptide and operably-linked regulatory sequences, the protein expressed from the resulting mutagenized polynucleotide sequences may maintain the Ppro thioesterase biological function but an ester synthase activity is observed in the recombinant microorganism. Because of the ester synthase activity, an improved yield of fatty esters, an improved compositions comprising a modified mixture of fatty ester products (in terms of chain length, saturation, and the like) is observed upon culture of the recombinant microorganism under con-

ditions effective to express the mutant Ppro polynucleotide. In another embodiment, the mutagenesis of one or more Ppro polynucleotide sequences generally results in expression of a Ppro polypeptide product that may retain the same biological function as the wild type or parent Ppro polypeptide even though the mutant Ppro polypeptide demonstrates a modified biological function. For example, when preparing a group of recombinant microorganisms by mutagenesis of one or more polynucleotide sequences including the open reading frame encoding a Ppro polypeptide and operably-linked regulatory sequences, the protein expressed from the resulting mutagenized polynucleotide sequences may maintain the Ppro thioesterase biological function but a ester synthase activity is observed in the recombinant microorganism.

Hot Spots

The disclosure is also based, at least in part, on the identification of certain structurally conserved "hot spots" among variant ester synthase polypeptides such as '377 polypeptides. Hot spots are regions where a high number of mutations were observed that lead to a higher titer of fatty ester product and/or a lower production of beta hydroxy esters in '377 polypeptides. Notably, such regions are seen in variant ester synthase polypeptides that exhibit a greater ester synthase activity as compared to the wild type polypeptide that lacks those hot spots. Hot spots include amino acid regions 39-44; 76-80; 98-102; 146-150; 170-207, in particular 182-207 (e.g., showing the highest number of mutations); 242-246; 300-320; 348-357; and 454-458.

Motifs

The disclosure is also based, at least in part, on the identification of certain structurally conserved motifs among thioesterase polypeptides with enhanced ester synthase activity, wherein a thioesterase polypeptide, such as Ppro, that comprises one of these motifs has greater ester synthase activity as compared to the wild type thioesterase polypeptide which lacks the motif. Accordingly, the disclosure features variant thioesterase polypeptides, such as Ppro, that comprise a motif at amino acid region 73-81 of SEQ ID NO: 51, with the substitutions indicated below.

Leu-Gly-[Ala or Gly]-[Val or Ile or Leu]-Asp-[Ala or Gly]-Leu-Arg-Gly

LG[AG][VIL]D[AG]LRG (SEQ ID NO: 66)

The potential substitutions include alanine (A) or glycine (G) at amino acid position 75; valine (V), isoleucine (I) or leucine (L) at amino acid position 76; and alanine (A) or glycine (G) at amino acid position 78.

Recombinant Host Cells and Recombinant Host Cell Cultures

The recombinant host cells of the disclosure comprise one or more polynucleotide sequences that comprise an open reading frame encoding a polypeptide having ester synthase activity, e.g., any polypeptide which catalyzes the conversion of an acyl-thioester to a fatty ester, together with operably-linked regulatory sequences that facilitate expression of the polypeptide having ester synthase activity in a recombinant host cell. In one preferred embodiment, the polypeptide having improved ester synthase activity is a variant or mutant of '377. In another preferred embodiment, the polypeptide having improved ester synthase activity is a variant or mutant of Ppro. In a recombinant host cell of the disclosure, the open reading frame coding sequences and/or the regulatory sequences may be modified relative to the corresponding wild-type coding sequence of the '377 or Ppro polypeptide. A fatty ester composition is produced by culturing a recombinant host cell in the presence of a carbon source under conditions effective to express the variant ester

synthase polypeptide (e.g., '377) or the variant thioesterase polypeptide with ester synthase activity (e.g., Ppro). Expression of mutant or variant '377 polypeptides results in production of fatty ester compositions with increased yields of fatty esters and decreased yields of beta hydroxyl esters. Expression of mutant or variant Ppro polypeptides results in production of fatty ester compositions with increased yields of fatty esters.

In order to illustrate the disclosure, the applicants have constructed host strains that express variant ester synthase polypeptide sequences (see Examples, *infra*). Examples of variant ester synthase polypeptide sequences that when expressed in a recombinant host cell result in a higher titer of fatty esters include but are not limited to, sequences expressed in host strains 9B12 (SEQ ID NO: 4), pKEV022 (SEQ ID NO: 10), KASH008 (SEQ ID NO: 16), KASH280 (SEQ ID NO: 33) and KASH281 (SEQ ID NO: 35). Examples of variant ester synthase polypeptide sequences that when expressed in a recombinant host cell result in a higher titer of fatty esters and a decrease in β -OH ester production include polypeptide sequences expressed in host strains pKEV28 (SEQ ID NO: 12), and KASH008 (SEQ ID NO: 16), KASH285 (SEQ ID NO: 37), KASH286 (SEQ ID NO: 39), KASH288 (SEQ ID NO: 41) and KASH289 (SEQ ID NO: 43).

Examples of variant Ppro polypeptide sequences that when expressed in a recombinant host cell result in a higher titer of fatty esters include but are not limited to, sequences expressed in host strains PROF1 (SEQ ID NO: 53), PROF2 (SEQ ID NO: 55), P1B9 (SEQ ID NO: 58), N115F (SEQ ID NO: 61), Vc7P4H5 (SEQ ID NO: 63), Vc7P5H9 (SEQ ID NO: 65) and Vc7P6F9 (SEQ ID NO: 31).

The recombinant host cell may produce a fatty ester, such as a fatty acid methyl ester (FAME), a fatty acid ethyl ester (FAEE), a wax ester, or the like. The fatty acid esters are typically recovered from the culture medium. The fatty acid ester composition produced by a recombinant host cell can be analyzed using methods known in the art, for example, GC-FID, in order to determine the distribution of particular fatty acid esters as well as chain lengths and degree of saturation of the components of the fatty ester composition. Methods of Making Recombinant Host Cells and Cultures

Various methods well known in the art can be used to genetically engineer host cells to produce fatty esters and/or fatty ester compositions. The methods can include the use of vectors, preferably expression vectors, comprising a nucleic acid encoding a mutant or variant '377 ester synthase or Ppro thioesterase with ester synthase activity, as described herein. Those skilled in the art will appreciate a variety of viral and non-viral vectors can be used in the methods described herein.

In some embodiments of the present disclosure, a "higher" titer of fatty esters in a particular composition is a higher titer of a particular type of fatty acid ester or a combination of fatty acid esters produced by a recombinant host cell culture relative to the titer of the same fatty acid ester or combination of fatty acid esters produced by a control culture of a corresponding wild-type host cell. In some embodiments, a mutant or variant '377 ester synthase polynucleotide (or gene) sequence or a mutant or variant Ppro polynucleotide (or gene) sequence is provided to the host cell by way of a recombinant vector, which comprises a promoter operably linked to the polynucleotide sequence. In certain embodiments, the promoter is a developmentally-regulated, an organelle-specific, a tissue-specific, an inducible, a constitutive, or a cell-specific promoter. The recombinant vector typically comprises at least one sequence

selected from an expression control sequence operatively coupled to the polynucleotide sequence; a selection marker operatively coupled to the polynucleotide sequence; a marker sequence operatively coupled to the polynucleotide sequence; a purification moiety operatively coupled to the polynucleotide sequence; a secretion sequence operatively coupled to the polynucleotide sequence; and a targeting sequence operatively coupled to the polynucleotide sequence. The one or more polynucleotide sequences, comprising open reading frames encoding proteins and operably-linked regulatory sequences can be integrated into a chromosome of the recombinant host cells, incorporated in one or more plasmid expression system resident in the recombinant host cells, or both.

The expression vectors described herein include a polynucleotide sequence described herein in a form suitable for expression of the polynucleotide sequence in a host cell. It will be appreciated by those skilled in the art that the design of the expression vector can depend on such factors as the choice of the host cell to be transformed, the level of expression of polypeptide desired, etc. The expression vectors described herein can be introduced into host cells to produce polypeptides, including fusion polypeptides, encoded by the polynucleotide sequences as described herein. Expression of genes encoding polypeptides in prokaryotes, for example, *E. coli*, is most often carried out with vectors containing constitutive or inducible promoters directing the expression of either fusion or non-fusion polypeptides. Suitable expression systems for both prokaryotic and eukaryotic cells are well known in the art; see, e.g., Sambrook et al., "Molecular Cloning: A Laboratory Manual," second edition, Cold Spring Harbor Laboratory, (1989). In certain embodiments, a polynucleotide sequence of the disclosure is operably linked to a promoter derived from bacteriophage T5. In one embodiment, the host cell is a yeast cell. In this embodiment, the expression vector is a yeast expression vector. Vectors can be introduced into prokaryotic or eukaryotic cells via a variety of art-recognized techniques for introducing foreign nucleic acid (e.g., DNA) into a host cell. Suitable methods for transforming or transfecting host cells can be found in, for example, Sambrook et al. (supra).

For stable transformation of bacterial cells, it is known that, depending upon the expression vector and transformation technique used, only a small fraction of cells will take-up and replicate the expression vector. In order to identify and select these transformants, a gene that encodes a selectable marker (e.g., resistance to an antibiotic) can be introduced into the host cells along with the gene of interest. Selectable markers include those that confer resistance to drugs such as, but not limited to, ampicillin, kanamycin, chloramphenicol, or tetracycline. Nucleic acids encoding a selectable marker can be introduced into a host cell on the same vector as that encoding a polypeptide described herein or can be introduced on a separate vector. Cells stably transformed with the introduced nucleic acid can be identified by growth in the presence of an appropriate selection drug.

Examples of host cells that are microorganisms, include but are not limited to cells from the genus *Escherichia*, *Bacillus*, *Lactobacillus*, *Zymomonas*, *Rhodococcus*, *Pseudomonas*, *Aspergillus*, *Trichoderma*, *Neurospora*, *Fusarium*, *Humicola*, *Rhizomucor*, *Kluyveromyces*, *Pichia*, *Mucor*, *Myceliophthora*, *Penicillium*, *Phanerochaete*, *Pleurotus*, *Trametes*, *Chrysosporium*, *Saccharomyces*, *Stenotrophomonas*, *Schizosaccharomyces*, *Yarrowia*, or *Streptomyces*. In some embodiments, the host cell is a Gram-positive

bacterial cell. In other embodiments, the host cell is a Gram-negative bacterial cell. In some embodiments, the host cell is an *E. coli* cell. In other embodiments, the host cell is a *Bacillus lentus* cell, a *Bacillus brevis* cell, a *Bacillus stearothermophilus* cell, a *Bacillus licheniformis* cell, a *Bacillus alkalophilus* cell, a *Bacillus coagulans* cell, a *Bacillus circulans* cell, a *Bacillus pumilis* cell, a *Bacillus thuringiensis* cell, a *Bacillus clausii* cell, a *Bacillus megaterium* cell, a *Bacillus subtilis* cell, or a *Bacillus amyloliquefaciens* cell.

In still other embodiments, the host cell is a *Trichoderma koningii* cell, a *Trichoderma viride* cell, a *Trichoderma reesei* cell, a *Trichoderma longibrachiatum* cell, an *Aspergillus awamori* cell, an *Aspergillus fumigatus* cell, an *Aspergillus foetidus* cell, an *Aspergillus nidulans* cell, an *Aspergillus niger* cell, an *Aspergillus oryzae* cell, a *Humicola insolens* cell, a *Humicola lanuginosa* cell, a *Rhodococcus opacus* cell, a *Rhizomucor miehei* cell, or a *Mucor miehei* cell. In yet other embodiments, the host cell is a *Streptomyces lividans* cell or a *Streptomyces murinus* cell. In yet other embodiments, the host cell is an Actinomycetes cell. In some embodiments, the host cell is a *Saccharomyces cerevisiae* cell.

In other embodiments, the host cell is a cell from a eukaryotic plant, algae, cyanobacterium, green-sulfur bacterium, green non-sulfur bacterium, purple sulfur bacterium, purple non-sulfur bacterium, extremophile, yeast, fungus, an engineered organism thereof, or a synthetic organism. In some embodiments, the host cell is light-dependent or fixes carbon. In some embodiments, the host cell has autotrophic activity.

In some embodiments, the host cell has photoautotrophic activity, such as in the presence of light. In some embodiments, the host cell is heterotrophic or mixotrophic in the absence of light. In certain embodiments, the host cell is a cell from *Arabidopsis thaliana*, *Panicum virgatum*, *Miscanthus giganteus*, *Zea mays*, *Botryococcus braunii*, *Chlamydomonas reinhardtii*, *Dunaliella salina*, *Synechococcus* Sp. PCC 7002, *Synechococcus* Sp. PCC 7942, *Synechocystis* Sp. PCC 6803, *Thermosynechococcus elongatus* BP-1, *Chlorobium tepidum*, *Chloroflexus auranticus*, *Chromatium vinosum*, *Rhodospirillum rubrum*, *Rhodobacter capsulatus*, *Rhodopseudomonas palustris*, *Clostridium ljungdahlii*, *Clostridium thermocellum*, *Penicillium chrysogenum*, *Pichia pastoris*, *Saccharomyces cerevisiae*, *Schizosaccharomyces pombe*, *Pseudomonas fluorescens*, or *Zymomonas mobilis*.

Culture and Fermentation of Genetically Engineered Host Cells

As used herein, the term "fermentation" broadly refers to the conversion of organic materials into target substances by host cells, for example, the conversion of a carbon source by recombinant host cells into fatty acids or derivatives thereof by propagating a culture of the recombinant host cells in a media comprising the carbon source. As used herein, the term "conditions permissive for the production" means any conditions that allow a host cell to produce a desired product, such as a fatty acid ester composition. Similarly, the term "conditions in which the polynucleotide sequence of a vector is expressed" means any conditions that allow a host cell to synthesize a polypeptide. Suitable conditions include, for example, fermentation conditions. Fermentation conditions can comprise many parameters, including but not limited to temperature ranges, levels of aeration, feed rates and media composition. Each of these conditions, individually and in combination, allows the host cell to grow. Fermentation can be aerobic, anaerobic, or variations thereof

(such as micro-aerobic). Exemplary culture media include broths or gels. Generally, the medium includes a carbon source that can be metabolized by a host cell directly. In addition, enzymes can be used in the medium to facilitate the mobilization (e.g., the depolymerization of starch or cellulose to fermentable sugars) and subsequent metabolism of the carbon source.

For small scale production, the engineered host cells can be grown in batches of, for example, about 100 μ L, 200 μ L, 300 μ L, 400 μ L, 500 μ L, 1 mL, 5 mL, 10 mL, 15 mL, 25 mL, 50 mL, 75 mL, 100 mL, 500 mL, 1 L, 2 L, 5 L, or 10 L; fermented; and induced to express a desired polynucleotide sequence, such as a polynucleotide sequence encoding a polypeptide having ester synthase activity. For large scale production, the engineered host cells can be grown in cultures having a volume batches of about 10 L, 100 L, 1000 L, 10,000 L, 100,000 L, 1,000,000 L or larger; fermented; and induced to express a desired polynucleotide sequence. The fatty ester compositions described herein are found in the extracellular environment of the recombinant host cell culture and can be readily isolated from the culture medium. A fatty acid derivative may be secreted by the recombinant host cell, transported into the extracellular environment or passively transferred into the extracellular environment of the recombinant host cell culture. The fatty ester composition may be isolated from a recombinant host cell culture using routine methods known in the art.

Screening Genetically Engineered Host Cells

In one embodiment of the present disclosure, the activity of a mutant or variant '377 ester synthase polypeptide is determined by culturing recombinant host cells (comprising one or more mutagenized ester synthase such as '377 polynucleotide sequences), followed by screening to identify characteristics of fatty acid ester compositions produced by the recombinant host cells; for example, titer, yield and productivity of fatty acid esters; and percent beta hydroxy esters. In another embodiment, the activity of a mutant or variant Ppro polypeptide is determined by culturing recombinant host cells (comprising one or more mutagenized thioesterase such as Ppro polynucleotide sequences), followed by screening to identify characteristics of fatty acid ester compositions produced by the recombinant host cells; for example: titer, yield and productivity of fatty acid esters; and free fatty acids. Mutant or variant '377 ester synthase polypeptides or mutant or variant Ppro polypeptides and fragments thereof can be assayed for ester synthase activity using routine methods. For example, a mutant or variant '377 ester synthase polypeptide or Ppro polypeptide or fragment thereof is contacted with a substrate (e.g., an acyl-CoA, an acyl-ACP, a free fatty acid, or an alcohol) under conditions that allow the polypeptide to function. A decrease in the level of the substrate or an increase in the level of a fatty ester or a fatty ester composition can be measured to determine the ester synthase activity.

Products Derived From Recombinant Host Cells

As used herein, "fraction of modern carbon" or fM has the same meaning as defined by National Institute of Standards and Technology (NIST) Standard Reference Materials (SRMs 4990B and 4990C, known as oxalic acids standards HOxI and HOxII, respectively). The fundamental definition relates to 0.95 times the $^{14}\text{C}/^{12}\text{C}$ isotope ratio HOxI (referenced to AD 1950). This is roughly equivalent to decay-corrected pre-Industrial Revolution wood. For the current living biosphere (plant material), fM is approximately 1.1.

Bioproducts (e.g., the fatty ester compositions produced in accordance with the present disclosure) comprising biologically produced organic compounds, and in particular, the

fatty ester compositions produced using the fatty acid biosynthetic pathway herein, have been produced from renewable sources and, as such, are new compositions of matter. These new bioproducts can be distinguished from organic compounds derived from petrochemical carbon on the basis of dual carbon-isotopic fingerprinting or ^{14}C dating. Additionally, the specific source of biosourced carbon (e.g., glucose vs. glycerol) can be determined by dual carbon-isotopic fingerprinting (see, e.g., U.S. Pat. No. 7,169,588). The ability to distinguish bioproducts from petroleum based organic compounds is beneficial in tracking these materials in commerce. For example, organic compounds or chemicals comprising both biologically based and petroleum based carbon isotope profiles may be distinguished from organic compounds and chemicals made only of petroleum based materials. Hence, the bioproducts herein can be followed or tracked in commerce on the basis of their unique carbon isotope profile. Bioproducts can be distinguished from petroleum based organic compounds by comparing the stable carbon isotope ratio ($^{13}\text{C}/^{12}\text{C}$) in each sample. The $^{13}\text{C}/^{12}\text{C}$ ratio in a given bioproduct is a consequence of the $^{13}\text{C}/^{12}\text{C}$ ratio in atmospheric carbon dioxide at the time the carbon dioxide is fixed. It also reflects the precise metabolic pathway. Regional variations also occur. Petroleum, C3 plants (the broadleaf), C4 plants (the grasses), and marine carbonates all show significant differences in $^{13}\text{C}/^{12}\text{C}$ and the corresponding $\delta^{13}\text{C}$ values. Both C4 and C3 plants exhibit a range of $^{13}\text{C}/^{12}\text{C}$ isotopic ratios, but typical values are about -7 to about -13 per mil for C4 plants and about -19 to about -27 per mil for C3 plants (see, e.g., Stuiver et al., Radiocarbon 19:355 (1977)). Coal and petroleum fall generally in this latter range.

$$\delta^{13}\text{C}(\text{‰}) = \left[\frac{(^{13}\text{C}/^{12}\text{C})_{\text{sample}} - (^{13}\text{C}/^{12}\text{C})_{\text{standard}}}{(^{13}\text{C}/^{12}\text{C})_{\text{standard}}} \right] \times 1000$$

A series of alternative RMs have been developed in cooperation with the IAEA, USGS, NIST, and other selected international isotope laboratories. Notations for the per mil deviations from PDB is $\delta^{13}\text{C}$. Measurements are made on CO_2 by high precision stable ratio mass spectrometry (IRMS) on molecular ions of masses 44, 45, and 46. The compositions described herein include fatty ester compositions and products produced by any of the methods described herein. Specifically, fatty ester composition or product can have a $\delta^{13}\text{C}$ of about -28 or greater, about -27 or greater, -20 or greater, -18 or greater, -15 or greater, -13 or greater, -10 or greater, or -8 or greater. For example, the fatty ester composition or product can have a $\delta^{13}\text{C}$ of about -30 to about -15, about -27 to about -19, about -25 to about -21, about -15 to about -5, about -13 to about -7, or about -13 to about -10. In other instances, the fatty ester composition or product can have a $\delta^{13}\text{C}$ of about -10, -11, -12, or -12.3. Fatty ester compositions and products produced in accordance with the disclosure herein can also be distinguished from petroleum based organic compounds by comparing the amount of ^{14}C in each compound. Because ^{14}C has a nuclear half-life of 5730 years, petroleum based fuels containing "older" carbon can be distinguished from fatty ester compositions and bioproducts which contain "newer" carbon (see, e.g., Currie, "Source Apportionment of Atmospheric Particles", Characterization of Environmental Particles, J. Bufile and H. P. van Leeuwen, Eds., 1 of Vol. I of the IUPAC Environmental Analytical Chemistry Series (Lewis Publishers, Inc.) 3-74, (1992)).

The basic assumption in radiocarbon dating is that the constancy of ^{14}C concentration in the atmosphere leads to the constancy of ^{14}C in living organisms. However, because

of atmospheric nuclear testing since 1950 and the burning of fossil fuel since 1850, ^{14}C has acquired a second, geochemical time characteristic. Its concentration in atmospheric CO_2 , and hence in the living biosphere, approximately doubled at the peak of nuclear testing, in the mid-1960s. It has since been gradually returning to the steady-state cosmogenic (atmospheric) baseline isotope rate ($^{14}\text{C}/^{12}\text{C}$) of about 1.2×10^{-12} , with an approximate relaxation "half-life" of 7-10 years. (This latter half-life must not be taken literally; rather, one must use the detailed atmospheric nuclear input/decay function to trace the variation of atmospheric and biospheric ^{14}C since the onset of the nuclear age.) It is this latter biospheric ^{14}C time characteristic that holds out the promise of annual dating of recent biospheric carbon. ^{14}C can be measured by accelerator mass spectrometry (AMS), with results given in units of "fraction of modern carbon" (fM). The fatty ester compositions and products described herein include bioproducts that can have an fM ^{14}C of at least about 1. For example, the bioproduct of the disclosure can have an fM ^{14}C of at least about 1.01, an fM ^{14}C of about 1 to about 1.5, an fM ^{14}C of about 1.04 to about 1.18, or an fM ^{14}C of about 1.11 to about 1.124.

Another measurement of ^{14}C is known as the percent of modern carbon (pMC). For an archaeologist or geologist using ^{14}C dates, AD 1950 equals "zero years old". This also represents 100 pMC. "Bomb carbon" in the atmosphere reached almost twice the normal level in 1963 at the peak of thermo-nuclear weapons. Its distribution within the atmosphere has been approximated since its appearance, showing values that are greater than 100 pMC for plants and animals living since AD 1950. It has gradually decreased over time with today's value being near 107.5 pMC. This means that a fresh biomass material, such as corn, would give a ^{14}C signature near 107.5 pMC. Petroleum based compounds will have a pMC value of zero. Combining fossil carbon with present day carbon will result in a dilution of the present day pMC content. By presuming 107.5 pMC represents the ^{14}C content of present day biomass materials and 0 pMC represents the ^{14}C content of petroleum based products, the measured pMC value for that material will reflect the proportions of the two component types. For example, a material derived 100% from present day soybeans would give a radiocarbon signature near 107.5 pMC. If that material was diluted 50% with petroleum based products, it would give a radiocarbon signature of approximately 54 pMC. A biologically based carbon content is derived by assigning "100%" equal to 107.5 pMC and "0%" equal to 0 pMC. For example, a sample measuring 99 pMC will give an equivalent biologically based carbon content of 93%. This value is referred to as the mean biologically based carbon result and assumes all the components within the analyzed material originated either from present day biological material or petroleum based material. A bioproduct comprising one or more fatty esters as described herein can have a pMC of at least about 50, 60, 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or 100. In other instances, a fatty ester composition described herein can have a pMC of between about 50 and about 100; about 60 and about 100; about 70 and about 100; about 80 and about 100; about 85 and about 100; about 87 and about 98; or about 90 and about 95. In yet other instances, a fatty ester composition described herein can have a pMC of about 90, 91, 92, 93, 94, or 94.2.

Fatty Ester Compositions

Examples of fatty esters include fatty acid esters, such as those derived from short-chain alcohols, including FAEE and FAME, and those derived from longer chain fatty alcohols. The fatty esters and/or fatty ester compositions that

are produced can be used, individually or in suitable combinations, as a biofuel (e.g., a biodiesel), an industrial chemical, or a component of, or feedstock for, a biofuel or an industrial chemical. In some aspects, the disclosure pertains to a method of producing a fatty ester composition comprising one or more fatty acid esters, including, for example, FAEE, FAME and/or other fatty acid ester derivatives of longer chain alcohols. In related aspects, the method comprises a genetically engineered production host suitable for making fatty esters and fatty ester compositions including, but not limited to, FAME, FAEE, fatty acid propyl esters, fatty acid isopropyl esters, fatty acid butyl esters, monoglycerides, fatty acid isobutyl esters, fatty acid 2-butyl esters, and fatty acid tert-butyl esters, and the like.

Accordingly, in one aspect, the disclosure features a method of making a fatty ester composition that may comprise a decreased percentage of beta hydroxy esters and free fatty acids relative to a fatty ester compositions produced by a wild type ester synthase enzyme, such as '377. The variant '377 polypeptide or enzyme with improved fatty acid methyl ester activity has improved properties including, but not limited to, increased beta hydroxy esters, decreased beta hydroxyl esters, increased chain lengths of fatty acid esters, and decreased chain lengths of fatty acid esters. The method includes expressing in a host cell a gene encoding a polypeptide having ester synthase activity. In another aspect, the disclosure features a method of making a fatty ester composition that may or may not comprise free fatty acids, wherein the method includes expressing in a host cell a gene encoding a Ppro polypeptide having ester synthase activity. In some embodiments, the gene encoding the ester synthase polypeptide or thioesterase polypeptide with ester synthase activity is selected from the enzymes classified as EC 2.3.1.75 or EC 3.1.2, respectively, and any other polypeptides capable of catalyzing the conversion of an acyl thioester to fatty esters, including, without limitation, thioesterases, ester synthases, acyl-CoA:alcohol transacylases, alcohol O-fatty acid-acyltransferase, acyltransferases, and fatty acyl-coA:fatty alcohol acyltransferases, or a suitable variant thereof.

In certain embodiments, an endogenous thioesterase of the host cell, if present, is unmodified. In certain other embodiments, the host cell expresses an attenuated level of a thioesterase activity or the thioesterase is functionally deleted. In some embodiments, the host cell has no detectable thioesterase activity. As used herein the term "detectable" means capable of having an existence or presence ascertained. For example, production of a product from a reactant (e.g., production of a certain type of fatty acid esters) is detectable using the methods known in the art or provided herein. In certain embodiments, the host cell expresses an attenuated level of a fatty acid degradation enzyme, such as, for example, an acyl-CoA synthase, or the fatty acid degradation enzyme is functionally deleted. In some embodiments, the host cell has no detectable fatty acid degradation enzyme activity. In particular embodiments, the host cell expresses an attenuated level of a thioesterase, a fatty acid degradation enzyme, or both. In other embodiments, the thioesterase, the fatty acid degradation enzyme, or both, are functionally deleted. In some embodiments, the host cell can convert an acyl-ACP or acyl-CoA into fatty acids and/or derivatives thereof such as esters, in the absence of a thioesterase, a fatty acid derivative enzyme, or both. Alternatively, the host cell can convert a free fatty acid to a fatty ester in the absence of a thioesterase, a fatty acid derivative enzyme, or both. In certain embodiments, the method further includes isolating a fatty ester composition,

a fatty ester or a free fatty acid from the host cell or from the host cell culture. In preferred embodiments of the disclosure, the fatty acid derivative composition comprises a high percentage of fatty esters. In certain embodiments, the fatty ester or fatty ester composition is derived from a suitable alcohol substrate such as a short- or long-chain alcohol.

In general, the fatty ester or fatty ester composition is isolated from the extracellular environment of the host cell. In some embodiments, the fatty ester or fatty ester composition is spontaneously secreted, partially or completely, from the host cell. In alternative embodiments, the fatty ester or fatty ester composition is transported into the extracellular environment, optionally with the aid of one or more transport proteins. In still other embodiments, the fatty ester or fatty ester composition is passively transported into the extracellular environment.

In some embodiments, the disclosure comprises a method of producing a fatty acid ester or a fatty acid ester composition which composition may comprise a decreased percentage of beta hydroxy esters and/or free fatty acids by culturing a genetically engineered host cell under conditions that allow expression or overexpression of a mutant or variant '377 polypeptide. In an alternative embodiment, the disclosure comprises a method of producing a fatty acid ester or a fatty acid ester composition which composition may comprise an increased percentage of beta hydroxy esters and/or free fatty acids by culturing a genetically engineered host cell under conditions that allow expression or overexpression of a mutant or variant '377 polypeptide. In some embodiments, the method further comprises culturing the genetically engineered host cell in medium comprising carbon source (such as a carbohydrate) under conditions that permit production of a fatty acid ester or a fatty acid ester composition which may comprise an increased or decreased percentage of beta hydroxy esters and/or free fatty acids. In other embodiments, the disclosure comprises a method of producing a fatty acid ester or a fatty acid ester composition which composition may comprise free fatty acids by culturing a genetically engineered host cell under conditions that allow expression or overexpression of a Ppro polypeptide having ester synthase activity. In some embodiments, the method further comprises culturing the genetically engineered host cell in medium comprising carbon source such as a carbohydrate under conditions that permit production of a fatty acid ester or a fatty acid ester composition which may comprise free fatty acids.

In some embodiments, the disclosure provides an ester synthase polypeptide comprising the amino acid sequence of SEQ ID NO: 2, with one or more amino acid substitutions, additions, insertions, or deletions wherein the polypeptide has ester synthase activity. A mutant or variant '377 polypeptide has greater ester synthase activity than the corresponding wild type '377 polypeptide. For example, a mutant or variant '377 polypeptide is capable, or has an improved capacity, of catalyzing the conversion of thioesters, for example, fatty acyl-CoAs or fatty acyl-ACPs, to fatty acids and/or fatty acid derivatives. In particular embodiments, the mutant or variant '377 polypeptide is capable, or has an improved capacity, of catalyzing the conversion of thioester substrates to fatty acids and/or derivatives thereof, such as fatty esters, in the absence of a thioesterase activity, a fatty acid degradation enzyme activity, or both. For example, a mutant or variant '377 polypeptide can convert fatty acyl-ACP and/or fatty acyl-CoA into fatty esters in vivo, in the absence of a thioesterase or an acyl-CoA synthase activity.

In other embodiments, the disclosure provides a Ppro polypeptide comprising the amino acid sequence of SEQ ID

NO: 51, with one or more amino acid substitutions, additions, insertions, or deletions wherein the polypeptide has ester synthase activity. In certain embodiments, a Ppro polypeptide of the disclosure has greater ester synthase activity than the corresponding wild type Ppro polypeptide. For example, a Ppro polypeptide of the disclosure having ester synthase activity is capable, or has an improved capacity, of catalyzing the conversion of thioesters, for example, fatty acyl-CoAs or fatty acyl-ACPs, to fatty acids and/or fatty acid derivatives. In particular embodiments, the Ppro polypeptide having ester synthase activity is capable, or has an improved capacity, of catalyzing the conversion of thioester substrates to fatty acids and/or derivatives thereof, such as fatty esters, in the absence of a thioesterase activity, a fatty acid degradation enzyme activity, or both. For example, a Ppro polypeptide having ester synthase activity can convert fatty acyl-ACP and/or fatty acyl-CoA into fatty esters in vivo, in the absence of a thioesterase or an acyl-CoA synthase activity.

In some embodiments, the mutant or variant '377 polypeptide or mutant or variant Ppro polypeptide comprises one or more of the following conserved amino acid substitutions: replacement of an aliphatic amino acid, such as alanine, valine, leucine, and isoleucine, with another aliphatic amino acid; replacement of a serine with a threonine; replacement of a threonine with a serine; replacement of an acidic residue, such as aspartic acid and glutamic acid, with another acidic residue; replacement of residue bearing an amide group; exchange of a basic residue, such as lysine and arginine, with another basic residue; and replacement of an aromatic residue, such as phenylalanine and tyrosine, with another aromatic residue. In some embodiments, the mutant or variant '377 polypeptide or mutant or variant Ppro polypeptide having ester synthase activity has about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 40, 50, 60, 70, 80, 90, 100, or more amino acid substitutions, additions, insertions, or deletions. In some embodiments, the polypeptide variant has ester synthase and/or acyltransferase activity. For example, the mutant or variant '377 polypeptide polypeptide or mutant or variant Ppro polypeptide is capable of catalyzing the conversion of thioesters to fatty acids and/or fatty acid derivatives, using alcohols as substrates. In a non-limiting example, the mutant or variant '377 polypeptide polypeptide or mutant or variant Ppro polypeptide is capable of catalyzing the conversion of a fatty acyl-CoA and/or a fatty acyl-ACP to a fatty acid and/or a fatty acid ester, using a suitable alcohol substrate, such as, for instance, a methanol, ethanol, propanol, isopropanol, butanol, pentanol, hexanol, heptanol, octanol, decanol, dodecanol, tetradecanol, or hexadecanol.

EXAMPLES

The following specific examples are intended to illustrate the disclosure and should not be construed as limiting the scope of the claims. The applicants have employed a one-gene-system for production of fatty acid esters in recombinant host cells. This includes expressing in suitable host cells mutant or variant '377 polypeptides that have been engineered to have improved ester synthase activity in order to produce fatty ester compositions at increased titer and yield. The fatty ester compositions obtained include FAME as well as increased and decreased amounts of beta hydroxy esters. The same strategy was used for making Ppro polypeptides with enhanced ester synthase activity for the production of fatty esters at increased titer and yield.

Protocols:

All protocols use 96 well plates, master block, 2 mL (Greiner Bio-One, Monroe, NC or Corning, Amsterdam, The Netherlands) for growing cultures and Costar plates for extracting fatty acid species from the culture broth. The protocols below include the fermentation conditions and can be used to evaluate fatty acid species production. Alternative protocols can also be used to evaluate fatty acid species production.

Protocol 1 (FA4P, 32° C.):

From an LB culture growing in 96 well plate: 30 µL LB culture was used to inoculate 270 µL FA2P (Table 4), which was then incubated for approximately 16 hours at 32° C. shaker. 30 µL of the overnight seed was then used to inoculate 300 µL FA4P+2% MeOH+1 mM IPTG (Table 4). The cultures were then incubated at 32° C. shaker for 24 hours, when they were extracted following the standard extraction protocol detailed below (Protocol 5).

Protocol 2 (BP3G2P, 32° C.):

Library glycerol stocks were thawed and 10 µL transferred to 150 µL LB+100 µg/mL spectinomycin in 96-shallow-well plates, which were incubated at 32° C. for 20 hours. 20 µL of this culture was used to inoculate 280 µL BS1G4Pmedia (Table 4), which was incubated at 32° C. shaking at 250 rpm for 20 hours. 20 µL of this culture was used to inoculate 380 µL BP3G2P media (Table 4). Cultures were incubated at 32° C. shaking for 24 hours, when they were either subjected to the Nile Red assay following the standard protocol detailed below or extracted following the standard extraction protocol detailed below (Protocol 5).

Protocol 3 (FA2, 32° C.):

From an LB culture growing in a 96 well plate: 30 µL of LB culture was used to inoculate 270 µL FA2, which was then incubated for approximately 16 hours at 32° C. on a shaker. 30 µL of the overnight seed was then used to inoculate 300 µL FA2+2% MeOH+1 mM IPTG. The cultures were then incubated at 32° C. on a shaker for 24 hours, when they were extracted following the standard extraction protocol detailed below (Protocol 5).

Protocol 4 (FA2P, 32° C.):

From an LB culture growing in 96 well plate: 30 µL LB culture was used to inoculate 270 µL FA2P, which was then incubated for approximately 16 hours at 32° C. shaker. 30 µL of the overnight seed was then used to inoculate 300 µL FA2P+2% MeOH+1 mM IPTG. The cultures were then incubated at 32° C. shaker for 24 hours, when they were extracted following the standard extraction protocol detailed below (Protocol 5).

Protocol 5 (Fatty Acid Species Standard Extraction):

To each well to be extracted 40 µL 1M HCl, then 300 µL butyl acetate with 500 mg/L C11-FAME as internal standard was added. The 96 well plate was then heat-sealed using a plate sealer (ALPS-300; Abgene, ThermoScientific, Rockford, IL), and shaken for 15 minutes at 2000 rpm using MixMate (Eppendorf, Hamburg, Germany). After shaking, the plate was centrifuged for 10 minutes at 4500 rpm at room temperature (Allegra X-15R, rotor SX4750A, Beckman Coulter, Brea, CA) to separate the aqueous and organic layers. 50 µL of the organic layer was transferred to a 96 well plate (96-well plate, polypropylene, Corning, Amsterdam, The Netherlands). The plate was heat sealed then stored at -20° C. until it was evaluated by GC-FID by employing standard automated methods.

Protocol 6 (Fatty Acid Species Standard Nile Red Assay):

After 24 hour fermentation, Nile Red assay was performed by adding 70 µL of fermentation broth to 130 µL of 1.54 µg/mL Nile Red in 84.6% water and 15.4% acetonitrile

solution (for a final assay concentration of 1 µg/mL Nile Red) in a Greiner MicroLonFluotrac 200 plate and mixed by pipetting up and down. Relative fluorescence units were measured at excitation of 540 nm and emission of 630 nm using the SpectraMax M2.

Protocol 7 (Building a Saturation Library):

Standard techniques known to those of skill in the art were used to prepare saturation libraries. For example, the vector backbone may be prepared by using restriction endonucleases in the vector, while the creation of diversity in the DNA insert can be generated using degenerate primers. For example, the cloning of the vector backbone and a DNA insert with diversity can be performed using InFusion Cloning System (Clontech Laboratories, Inc., Mountain View, CA) according to the manufacturer's protocol.

Protocol 8 (Building a Combination Library):

Mutations identified as beneficial were combined to provide Ppro or 377 variants with further improvements in the production of fatty acid species or FAME. Standard techniques known to those of skill in the art were used to prepare the combination libraries. For example, the vector backbone can be prepared by using restriction endonucleases in the vector, while the creation of diversity in the DNA insert can be generated using primers to introduce the desired mutations. For example, the cloning of the vector backbone and a DNA insert with diversity can be performed using InFusion Cloning System (Clontech Laboratories, Inc., Mountain View, CA), according to manufacturer's protocol. For example, combination libraries can be generated using the transfer PCR (tPCR) protocol (Erijman et al., 2011. *J. Structural Bio.* 175, 171-177).

Protocol 9 (Library Screening):

Once the library diversity was generated in a saturation library or combination library, it was screened using one of the methods described above (Protocols 1-4). When screening ester synthase variants, three types of hits were identified: (1) increased fatty acid ester species titer ("FAS" titer); (2) increased amount of FAME produced; e.g., C14-FAME; and/or (3) decreased amount of beta hydroxyl ester produced. The mutations in the 377 variants within each hit were identified by sequencing using standard techniques routinely employed by those of skill in the art. The mutations in the 377 variants within each hit were identified by sequencing using standard techniques routinely employed by those of skill in the art. Tables 1 and 2 list the mutations ("hits") identified as beneficial in saturation libraries and Table 5 lists mutations ("hits") identified as beneficial in combination libraries. When screening Ppro variants, two types of hits were identified (1) increased fatty acid ester species; (2) increase in the amount of FAME produced. The mutations in the Ppro variants within each hit were identified by sequencing and using standard techniques routinely employed by those of skill in the relevant field. Tables 8 through 11 list the mutations ("hits") identified as beneficial in saturation libraries and Tables 12 and 14 list mutations ("hits") identified as beneficial in combination libraries.

Example 1

Saturation Library Prepared Using WS377 as a Template

A full saturation library of the ester synthase from *Marinobacter hydrocarbonoclasticus* ("377"), was built and screened for variants that showed improvement over the wild type WS377 (Protocol 7). The plasmid used to make the full saturation library was designated pKEV027 which

expressed SEQ ID NO: 1 (i.e., the wild type nucleic acid sequence of WS377). The full saturation library was screened in an *E. coli* strain (BD64) that was engineered to block beta oxidation (via deletion of FadE) and overexpress a number of fatty acid biosynthetic pathway enzymes (e.g., FabA, FabB, FabF, FabG, FabI, FabH, FabV, FabZ, accABCD) through standard manipulation techniques. However, any suitable host strain can be used herein. The libraries were screened using one of the standard protocols (Protocols 1-2) described above. The improvements were originally classified as either improving titer of FAME or reducing the fraction of beta hydroxy esters (referred to herein as “ β -OH esters” or “ β -OH FAME” or “beta-OH” or “beta-hydroxy”) without necessarily affecting FAME titer. The results from screening saturation libraries are shown in Tables 1 and 2 below. Table 1 presents results for mutations that all showed increased fatty ester titers when using '377 as the template (SEQ ID NO: 1). Table 2 presents results for mutations that led to a decrease in % β -OH FAME using '377 as the template. Notably, some mutants showed both, an improved titer of FAME as well as a decrease in β -OH FAME, which is beneficial if the final desired product is mostly FAME. Some mutants showed an improved titer of FAME as well as an increase in β -OH FAME. In addition, some mutants resulted in an increase or decrease of short-chain esters.

As can be seen in Table 1 below, mutations from the '377 saturation libraries correlated with improved fatty ester titer in all mutants, while the production of β -OH esters and shorter chain length ester varied from mutant to mutant. For example, a substitution of the amino acid at position 15 (Serine to Glycine) changed the total FAME titer from baseline 1 to 2.402. Thus, the titer increased by 2.4 fold compared to baseline (i.e., wild type). In this particular mutant, β -OH esters increased by about 2.66 fold. The production of esters with a carbon chain length of 14 over esters with a carbon chain length of 16 increased by 1.48 fold over baseline (i.e., wild type). On the other hand, a substitution of the amino acid position 393 (Glutamate to Glycine) increased the FAME titer by 2.3 fold and decreased the β -OH esters to 70% of control while C14/C16 ester production stayed about the same.

TABLE 1

Mutation	Average		
	Total FAME	% β -OH	C14/C16
S15G	2.402	2.664	1.476
Q26T	1.356	2.233	1.163
L39M	1.64	2.82	0.936
L39S	2.228	2.368	1.127
L39A	1.876	2.412	0.94
R40S	1.328	2.587	1.259
D41H	2.225	2.344	1.312
D41G	2.031	3.474	1.217
D41A	1.792	2.38	2.074
V43K	2.109	1.617	0.421
V43S	1.768	2.283	1.053
T44F	2.007	1.577	0.872
A73Q	2.195	1.877	0.915
V76L	1.553	1.255	0.76
D77A	1.952	1.645	0.887
K78W	1.758	1.469	0.826
K78F	1.794	1.984	0.767
I80V	2.961	2.903	1.028
R93T	1.649	1.587	0.734
G101L	1.787	1.294	1.288
I102R	1.718	1.418	0.928

TABLE 1-continued

	Mutation	Average		
		Total FAME	% β -OH	C14/C16
5	N110R	1.582	2.038	0.957
	P111S	2.05	1.09	1.176
	P111G	1.855	1.228	1.154
	G126D	1.963	1.065	0.751
10	R131M	2.139	1.857	1.115
	V155G	2.267	0.865	1.127
	N164R	1.692	1.114	0.806
	V171R	2.657	2.152	0.815
	V171E	1.472	1.065	1.042
	R172W	1.499	6.244	0.922
15	D182G	1.89	1.18	1.147
	E184L	2.352	1.055	1.157
	E184F	1.589	1.026	1.108
	E184S	1.566	0.937	1.147
	E184R	1.733	1.185	1.242
	E184G	1.647	1.463	1.138
20	A185M	1.518	1.134	1.131
	A185L	1.754	2.777	1.176
	P188R	2.519	1.138	1.108
	A190R	1.86	1.204	1.314
	A190W	1.559	1.645	1.196
	S192V	3.748	2.086	1.148
	S192A	1.518	0.998	0.902
25	Q193R	1.735	1.2	1.344
	Q201W	1.748	1.357	1.176
	Q201V	1.79	1.289	1.32
	Q201A	1.929	1.154	1.191
	A202L	2.571	1.224	1.192
30	D203R	1.998	1.042	1.552
	P206F	1.466	1.726	1.043
	G212L	1.919	1.179	1.167
	V219L	2.017	1.396	0.737
	T242R	1.879	0.894	1.072
	T242K	1.678	2.272	1.115
	A243R	2.56	1.232	0.912
35	R246Q	3.571	0.692	0.412
	R246L	2.083	0.746	0.711
	V272A	1.984	1.511	0.78
	Q287S	1.723	1.275	0.646
	D292F	1.827	1.162	0.814
	P294G	1.649	1.692	0.874
	G298G	1.511	1.383	0.796
40	I303G	3.36	0.388	0.21
	I303W	2.075	0.578	0.58
	R304W	1.578	0.425	0.354
	F317W	2.208	2.076	1.529
	I319G	2.515	0.89	0.721
	A323G	1.766	1.188	0.751
45	D328F	1.816	1.055	0.671
	Q334S	1.573	0.935	0.821
	Q348A	1.442	3.247	2.463
	P351G	1.875	1.211	1.312
	S353T	2.123	2.007	0.957
	M360W	2.303	0.634	0.582
50	M360S	1.502	0.397	0.394
	Y366W	1.506	0.969	1.175
	Y366G	1.407	1.378	0.91
	G375A	1.96	1.821	1.24
	M378A	1.877	1.616	0.834
	E393G	2.305	0.723	0.932
55	T395E	1.655	1.224	0.719
	V409L	2.314	0.796	0.295
	I420V	1.474	1.119	0.828
	S442G	1.947	1.166	1.226
	A447C	1.775	1.819	1.132
	A447L	1.545	1.341	1.021
	A447I	1.867	1.459	1.117
60	L454V	1.804	1.137	0.672
	R2S, Q348A	1.628	3.976	2.475
	R2R, S235S	1.178	1.949	1.213
	G4R, N331G	2.262	2.431	1.324
	TSP, S186T	1.6	2.837	1.198
	TSS, R98D	2.287	2.01	1.277
65	TSP, G310H	1.711	2.141	1.217
	L6Q, Q193S	1.529	2.069	2.878

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TABLE 1-continued

Mutation	Average		
	Total FAME	% β -OH	C14/C16
L6Q, A279V	1.414	3.573	1.003
D7N, T170R	3.163	1.079	0.853
G33S, S442E	2.28	1.453	1.199
R150P, A279G	1.702	1.434	0.769
E184G, K472T	2.619	1.022	1.269
V187G, *474Y	1.678	1.481	1.028
P188R, A197V	2.944	1.078	1.146
M195G, A197T	1.291	4.127	1.223
R207A, P366P	1.877	1.164	0.685
D307V, T470P	1.858	1.094	1.026
24 bp repeat	1.996	2.34	0.944
24bp repeat, R177V	2.553	2.516	1.518
24bp repeat, V171R	2.724	2.338	0.821

Some mutations in the tables have stars, for example, *474Y, which means that the position has a stop codon.

The applicants discovered a 24 bp repeat in one of the variant mutant nucleic acid sequences (i.e., SEQ ID NO: 28). The 24 bp repeat is a duplication of nucleic acid region ATGAAACGTCTCGGAACCCTGGAC (SEQ ID NO: 27) and located at the start of the gene prior to the ATG start codon. Thus, SEQ ID NO: 28 is a variant '377 nucleic acid sequence (this sequence is the result of SEQ ID NO: 27 and SEQ ID NO: 1 combined with no additional mutations) that codes for the amino acid sequence of SEQ ID NO: 29. SEQ ID NO: 29 gave surprisingly good titer as shown in Table 1. However, two additional mutants were detected that included SEQ ID NO: 29 as a template sequence with additional mutations in it, wherein the mutations increased the titer even further (see Table 1). (The numbering of mutations does not include the 24 bp repeat.) Notably, the 24 bp repeat led to an increase in titer of FAS and % beta-OH FAME, as seen in Table 1 which depicts the titer obtained when expressing SEQ ID NO: 29 in a host strain.

As can be seen in Table 2 below, some of the mutations from the '377 saturation libraries correlated with a decrease in % β -OH FAME while still increasing the total FAME tighter to a lesser degree compared in Table 1 (supra). However, some mutants significantly decreased % β -OH FAME while still significantly increasing total FAME titer.

TABLE 2

Mutations	Average		
	Total FAME	% β -OH	C14/C16
T24W	1.141	0.533	1.024
V69W	1.154	0.661	0.657
S105G	1.146	0.875	1.109
P111D	1.271	0.802	0.822
D113V	1.242	0.687	0.629
D113A	1.279	0.740	0.580
S115R	1.118	0.880	0.925
C121S	1.088	0.756	0.969
H122S	1.186	0.905	0.780
L127G	1.455	0.644	0.507
L134T	1.350	0.495	0.611
T136G	1.090	0.591	0.472
I146K	1.523	0.381	0.342
I146R	1.328	0.199	0.220
S147A	1.233	0.796	0.945
T158R	1.694	0.749	0.855
M165K	1.260	0.889	0.569

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TABLE 2-continued

Mutations	Average		
	Total FAME	% β -OH	C14/C16
V171H	1.168	0.677	0.902
V171W	1.168	0.091	0.812
R172S	1.590	0.505	0.897
P173W	1.189	0.647	0.652
S192L	1.215	0.898	1.176
A228G	1.147	0.787	0.503
V234C	1.059	0.185	0.775
H239G	1.318	0.833	0.874
Q244G	1.397	0.601	0.923
R246W	1.191	0.312	0.323
R246Q	3.571	0.692	0.412
R246G	1.660	0.669	0.431
R246V	1.130	0.454	0.838
R246L	2.083	0.746	0.711
R246A	1.470	0.757	0.559
Q250W	1.061	0.850	0.860
Q253G	1.186	0.772	0.877
L254R	1.409	0.647	0.900
L254T	1.089	0.704	0.868
K258R	1.098	0.657	0.759
S264D	1.220	0.876	0.549
S264V	1.411	0.730	0.489
S264W	1.772	0.646	0.491
G266S	1.121	0.846	0.613
S267G	1.214	0.262	0.263
Y274G	1.115	0.789	0.556
A285R	1.273	0.864	0.893
V301A	1.500	0.755	0.440
N302G	1.355	0.254	0.182
I303G	3.360	0.388	0.210
I303W	2.075	0.578	0.580
I303R	1.257	0.373	0.857
R304W	1.578	0.425	0.354
A306G	1.103	0.808	1.164
D307G	1.079	0.645	1.095
D307L	1.197	0.817	1.279
D307R	1.537	0.679	0.848
D307V	1.273	0.585	0.984
E309A	1.190	0.544	0.775
E309G	1.392	0.735	0.667
E309S	1.249	0.730	0.768
G310R	1.801	0.886	1.146
G310V	1.098	0.732	0.879
T311S	1.163	0.634	0.297
T313S	1.208	0.587	0.903
Q314G	1.140	0.347	0.694
I315F	1.345	0.178	0.182
S316G	1.079	0.000	0.413
A320C	1.387	0.777	0.595
A327I	1.057	0.373	0.889
K349C	1.145	0.781	0.481
K349Q	1.105	0.637	0.693
K349H	1.243	0.663	0.458
K349A	1.200	0.484	0.472
K352I	1.207	0.576	0.418
K352N	1.084	0.783	0.679
T356W	1.153	0.345	0.727
T356G	1.135	0.688	1.019
M360R	1.238	0.375	0.487
M360S	1.502	0.397	0.394
M360W	2.303	0.634	0.582
M360Q	1.062	0.735	0.865
M363W	1.218	0.000	0.497
P365G	1.102	0.611	0.376
I367M	1.274	0.704	0.768
M371R	1.297	0.073	0.554
G373R	1.440	0.720	0.436
G375S	1.146	0.609	0.816
G375V	1.429	0.770	0.833
T385G	1.215	0.875	0.609
E393R	1.290	0.234	0.478
E393W	1.768	0.817	0.852
E393G	2.305	0.723	0.932
E404K	1.350	0.762	0.603
E404R	1.089	0.793	0.529

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TABLE 2-continued

Mutations	Average		
	Total FAME	% β -OH	C14/C16
V409L	2.314	0.796	0.295
S410G	1.576	0.786	0.653
L411A	1.274	0.735	1.060
C422G	1.067	0.581	0.572
S424G	1.755	0.538	0.657
S424Q	1.268	0.395	0.629
D455E	1.096	0.737	0.749
E458W	1.428	0.654	0.503
I461G	1.259	0.927	0.584
K1T, L1_355_L3	1.183	0.874	0.732
K1N, I146L	1.269	0.695	0.692
L30H, K472*	1.054	0.872	1.139
D41Y, D307F	1.604	0.664	0.633
E99Q, D307N	1.610	0.801	0.969
V149L, N430G	1.232	0.569	0.900
R150P, A190P, E286H	1.186	0.614	0.338
V171F, R469R	1.085	0.619	0.713
L457Y, R467T	1.111	0.544	0.594

Example 2

Combination Library Prepared Using WS377 as a Template

Standard techniques were employed to prepare combination libraries (Protocol 8). The mutations tested in combination libraries (Tables 3A and 3B) were originally identified in the full saturation library of '377 (as described in Example 1). The plasmid used to make the combination libraries was pKEV38, which was constructed by removing the accABCD_{-birA} from pKEV027. Without wanting to be bound by theory, the removal of the accABCD_{-birA} is believed to reduce the intermediate levels of the fatty acids in the biosynthetic pathway such that the affinity of the mutant enzyme for the substrate is more easily detectable. Cells were plated on 2NBT+Nile Red agar (Table 4). Colonies were observed using the Dark Reader DR88X Transilluminator at an excitation range of 400 nm-500 nm and an amber filter permissive at 580-620 nm, handpicked into LB+100 μ g/mL spectinomycin, grown overnight, and screened using one of the standard protocols (Protocols 1-2) described above. As shown in FIG. 4, one of the variants from this library showed a dramatic increase in FAS over the control (BD64/pKEV038, SEQ ID NO: 2). FIG. 4 shows the results of a plate fermentation of combination library hits. Several library hits outperformed the control (i.e., BD64/pKEV038) showing that '377 variants can support high production of FAS. The plasmid of the highest-producing variant, named pSHU10 (expressing variant '377 of SEQ ID NO: 14), was isolated and used as template for subsequent combination libraries.

The combination libraries were screened in strain BD64 (supra) by using one of the standard protocols (Protocols 1-2; supra). The results from screening '377 combination libraries are shown in Table 5.

Table 3A below shows the amino acid substitutions that were introduced in the first round of combination libraries.

TABLE 3A

Mutations	
T5S; S15G; G33S; L39S; L39A; D41A; V43K; I102R; P111S; R131M; V171R; P188R; S192V; Q201V; R246Q; R246L; I303W; I303G; F317W; I319G; S353T; E393G; V409L; S442G	

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Table 3B below shows the amino acid substitutions that were introduced in the second round of combination libraries.

TABLE 3B

5	K1T	G101L	R172S	D203R	A285R	A327I	T385G
	K1N	I102R	R172W	P206F	E286H	D328F	E393G
	G4R	S105G	P173W	R207A	Q287S	N331G	E393R
10	S5P	N110R	R177V	G212L	D292F	Q334S	E393W
	S5T	S111P	A179V	G212A	P294G	Q348A	T395E
	L6Q	S111G	D182G	V219L	V301A	Q348R	E404K
	D7N	S111D	E184L	A228G	N302G	K349C	E404R
	G15S	D113V	E184G	V234C	I303W	K349Q	L409V
	T24W	D113A	E184R	H239G	I303G	K349H	S410G
15	Q26T	S115R	E184F	T242R	I303R	K349A	L411A
	L30H	C121S	E184S	T242K	R304W	P351G	I420V
	G33S	H122S	A185L	A243R	A306G	K352I	C422G
	L39A	G126D	A185M	Q244G	D307F	K352N	S424G
	L39M	L127G	S186T	R246Q	D307N	T353S	S424Q
20	L39S	R131M	V187G	R246L	D307G	T356G	N430G
	R40S	L134T	V187R	R246G	D307L	T356W	G442S
	D41A	T136G	R188P	R246W	D307R	Q357V	G442E
	D41H	I146K	A190R	R246V	D307V	M360R	M443G
	D41G	I146R	A190P	R246A	E309A	M360W	A447C
	D41Y	I146L	A190W	Q250W	E309G	M360S	A447L
25	V43K	S147A	S192V	Q253G	E309S	M360Q	A447I
	V43S	V149L	S192L	L254R	G310R	M363W	L454V
	T44F	R150P	S192A	L254T	G310H	P365G	D455E
	V69W	V155G	Q193S	K258R	G310V	Y366W	L457Y
	A73Q	T158R	Q193R	S264D	T311S	Y366G	E458W
30	V76L	N164R	M195G	S264V	T313S	I367M	I461G
	D77A	M165K	A197V	S264W	Q314G	M371R	R467T
	K78W	T170R	A197T	G266S	I315F	G373R	K472T
	K78F	R171V	L200R	S267G	S316G	G375S	K472*
	I80V	R171H	Q201V	V272A	W317F	G375V	
	R93T	R171E	Q201W	Y274G	I319G	G375A	
35	R98D	R171F	Q201A	A279G	A320C	M378A	
	E99Q	R171W	A202L	A279V	A323G	V381F	

Table 4 below shows the media names and formulations that were used.

TABLE 4

Media Name	Formulation		
45 2NBT + Nile Red agar	1	X	P-lim Salt Soln w/(NH ₄) ₂ SO ₄
	1	mg/ml	Thiamine
	1	mM	MgSO ₄
	1	mM	CaCl ₂
	30	g/L	glucose
	1	X	TM2
50	5	g/L	casaminoacids
	15	g/L	Agarose
	2%		Methanol
	0.5	mg/L	Nile Red
	10	mg/L	Fe Citrate
	100	mg/L	spectinomycin
	100	mM	BisTris (pH 7.0)
55 BS1G4P	1	X	P-lim Salt Soln w/(NH ₄) ₂ SO ₄
	1	mg/ml	Thiamine
	1	mM	MgSO ₄
	0.1	mM	CaCl ₂
	10	g/L	glucose
	1	X	TM2
60	10	mg/L	Fe Citrate
	100	μ g/mL	spectinomycin
	100	mM	BisTris (pH 7.0)
BP3G2P	0.5	X	P-lim Salt Soln w/(NH ₄) ₂ SO ₄
	1	mg/ml	Thiamine
	1	mM	MgSO ₄
65	0.1	mM	CaCl ₂
	50	g/L	glucose

TABLE 4-continued

Media Name	Formulation		
FA2P	1	X	TM2
	10	mg/L	Fe Citrate
	100	μg/mL	spectinomycin
	100	mM	BisTris (pH 7.0)
	2.5	g/L	(NH ₄) ₂ SO ₄
	2%		MeOH
	1	mM	IPITG
	1	X	P-lim Salt Soln
	2	g/L	NH ₄ Cl
	1	mg/ml	Thiamine
FA4P	1	mM	MgSO ₄
	0.1	mM	CaCl ₂
	30	g/L	glucose
	1	X	TM2
	10	mg/L	Fe Citrate
	100	mM	BisTris (pH 7.0)
	0.5	X	P-lim Salt Soln
	2	g/L	NH ₄ Cl
	1	mg/ml	Thiamine
	1	mM	MgSO ₄
FA2 (2 g/L NH ₄ Cl) for 96 well plate screen	0.1	mM	CaCl ₂
	50	g/L	glucose
	1	X	TM2
	10	mg/L	Fe Citrate
	100	mM	BisTris (pH 7.0)
	1	X	Salt Soln
	1	g/L	NH ₄ Cl
	1	mg/ml	Thiamine
	1	mM	MgSO ₄
	0.1	mM	CaCl ₂
FA2 (2 g/L NH ₄ Cl) with Triton for shake flask validation	30	g/L	glucose
	1	X	TM2
	10	mg/L	Fe Citrate
	100	mM	BisTris (pH 7.0)
	1	X	Salt Soln
	1	g/L	NH ₄ Cl
	1	mg/ml	Thiamine
	1	mM	MgSO ₄
	0.1	mM	CaCl ₂
	30	g/L	glucose
	1	X	TM2
	10	mg/L	Fe Citrate
	0.05%		Triton X-100
	100	mM	BisTris (pH 7.0)

Table 5 below depict the secondary screen, showing the mutations over pSHU10 (which contains mutations T5S, S15G, P111S, V171R, P188R, F317W, S353T, V409L, S442G) and the normalized GC-FID results. The results depicted are relative to control (i.e., pSHU10, SEQ ID NO: 14). Each line on the table represents a different combination mutant that was made. All mutants were measured to have distinct properties in that they either affected the total FAME produced, the total FAS produced, the percent FAME produced, or the ester species in the final composition (with C12/C14 or C14/C16 chain length), or a combination of these properties. As can be seen in Table 5, some mutants behaved similar to other mutants in terms of their properties.

For example, as shown on the first line in Table 5 below, the mutant containing mutations A190P, N302G, T313S, and Q348R produced a higher percent FAME (and reduction of β-OH FAME), and also increased the production of C14/C16 chain length esters, meaning it produced more of C14 esters than C16 esters in the final product. This is desirable since shorter chain esters can be used for many products including biodiesel. The results also show that the applicants were able to alter the final chain length of ester species in the final composition.

The following combinations of mutations relate to the results shown herein; their performance is relative to the control (pSHU10; SEQ ID NO: 14). KASH008 (SEQ ID NO: 16) increased the % FAME by 1.4 fold, while increasing titer by 1.06 fold. KASH032 (SEQ ID NO: 18) increased % FAME by 1.29 fold, while affecting titer. KASH040 (SEQ ID NO: 20) increased % FAME by 1.16 fold, while the titer decreased to 0.91 times the control. KASH078 (SEQ ID NO: 26) increased the % FAME by 1.25 fold, while decreasing titers to 0.76 of control. KASH060 (SEQ ID NO: 22) and KASH061 (SEQ ID NO: 24) are of special interest because they affected not only the % FAME, but they also produced significantly shorter ester chain products, as was shown by an increase in C12/C14 (6.8 fold for KASH060 and 3.1 fold for KASH061) and C14/C16 (4.0 fold for KASH060 and 3.2 fold for KASH061).

TABLE 5

Mutation	Total FAME	Total FAS	% FAME	C12/C14	C14/C16
A190P, N302G, T313S, Q348R	0.545	0.475	1.146	0.945	1.486
A190P, S264V, A327I, Q348A	0.444	0.807	0.556	2.441	3.171
A197T, V219L, R467L	0.409	0.473	0.865	1.491	1.821
A197V, Q250W, G442S	0.663	0.647	1.024	1.130	1.610
A228G, D328F, L409V	0.922	1.385	0.701	1.254	1.918
A279V, K472T	0.904	1.286	0.732	0.643	1.733
A327I	0.682	0.923	0.738	1.225	2.259
A327I, Q348R, V381F, I420V	0.473	0.974	0.485	2.536	3.363
A73Q, G101L, H122S, R172S, Q201W, Q357V	0.947	0.881	1.103	0.448	1.539
A73Q, G101L, H122S, V187G, Q201V, N331G, A447L	0.530	0.525	1.009	0.838	1.651
A73Q, H122S, V187G, Q201A, V234C, G375S, A447L	0.484	0.390	1.242	0.647	1.231
A73Q, Q201W, I315F, G375V, A447I	0.437	0.350	1.242	0.081	0.951
C121S, A228G, A279V, K472*	0.912	0.989	0.962	0.799	1.389
C121S, H122Q, S147A, L200R	0.654	0.608	1.122	0.244	1.164
C121S, R171H, I303R	0.901	0.805	1.199	2.550	1.812
C121S, S147A	1.016	1.172	0.945	2.072	1.144
D113V, E184F, G212L, H239G, V272A, T395E, N430G	0.528	0.506	1.044	1.278	1.557
D113V, E184F, N430G, I461G	0.818	0.784	1.042	0.923	1.411
D113V, E184R, D203E, D307N, N430G, I461G	0.918	0.957	0.965	0.867	1.652
D307F	0.991	1.122	0.886	0.835	1.563
D307N	0.379	0.393	0.973	1.675	1.700
D328F, K472T	1.068	1.580	0.742	2.931	1.740

TABLE 5-continued

Mutation	Total FAME	Total FAS	% FAME	C12/C14	C14/C16
D328F, T356G, L409V, A468T, K472T	0.971	1.303	0.772	0.786	1.645
D41G, Q193R, Q244G, G310V, Y366G, S424G	0.669	0.569	1.176	0.000	0.576
D41H, I80V, D182G, Q193S, R207A, S267G, G310V, S424Q	0.535	0.414	1.296	1.056	0.598
D41Y, I80V, A320C	0.579	0.558	1.043	1.109	1.219
D41Y, N110R, Q193S, Y366W	0.641	1.214	0.528	1.179	2.665
D455E	0.671	0.842	0.800	1.188	2.183
D77A, A190W, T242K, N302G, T313S	0.560	0.436	1.285	2.048	0.554
D77A, L127G, V155G, A190R, D455E	0.304	0.237	1.292	0.670	1.216
D77A, L127G, V155G, V381F	0.699	0.646	1.084	1.291	1.415
D77A, N302G, T313S, Q348R, M363W, V381F, I420V	0.357	0.300	1.192	7.037	1.598
D77A, S105G, A190P	0.593	0.657	0.925	1.077	1.678
D77A, T170M, R177V, T242R, S264D	0.832	1.206	0.689	1.095	1.851
D77A, V155G, A190P, T242K	0.640	0.589	1.088	1.105	1.726
D7N, D41Y, R207A, G310V, Y366G	0.762	0.649	1.174	0.886	0.767
D7N, N164R, D182G, R207A, G310R, P351G, S424Q, T470A	0.509	0.405	1.258	0.000	0.391
D7N, Q193R, Y366W, E393G, S424Q, E458G	0.655	0.603	1.086	1.653	0.719
D7N, Q193R, Y366W, E393G, S424Q, E458W	0.658	0.561	1.172	0.637	0.538
E184F, A323G, T395E, I461G	0.960	1.009	0.952	0.868	1.404
E184G, I461G	0.994	1.164	0.854	0.829	1.631
E184G, M195G, G212A, T311S, N430G, I461G	0.870	0.933	0.932	0.809	1.705
E184L, M195G, G212A, H239G, T311S, K352I	0.769	0.654	1.178	0.943	1.229
E184R, I461G	0.375	0.438	0.857	0.930	2.234
E184S, G212A, V272A, I367M, I461G	0.707	0.635	1.118	1.175	1.660
E184S, G212L, H239G, K352N, I461G	0.734	0.763	0.964	1.112	1.735
E458W	0.894	1.126	0.794	1.125	1.692
E99Q, A228G, A279G, I303R, L409V, K472T	0.788	0.709	1.193	2.434	1.496
E99Q, C121S, T356G, K472T	1.360	1.552	0.946	2.043	1.292
E99Q, I303W, T356G, K472T	1.286	1.260	1.096	1.966	1.049
E99Q, R171V, M443G	0.734	0.790	0.971	0.805	1.602
E99Q, S147A, R171W, L200R, K472*	1.019	1.540	0.696	1.224	1.863
G101L, G375A, S410G	0.642	0.577	1.113	0.604	0.973
G101L, H122S, V187R, G375S, A447I	0.280	0.237	1.187	0.520	1.544
G101L, Q201V, I315F, G375V, A447L	0.481	0.400	1.199	0.730	1.359
G126D, A202L, A306G, S316G, M360S	0.528	0.410	1.286	0.248	0.549
G126D, A306G, M360W	0.443	0.353	1.258	0.000	0.972
G126D, M378A	0.489	0.456	1.076	0.364	1.146
G126D, R188P, A306G, M360W	0.526	0.439	1.207	0.000	1.109
G15S, D113A, E184G, T395E, N430G, I461G	0.921	0.865	1.065	0.867	1.348
G15S, E184G, G212A, K352N, I461G	1.006	1.045	0.962	0.856	1.465
G15S, E184L, H239G, V272A, I461G	0.788	0.701	1.124	1.109	1.713
G15S, E184L, T311S, K352N, I461G	0.769	0.646	1.193	1.202	1.150
G15S, H239G, I461G	0.465	0.468	0.995	1.504	1.768
G15S, I461G	1.159	1.314	0.883	0.819	1.431
G15S, M165K, E184F, G212A	0.649	0.767	0.846	1.151	1.541
G15S, M165K, E184S, A323G, I367M, N430G, I461G	0.554	0.484	1.145	1.915	1.062
G15S, M195G, H239G, I461G	0.783	0.698	1.120	1.166	1.523
G15S, R93T, T136G, G212L, H239G, I461G	0.594	0.528	1.125	0.788	1.466
G15S, V272A, I461G	0.536	0.468	1.144	1.304	1.875
G212A, D307F, K352N, I461G	0.794	0.693	1.146	1.172	1.432
G33S, I102R	0.777	1.150	0.675	0.742	1.702
G33S, I102R, M360W, M378A	0.577	0.528	1.101	0.298	1.542
G33S, L454V	0.459	0.577	0.797	0.844	2.063
G33S, P173R, R188P, R246P, A306G	0.414	0.354	1.168	0.449	2.310
G33S, R188P, M360S, L454V	0.466	0.378	1.232	0.139	1.564
G33S, R246V, A306G, M360R	0.597	0.457	1.306	0.000	0.599
G33S, T470S	0.680	1.044	0.654	0.792	1.749
G373R	1.159	1.332	0.943	2.117	1.109
G375S, A447I	0.765	0.863	0.886	0.753	1.631
G442E	0.566	0.688	0.824	0.944	1.738
G4R, G33S, G126D, P173S, R188P, R246V, Q334S, M360W, L411P	0.524	0.400	1.311	0.000	0.382
G4R, P173W, Q175E, R246Q, K258R, M360S	0.171	0.133	1.286	0.000	0.434
G4R, R188P, R246Q	0.603	0.500	1.209	0.000	1.093
H122S, G375S	0.482	0.405	1.193	0.717	1.483
H122S, I315F	0.461	0.423	1.091	0.724	1.575
H122S, L254T, N331G	0.436	0.371	1.173	0.441	1.113
H122S, Q201V	0.522	0.898	0.590	0.741	4.476
H122S, R172S, Q201W, A447I	0.567	0.496	1.140	1.294	2.362
H122S, V187R, L254T, N331G	0.389	0.329	1.176	0.357	1.554
H122S, V234C, L254R, I315F	0.456	0.365	1.242	0.101	1.069
H239G, K352I, T395E, I461G	0.795	0.703	1.131	1.024	1.636
H239G, T311S, K352N, I461G	0.913	0.782	1.168	1.066	1.125

TABLE 5-continued

Mutation	Total FAME	Total FAS	% FAME	C12/C14	C14/C16
I102L, R246W, K258R, A306G, M360R, L411A, L454V	0.559	0.464	1.205	0.000	0.493
I102R, P173W, A202L, Q334S, M360R	0.495	0.405	1.225	0.175	0.795
I102R, R246A, Q334S, M360R	0.580	0.451	1.286	0.000	0.519
I102R, R246L, K258R, A306G, M360R, L411A, L454V	0.664	0.517	1.286	0.000	0.360
I102R, R246Q, K258R, M360S	0.709	0.448	1.502	0.052	0.194
I102R, R246W, L454V	0.772	0.668	1.157	0.000	0.940
I146K, V219L, D307N	0.357	0.305	1.170	2.470	1.551
I146K, V219L, D307N, G442S	0.356	0.286	1.248	3.212	0.932
I146L, A185L, D307R, G442S	0.358	0.276	1.294	2.393	0.529
I146L, V219L, D307R, G442S	0.361	0.284	1.272	4.822	0.565
I146R, A197T, D307G	0.281	0.221	1.274	4.844	0.902
I146R, A197T, D307N, G442E	0.315	0.255	1.234	3.768	1.213
I303R, D328F	0.619	0.531	1.212	1.228	1.882
I303W	1.192	1.188	1.088	1.898	1.235
I315F	0.529	0.483	1.099	0.715	1.406
I315F, N331G, G375S, A447C	0.606	0.532	1.135	0.389	1.283
I420V	0.567	0.864	0.656	1.033	2.097
I461G	0.914	1.151	0.794	0.770	1.821
K472T	0.974	1.088	0.894	1.215	1.442
K78F, R131M, S192V, G266S, K349H	0.754	0.540	1.327	0.660	0.537
K78F, S192V, A243R, K349H	1.501	1.055	1.417	0.695	0.629
K78W, S111G, C422G	0.788	0.538	1.387	0.839	0.495
L127G, A190W, D203R, A327I, I420V, D455E	0.547	0.565	0.968	1.198	1.590
L127G, A190W, S264D, V381F	0.292	0.260	1.124	0.994	1.950
L134T	0.745	1.165	0.639	1.364	2.156
L200R, A228G, M443G	0.537	0.569	0.980	0.610	1.696
L200R, A228G, T356W, G373R	1.224	1.167	1.126	1.326	0.904
L200R, I303R, Q314G, T356G	1.280	1.092	1.163	0.651	0.703
L30H, Q201V, V234C, G265D, R304W, G375V	0.271	0.217	1.242	0.784	0.696
L30H, R304W, I315F	0.507	0.397	1.282	0.626	0.650
L30H, V187G, Q201V, L254T, I315F, Q357V, G375V	0.456	0.353	1.294	0.746	0.710
L30H, V234C, S410G, A447L	0.862	1.035	0.833	1.022	1.401
L39A, D77A, S105G, V155G, T242K, V381F, T470S	0.595	0.557	1.071	4.219	1.897
L39A, D77A, T242R, S264D, A327I, M363W, D455E	0.392	0.307	1.276	0.332	0.710
L39A, L127G, A190W, T242R, S264V, V381F, D455E	0.382	0.313	1.221	0.000	1.181
L39A, T313S, Q348R, V381F	0.434	0.547	0.794	2.921	1.846
L39M, D77A, L127G, A190W, T313S	0.437	0.366	1.200	0.158	1.553
L39M, D77A, S105G, V155G, S264V, A327I, Q348R, D455E	0.542	0.628	0.862	2.186	2.129
L39M, L127G, A190R, D203R, Q348R, M363W, I420V	0.462	0.469	0.987	1.428	1.732
L39M, L127G, A190W, V381F, D455E	0.490	0.591	0.836	1.222	2.070
L39M, R177V, A190R, I420V	0.428	0.565	0.768	1.079	1.827
L39M, R177V, A190W, S264D, N302G, A327I	0.544	0.424	1.285	0.000	0.902
L39M, S105G, A190P, T242R, T313S, M363W	0.954	0.819	1.163	0.549	0.790
L39M, S105G, L127G, A190W, D203R, Q348R	0.480	0.525	0.917	8.297	2.700
L39S, A190P, T242R, N302G, T313S, A327I, Q348R	0.500	0.389	1.276	0.000	0.787
L39S, R177V, D203R, T242R, S264D, A327I, M363W	0.485	0.411	1.181	0.000	1.013
L39S, S105G, R177V, N302G, A327I, V381F	0.324	0.255	1.269	0.000	0.980
L39S, V155G, A190P, A327I, Q348R, V381F	0.239	0.287	0.837	1.552	2.340
L39S, V155G, A190R, T242R, S264D, A327I, Q348R	0.419	0.431	0.976	1.159	1.695
L411A	0.828	1.016	0.764	1.327	0.522
L454V	0.799	1.000	0.798	0.572	1.528
L6Q, K78W, S111G, T158R, S192L, E309S, L457Y	0.167	0.104	1.526	0.000	0.282
L6Q, K78W, S111P, R131M, A243R, G266S, C422G	1.111	0.728	1.445	0.458	0.253
L6Q, R40S, S192A, K349Q, T385G, C422G	0.786	0.496	1.500	0.027	0.171
M165K, E184L, I461G	0.672	0.594	1.133	1.490	1.269
M165K, I367M, I461G	0.544	0.471	1.155	1.572	1.386
M360Q	0.844	0.961	0.880	0.227	1.282
M360Q, L454V	0.259	0.219	1.236	0.000	11.652
M360S	0.596	0.548	1.090	0.159	0.956
M360S, M378A, L411A	0.596	0.514	1.160	0.000	1.039
M363W, V381F, D455E	0.711	1.021	0.697	14.861	2.109
M378A	0.525	0.554	0.947	0.553	1.676
M378A, L411A	0.626	0.727	0.860	0.482	1.566
N110R, L134T, N164R, Q193R	0.526	0.805	0.655	1.535	1.835

TABLE 5-continued

Mutation	Total FAME	Total FAS	% FAME	C12/C14	C14/C16
N164R, G310V, P351G, S424Q, A468T	0.615	0.519	1.186	1.215	0.870
N331G	0.630	0.667	0.942	0.811	1.503
P173W, R246V	0.515	0.393	1.311	0.000	0.880
P294G, G310H, A320C, P351G, E393G, S424G	0.585	0.454	1.289	1.376	0.633
Q193R, G310H, E458W	0.778	0.909	0.864	1.161	1.435
Q193R, P351G, E393W, S424Q, E458W	0.552	0.449	1.228	0.000	0.873
Q193S, Q244G, G310H, P351G, E458W	0.789	0.727	1.085	1.045	1.192
Q193S, Q244G, P294G, G310R, S424G	0.764	0.637	1.199	1.015	0.859
Q201A, G375S, S410G, A447L	0.930	0.964	0.959	0.792	1.251
Q201A, I315F, G375A, A447L	0.527	0.464	1.139	1.059	1.250
Q201A, R304W, S410G	0.508	0.402	1.265	0.708	0.831
Q201V, I315F, A447I	0.622	0.591	1.047	1.053	1.398
Q201V, I315F, G375S, A447C	0.652	0.568	1.149	0.739	0.981
Q201W, R304W, S410G	0.643	0.524	1.227	0.669	0.837
Q244G, G310R, A320C, P351G, S424Q	0.664	0.534	1.245	1.226	0.610
Q244G, G310V, P351G, Y366W, E458W	0.849	0.892	0.951	1.220	1.673
Q244G, P294G, G310H, P351G, Y366W, E393W, S424G	0.507	0.457	1.108	1.205	1.023
Q244G, S267G, G310V, A320C, Y366W	1.052	0.905	1.161	0.820	0.649
Q244G, Y366G, S424Q	0.601	0.491	1.224	0.895	0.724
Q26P, R171E, A228G, L268F	0.540	0.964	0.635	4.918	3.527
Q26T, C121S	0.740	0.886	0.837	0.599	1.366
Q26T, C121S, R171F, A228G, K472*	0.892	0.973	0.996	1.552	1.839
Q26T, C121S, R171W, A228G, K472*	0.542	0.553	0.983	1.047	2.189
Q26T, E99Q, R171W, I303W	0.618	0.563	1.184	0.529	1.472
Q26T, E99Q, S147T, R171V, S186T, I303G, K472T	0.285	0.316	0.958	2.050	2.607
Q26T, K472*	0.826	1.343	0.618	0.880	2.071
Q26T, L200R, I303R, K472*	0.909	1.148	0.825	0.997	1.672
Q26T, Q314G	1.151	1.386	0.830	0.921	1.448
Q26T, R171E, I303W, K472T	0.618	0.602	1.054	0.527	1.840
Q26T, R171F, Q253G, T356G	0.890	1.075	0.831	1.079	1.930
Q26T, R171H, S186T, I303G, K472T	0.709	0.697	1.064	0.780	1.558
Q26T, R171V, A228G, I303R, K472T	0.792	0.995	0.807	1.668	2.166
Q26T, S147A, R171E, I303R	0.632	0.614	1.027	0.844	1.988
Q26T, V69L, E99Q, C121S, R171L, A228G, I303W, Q314R, D328F	0.479	0.460	1.130	0.000	1.484
Q26T, V69W, C121S	0.671	0.608	1.097	0.383	1.046
Q26T, V69W, E99Q, A279G, T356G, K472*	0.771	0.712	1.077	0.454	0.984
Q26T, V69W, E99Q, K472*	0.685	0.694	1.019	0.565	1.141
Q26T, V69W, E99Q, R171F, D328F	0.650	0.549	1.174	0.071	1.241
Q26T, V69W, S186T, L200R, I303W	1.307	1.374	1.030	1.840	0.943
Q348R	0.447	0.961	0.467	2.033	3.548
Q348R, M363W	0.647	1.004	0.646	1.613	2.385
R171F, I303W	0.883	0.741	1.236	0.407	1.274
R171F, M443G	0.557	0.450	1.228	0.494	1.719
R171F, Q314G, G373R	0.710	0.573	1.251	0.424	1.208
R171H, S186T, A228G, I303G, D328F	0.594	0.479	1.230	0.432	1.791
R171V, L200R, Q253G, I303W, M443G, K472T	0.961	0.804	1.186	0.339	1.358
R171V, L200R, T356W, G373R, K472T	0.800	0.930	0.957	2.087	1.442
R171W, I303W, K472*	0.864	0.765	1.175	1.049	1.629
R171W, K472T	0.897	1.313	0.712	0.975	2.490
R172S, V187G	0.557	0.609	0.945	1.058	1.874
R172S, V187G, Q201W, R304W	0.639	0.529	1.211	0.691	1.149
R172W, V234C, A447L	0.544	0.848	0.646	1.245	2.495
R188P	0.583	0.648	0.897	0.491	1.794
R188P, A202L, R246G, K258R, Q334S, M360R	0.416	0.317	1.311	0.000	0.512
R188P, A202L, S316G, M360Q	0.513	0.401	1.286	0.000	1.088
R188P, L411A	0.526	0.539	0.979	0.672	1.468
R207A, Q244G, P294G, Y366W, E393W, S424G	0.873	0.751	1.161	0.932	0.952
R246L, K258R, M360R, L454V, A468S	0.521	0.397	1.311	0.000	0.390
R246V, A306G, M360R, L411V, L454V	0.743	0.566	1.311	0.000	0.419
R246V, K258R, M360R	0.590	0.520	1.143	0.000	0.778
R304W, I315F	0.422	0.330	1.280	0.977	0.648
R467T	0.440	0.526	0.848	1.399	1.791
R87W, K349Q, C422G	0.349	0.217	1.522	0.333	0.281
R93T, E184F, I461G	0.474	0.491	0.974	1.611	1.770
R93T, G212A, K352I, I461G	0.869	0.917	0.955	0.999	1.782
R93T, I461G	1.124	1.474	0.762	0.791	1.740
R98D, I146K, T170R, A185L, D307L, R467T	0.354	0.292	1.208	3.175	1.166
R98D, I146L, A185M, A197V, Q250W, D307G	0.312	0.253	1.236	3.559	1.030
R98D, I146R, T170R, D307F, T353S	0.331	0.271	1.224	3.074	1.073
S105G, A190P, Q348R	0.415	0.616	0.831	1.917	2.888
S105G, A190P, S264D, M363W	0.403	0.311	1.295	0.000	1.122
S105G, L127G	0.413	0.358	1.155	0.438	1.411

TABLE 5-continued

Mutation	Total FAME	Total FAS	% FAME	C12/C14	C14/C16
S105G, L127G, D203R, T242R, A327I, Q348A, V381F	0.364	0.446	0.818	2.635	2.283
S105G, L127G, T242K, Q348A, I420V	0.320	0.313	1.024	1.097	1.840
S105G, V155G, A190P, T242K	0.555	0.446	1.244	1.520	1.507
S105G, V155G, R177V, A190W, T242R, Q348A	0.464	0.488	0.950	3.447	1.568
S111G, E309A, C422G	1.407	0.951	1.402	0.605	0.299
S111P, S192L, E309S, K349Q	1.111	0.858	1.230	0.681	0.399
S115R, I146R, D307F, R467T	0.325	0.255	1.272	5.998	0.542
S147A, A228G, A279V, K472*	0.920	1.143	0.806	0.887	1.905
S147A, D328F, G373R, K472T	0.883	0.822	1.120	0.245	1.198
S147A, I303R, D328F, K472*	0.400	0.349	1.232	2.937	2.254
S147A, R171E, A279G, I303W	0.624	0.546	1.187	1.400	1.972
S147A, R171E, I303R, T356G	0.980	1.052	1.008	4.987	2.253
S147A, S186T, I303R, D328F	0.504	0.428	1.226	0.819	1.923
S147A, T356G	0.669	0.751	0.901	0.660	1.404
S186T, A228G, T231S	0.662	0.914	0.760	0.610	2.169
S192A, L457Y	0.612	0.654	0.916	0.962	1.490
S264D, A327I	0.612	0.729	0.840	1.113	1.916
S264D, Q348R, M363W, V381F, D455E	0.483	0.905	0.536	2.198	3.734
S264V, A327I, D455E	0.768	0.960	0.800	1.008	1.472
S264W	0.690	0.804	0.855	0.584	1.686
S316G, L411A, L454V	0.606	0.525	1.155	0.000	0.842
S5P, D77A, S105G, A190P, S264W, N302G, Q348A, I420V	0.329	0.259	1.267	0.669	1.892
S5P, D77A, V155G, A190R, T242K, N302G, A327I, A426V	0.435	0.328	1.326	0.000	0.976
S5P, D77A, V155G, S264D, A327I, I420V	0.501	0.474	1.078	0.695	1.865
S5P, D77A, V155G, S264V, A327I	0.885	1.117	0.788	1.552	1.805
S5P, G15D, L39S, A190R, S264D, Q348R, V381F	0.482	1.085	0.444	2.533	3.606
S5P, L39A, D77A, A190W, D203R, S264W, N302G, A327I, Q348A, I420V	0.752	0.721	1.039	1.437	1.514
S5P, L39A, D77A, V155G, A190P, T242K, L268M	0.652	0.770	0.849	1.549	1.840
S5P, L39A, V155G, A190W, T242R, N302G, T313S, A320C, Q348A	0.591	0.580	1.019	2.749	1.365
S5P, L39A, V155G, Q348R, I420V	0.496	1.177	0.421	2.953	3.910
S5P, L39M, D77A, A190P, T242K, S264D, Q348R, M363W, I420V	0.610	0.602	1.013	0.874	1.485
S5P, L39M, D77A, S105G, A190P, S264W, Q348A	0.359	0.568	0.799	2.419	2.998
S5P, L39M, G50S, D77A, V155G, A190P, T242R, S264D, A327I, V381F, D455E	0.600	0.599	0.996	1.610	1.898
S5P, L39M, S264D, I420V, D455E	0.604	0.643	0.939	1.494	3.014
S5P, L39M, V155G, A190W, D203R, T242R, S264W, N302G, T313S, Q348A, R469L	0.549	0.490	1.172	3.399	1.640
S5P, L39S, D77A, V155G, A190P, T242K, M363W, V381F	0.486	0.393	1.238	0.000	1.277
S5P, L39S, S105G, A190R, N302G, M363W	0.441	0.339	1.295	0.000	0.472
S5P, R177V, D203R, T242R, S264D, N302G, A327I, Q348A, M363W, I420V	0.592	0.498	1.182	1.497	1.203
S5P, S264V, Q348A, I420V	0.261	0.432	0.605	2.409	2.984
S5P, S264W, N302G, T313S	0.469	0.365	1.285	0.000	0.524
S5P, V155G, S264V, G265D	0.622	0.525	1.183	0.827	1.540
S5P, V381F	0.503	1.148	0.437	15.293	3.054
S5T	0.876	1.151	0.761	2.978	1.548
S5T, A190W, T242K, V381F	0.476	0.555	0.859	2.342	1.950
S5T, D77A	0.722	0.804	0.899	0.508	1.186
S5T, D77A, S105G, V155G, A190R, S264D	0.640	0.508	1.261	9.479	1.589
S5T, D77A, V155G, N302G, A327I	0.399	0.306	1.303	1.174	0.961
S5T, L127G, A190R, V381F	0.405	0.382	1.080	3.787	2.108
S5T, L127G, A327I, Q348A, D455E	0.485	0.736	0.656	2.040	2.847
S5T, L39A, A190P, S264D, A327I, Q348A, V381F, D455E	0.486	0.750	0.647	2.195	2.743
S5T, L39A, L127G, A190P, D203R, Q348R	0.347	0.352	0.986	0.000	1.415
S5T, L39A, L127G, D203R	0.704	0.665	1.055	0.283	1.068
S5T, L39M, A190P, D203R, S264W, N302G, I420V	0.463	0.356	1.295	0.000	0.425
S5T, L39M, D77A, S105G, S264W, N302G, T313S, Q348R, V381F	0.614	0.508	1.211	0.000	0.794
S5T, L39M, L127G, A190R, T242K, A327I, Q348R, V381F	0.442	0.477	0.922	2.084	1.958
S5T, L39M, L127G, D203R, T242R, Q348R, V381F	0.592	0.832	0.708	1.814	2.556
S5T, L39M, R177V, A190P, T242R	0.511	0.522	0.978	2.163	3.546
S5T, L39S, D77A, A190P	0.620	0.634	0.976	0.000	1.243

TABLE 5-continued

Mutation	Total FAME	Total FAS	% FAME	C12/C14	C14/C16
S5T, L39S, D77A, S264V, Q348R, V381F, I420V	0.476	0.483	0.978	0.000	1.463
S5T, L39S, D77A, T313S, Q348A, V381F, I420V	0.377	0.733	0.518	3.052	3.172
S5T, L39S, D77A, V155G, A190P, N302G, Q348R, V381F	0.394	0.306	1.284	0.886	1.282
S5T, N302G	0.621	0.490	1.269	0.166	0.692
S5T, S105G, A190P, S264D, Q348A	0.445	0.603	0.741	2.077	2.668
S5T, T242K, N302G, V381F	0.486	0.383	1.269	0.000	0.836
S5T, V155G, N302G, A327I, Q348A, D455E	0.470	0.372	1.255	3.417	1.789
S5T, V155G, N302G, Q348R	0.497	0.391	1.269	2.115	1.577
S5T, V155G, P166S, Q348A, V381F	0.314	0.651	0.479	6.776	3.952
T170R, T353S, E404R	0.733	0.992	0.744	0.913	2.173
T24W, D307N, G442E	0.353	0.331	1.067	2.003	1.690
T24W, D307N, T353S, E404K, G442S	0.746	0.659	1.132	1.526	1.666
T24W, R98D, A185L, A197V, D307V	0.404	0.336	1.201	2.744	1.495
T24W, R98D, A185M, A197T, D307L	0.363	0.329	1.102	2.617	1.579
T24W, R98D, A197T, D307G	0.426	0.363	1.172	2.189	1.579
T24W, R98D, S115R, I146L, A197V, V219L, T353S, G442E, R467T	0.904	0.732	1.235	2.231	1.040
T24W, T170R, A197T, D307L, T311S, G442S	0.913	0.745	1.227	1.561	1.357
T24W, T44F, I146L, D307N	0.947	0.755	1.254	2.211	1.086
T24W, T44F, R98D, A185L, R467T	0.376	0.335	1.128	2.119	1.709
T24W, T44F, R98D, T170R, A185L, V219L, R467T	0.400	0.392	1.023	2.218	1.768
T311S, A323G, I367M	0.505	0.446	1.133	1.633	1.220
T311S, K352N, T395E, N430G, I461G	0.807	0.736	1.096	1.059	1.506
T313S, I420V	0.800	1.036	0.772	0.877	1.459
T44F, D307F	0.496	0.529	0.948	1.440	1.705
T44F, I146R, A197T, G442E, R467T	0.356	0.316	1.127	2.923	1.128
T44F, I146R, Q287S, D307R, G442S	0.826	0.649	1.274	1.907	0.450
T44F, R98D, A185M, G442E	0.604	0.717	0.844	1.168	1.904
T44F, R98D, I146R, A197T, D307G, T311S, T353S, G442E	0.350	0.274	1.272	4.569	0.678
T44F, R98D, S115R, A197T	0.717	0.715	1.003	1.045	1.417
T44F, S115R, A185M, A197V, D307N, E404R	0.609	0.593	1.040	1.139	1.301
T44F, S115R, I146R, A197T, V219L, D307F	0.266	0.205	1.294	3.144	0.518
V149L, I315F, A447C	0.533	0.422	1.263	0.929	1.104
V149L, Q201W, A285R, Q357V, A447C	0.505	0.406	1.245	0.962	0.731
V155G, A190P	0.517	0.471	1.097	1.254	1.680
V155G, A190P, S264V, Q348R, I420V	0.371	0.376	0.989	17.841	2.093
V155G, A190W, S264D, I420V, D455E	0.802	0.851	0.943	3.273	1.703
V155G, D203R, N302G, I420V	0.792	0.608	1.295	1.038	0.969
V155G, M363W	0.628	0.517	1.216	10.265	1.556
V155G, V381F, I420V, D455E	0.408	0.554	0.880	1.907	2.389
V187G, I315F, G375A, A447L	0.606	0.539	1.126	0.960	1.331
V187G, Q201R, V234C, I315F, Q357V, A447I	0.556	0.465	1.190	0.270	1.186
V187G, Q201V, L254R, A285R, S410G, A447L	0.535	0.538	1.006	0.510	1.467
V187G, Q201W, I315F, Q357V, A447I	0.545	0.492	1.114	0.484	1.801
V187R, Q201V, I315F, G375A	0.377	0.305	1.229	0.161	0.809
V187R, Q201V, L254R, R304W, N331G, G375A	0.516	0.413	1.242	0.376	0.768
V219L	0.327	0.347	0.952	1.786	1.657
V234C, I315F, Q357V, G375V	0.542	0.441	1.225	0.090	1.181
V381F, I420V	0.617	1.147	0.538	1.503	2.596
V43S, M165K, E184S, G212A	0.957	1.228	0.779	0.809	1.414
V43S, N430G	0.888	1.429	0.623	0.800	2.242
V43S, R93T, E184S, T395V, I461G	0.809	0.785	1.029	1.010	1.450
V69W	0.702	0.709	0.990	0.304	1.214
V69W, C121S, R171E, K472*	0.719	0.742	1.015	1.734	2.123
V69W, C121S, S147A	1.407	1.333	1.131	0.448	0.887
V69W, C121S, S186T, A279G, K472*	1.286	1.225	1.126	0.959	1.033
V69W, E99Q, Q253G, A279G, K472T	1.161	1.142	1.034	0.453	1.051
V69W, Q314G	0.684	0.593	1.200	0.298	0.847
V69W, R171E, S186T, I303W	0.915	0.763	1.190	0.691	1.652
V69W, R171E, S186T, I303W, K472*	0.933	0.838	1.193	1.431	1.585
V69W, R171W	1.147	1.263	0.983	3.002	1.738
V76L, K258R, S316G, M360R	0.713	0.555	1.286	0.000	0.657
V76L, L454V	0.736	1.056	0.698	0.814	1.770
V76L, R246L, Q334S, M360R	0.511	0.407	1.258	0.393	0.485
Y366G	0.720	0.907	0.793	1.015	1.657

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Example 3

Ester Synthase Mutant Strains with Beneficial Properties

As described in Examples 1 and 2, a number of variant or mutant forms of '377 were identified with beneficial properties. In this experiment, a particular '377 mutant designated herein as 9B12 (SEQ ID NO: 4), which contains three different mutations over wild-type 377 (D7N, A179V, V381F), was cloned into pKEV027, replacing the wild-type '377 gene (SEQ ID NO: 1) coding for the wild type '377 polypeptide (SEQ ID NO: 2) and creating pKEV018. When transformed into strain BD64 (supra), strain KEV040 was created. The plasmid pKEV018 was further mutated using saturation mutagenesis at codon 348 of the gene (start codon=0) to produce the plasmid pKEV022, which contains four different mutations over wild-type '377 (D7N, A179V, Q348R, V381F). Separately, several previously identified mutations were combined with the pKEV018 plasmid to produce plasmid pKEV028 (which contains eight mutations over wild-type '377 (D7N, R87R "silent mutation", A179V, V187R, G212A, Q357V, V381F, M443G). pKEV022 and pKEV028 were transformed into strain BD64 which resulted in the strains designated KEV075 (SEQ ID NO: 10), and KEV085 (SEQ ID NO: 12), respectively. These mutant strains were observed to produce either a significant increase in FAME titer or a reduction in the percentage of "β-OH" FAME or both (e.g., mutant strains 9B12 and KEV085 achieved both).

Strains KEV080 (BD64/pKEV027); KEV040; KEV075 and KEV085 were screened using one of the standard protocols (Protocols 1-2) described above. FIG. 5 shows the relative FAME titers and % β-OH FAME of mutant strains KEV040, KEV075 and KEV085 compared to a strain which contains wild-type '377 ester synthase. Herein, FIG. 5 illustrates that those three variants were observed to have significantly higher titer FAME than the control (wild type); that the % β-OH FAME varied from one variant to another; and that strains 9B12 (KEV040) and KEV085 produced lower % β-OH FAME than the control (wild type).

Example 4

Ester Production by Recombinant Host Cells Engineered to Express a Variant '377 Ester Synthase and a Variant Ppro

As demonstrated in Examples 1, 2 and 3, an ester synthase from *Marinobacter hydrocarbonoclasticus* (WS377) was engineered to produce higher levels of fatty acid esters and a lower percentage of β-OH esters. In carrying out the present disclosure, wild type WS377 (SEQ ID NO: 1) was engineered resulting in new strains, such as 9B12, that produce higher levels of FAME directly from acyl-ACP.

The 'tesA of *Photobacterium Profundum* (Ppro) was engineered to have ester synthase activity. One variant of Ppro Vc7P6F9 (SEQ ID NO: 31) in plasmid pBD207 was tested in BD64 (designated herein as becos. 128). Similarly, the 9B12 variant of '377 (SEQ ID NO: 4) in pKEV18 was tested in BD64 (designated herein as KEV040). Cultures of becos.128 (BD64/pBD207) and KEV040 (BD64/pKEV018) were tested in minimal media with 2% alcohol. After production for about 24 hours, they were extracted and analyzed by GC/FID or GC/MS for identification of the fatty acyl ester (following the protocols listed above).

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FIG. 6 shows the production of fatty acyl esters from KEV040 (ester synthase from *M. hydrocarbonoclasticus*) (shown as Wax in FIG. 6) and fatty acid species from becos.128 ('tesA from *P. profundum*) (shown as Ppro in FIG. 6) when fed methanol, ethanol, 1-propanol, isopropanol, 1-butanol or glycerol. As negative control, water was added (lack of exogenous alcohol means that any ester production would rely on endogenous production of alcohol by *E. coli*). While no free fatty acid was produced by KEV040, becos.128 produced a mixture of fatty acid esters and fatty acids. FIG. 8 shows the amounts of fatty acid esters and fatty acid produced by becos.128. It should be noted that the in vitro results presented in Holtzapple and Schmidt-Dannert (see Journal of Bacteriology (2007) 189 (10): 3804-3812) showed that ester synthase from *M. hydrocarbonoclasticus* ('377) has a preference for longer chain alcohols producing primarily waxes. Holtzapple's data suggests that '377 would not be capable of using shorter chain alcohols. In contrast, the applicants work was conducted in vivo and established that the variant '377 enzyme can utilize shorter chain alcohols (C1, C2, C3 and C4), leading to shorter chain esters that can be used for a variety of valuable products (e.g., biodiesel, surfactants, etc.). The applicants used acyl-ACP as a substrate, which is believed to bring about a reduction of the futile cycle involvement (i.e., making the cell more energy efficient). In addition, the applicants showed that not all alcohols needed to have the OH group at position 1, for example, the OH group for propanol could be at position 1 or 2. FIG. 6 shows the production of fatty acyl esters with various chain lengths using a recombinant host cell genetically modified to express a single variant '377 ester synthase. This result was observed for engineered forms of both, a variant of tesA of *P. profundum* (Ppro) and a variant of the '377 ester synthase. It is noteworthy that the engineered mutants can convert not only primary alcohols but also secondary alcohols into a variety of different products, including FAME, FAEE, fatty acyl monoglycerides, and others. Recombinant bacteria, yeast, algae or other host cells can be used for the production of fatty esters.

Example 5

Ester Synthase Mutant Strains Expressing Combination Mutants

Standard techniques known to those of skill in the art were used to prepare combination libraries (Protocol 8). The mutations tested in combination libraries (Tables 3A and 3B) were originally identified in the full saturation library of '377 (as described in Example 1).

The combination libraries were screened in strain GLPH077 which is similar to BD64 but expressed slightly different levels of biosynthetic pathway enzymes (supra) and using one of the standard protocols (Protocols 1-2) (supra). The results from screening pKASH008 (SEQ ID NO: 16) combination libraries are shown in Table 6. As can be seen in Table 6, the resultant '377 ester synthase variants had 24 mutations compared to wild type 377. Two strains, KASH280 (SEQ ID NO: 33) and KASH281 (SEQ ID NO: 35) were used to test the variant '377. However, KASH280 expressed a '377 variant that differed from the one expressed in KASH280 by one substitution mutation. KASH280 expressed a '377 variant where a glutamine at position 348 was exchanged with an arginine (Q348R), allowing the strain to produce a higher titer FAME and a greater number of short chain esters. Conversely, KASH281 expressed a '377 variant where a lysine at position 349 was exchanged

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with a histidine (K349H), leading the strain to produce a lower titer FAME compared to KASH008. In addition KASH281 produced a lower titer of FAME and lesser number of short chain esters compared to KASH280. Overall, the comparison is to KASH008, which is better than pSHU10, which is better than the wild type (see examples above).

TABLE 6

Strain	Mutations over wild type	# Mutations over wild type	Total FAME	Total FAS	% FAME	C12/C14	C14/C16
KASH280	T5S, S15G, K78F, P111S, V171R, P188R, S192V, R207A, A243R, D255E, L257I, N259Q, L260V, H262Q, G265N, A285V, N288D, N289G, F317W, Q348R, S353T, V381F, V409L, S442G	24	1.59	0.66	0.42	1.65	2.36
KASH281	T5S, S15G, K78F, P111S, V171R, P188R, S192V, R207A, A243R, D255E, L257I, N259Q, L260V, H262Q, G265N, A285V, N288D, N289G, F317W, K349H, S353T, V381F, V409L, S442G	24	0.80	0.87	0.92	1.38	1.22

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TABLE 7

AA Position	Mutation	<i>M. adhaerens</i>	<i>M. algicola</i>	<i>M. sp. ELB17</i>
24	T24N	X	X	X
33	G33D			X
33	G33N		X	

Example 6

Ester Synthase Mutant Strains with Beneficial Properties from Natural Diversity

As described in Examples 1 and 2, a number of variant or mutant forms of '377 were identified with beneficial properties. An alignment of the ester synthase polypeptide from different *Marinobacter* organisms, such as *Marinobacter adhaerens* HP15 (YP_005886638.1, SEQ ID NO: 45), *Marinobacter algicola* DG893 (ZP_01893763.1, SEQ ID NO: 47), *Marinobacter sp. ELB17* (ZP_01736818.1, SEQ ID NO: 49) with the one from *Marinobacter hydrocarbonoclasticus* DSM 8798 (SEQ ID NO: 2) using standard alignment procedures was used to identify mutations that already exist in nature, as this would reveal specific mutations as well as amino acid positions that naturally have diversity and thus could be used to stabilize the variant protein. The alignment was carried out and the analysis identified natural diversity within the ester synthase homologues polypeptides. Table 7 below illustrates a summary of such a group of mutations. When those mutations are combined with an engineered ester synthase from *Marinobacter hydrocarbonoclasticus*, beneficial variants can be observed if they are screened using one of the standard protocols (Protocols 1-2) described above. Examples of such variants are described as SEQ ID NOS: 37, 39, 41 and 43.

Table 7 below shows a list of natural diversity observed by aligning *Marinobacter hydrocarbonoclasticus*, *M. adhaerens*, *M. algicola* and *M. sp. ELB17*.

TABLE 7-continued

AA Position	Mutation	<i>M. adhaerens</i>	<i>M. algicola</i>	<i>M. sp. ELB17</i>
44	T44K			X
44	T44A		X	
48	E48Q			X
48	E48A		X	
49	A49D		X	
49	A49T	X		
58	Y58C			X
58	Y58L	X		
70	I70V		X	
70	I70L	X		X
73	A73S		X	
73	A73T			X
73	A73G	X		
79	D79K	X	X	
79	D79H			X
123	V123I		X	
123	V123M			X
157	T157S		X	X
158	T158E		X	
158	T158K			X
161	E161G		X	
161	E161D	X		X
162	R162E		X	X
162	R162K	X		
163	C163I		X	
163	C163R	X		X
164	N164D	X		X
166	P166L		X	X
170	T170S	X	X	X
174	H174E	X	X	X
175	Q175R	X	X	
175	Q175S			X
176	R176T		X	X
179	A179S	X	X	
179	A179K			X
183	K183S	X	X	X
189	A189G	X		X

TABLE 7-continued

AA Position	Mutation	<i>M. adhaerens</i>	<i>M. algicola</i>	<i>M. sp. ELB17</i>
191	V191L		X	
191	V191I			X
196	D196E	X	X	X
212	G212M		X	
212	G212S			X
216	V216I		X	X
236	V236A		X	
236	V236K	X		X
237	L237I	X		X
243	A243G	X	X	X
255	D255E		X	X
257	L257I	X	X	
257	L257M			X
259	N259Q		X	
259	N259A			X
259	N259E	X		
260	L260V		X	
260	L260M			X
262	H262Q		X	
262	H262R			X
263	A263V	X		
265	G265N		X	
266	G266S			X
266	G266A	X		
285	A285V		X	
285	A285L	X		X
288	N288D		X	X
289	N289G		X	
289	N289E	X		
293	T293I			X
293	T293A	X		
306	A306S	X	X	
331	N331K		X	
331	N331T	X		X
334	Q334K		X	X
335	Q335S		X	
335	Q335C			X
335	Q335N	X		
338	T338H		X	
338	T338E			X
338	T338A	X		
353	S353K		X	X
393	E393G		X	
393	E393T			X
393	E393Q	X		
394	G394E		X	X
394	G394R	X		
402	R402K	X		X
413	A413T	X		X
461	I461L			X
461	I461V		X	

Example 7

Engineering of 'tesA of *P. profundum* (Ppro)

The 'tesA of *P. profundum* (Ppro) was engineered to act as an ester synthase. When screened in 96 well plates, the wild type Ppro produced about 15-18% FAME of the total acyl products (FAME+free fatty acids). Mutations were identified that increased the FAME content to 42-44% when the variants were tested on 96 well plates. When the same mutants were tested in a shake flask, they produced 62-65% FAME as compared to 30% FAME produced by the wild type enzyme. Mutations were identified based on standard techniques, such as error prone PCR or saturation libraries of Ppro (Protocol 7). The identified mutations were combined generating combination libraries, which were also tested. The libraries were then screened using one of the standard protocols (Protocols 1-4) described above.

Based on the screening of combination libraries, mutations were selected to be either (1) fixed for the next round

of combination library screening; (2) re-tested in the next round of combination library screening; or (3) not tested further at that time. Mutations that were identified as beneficial were fixed for future libraries. The process can be carried out by screening using 96 well plates, validation in 96 well plates, and follow-up validation in shake flasks.

The best performers in shake flasks can then be tested in fermenters to verify that there is improvement of the engineered Ppro that is consistent with scale-up. A few of the top hits from the combination libraries can be tested in shake flasks to confirm that the Ppro variants have improved performance at a larger scale than that of 96 well plates. A shake flask validation can be carried out as follows. Desired mutants are streaked on LB+spectinomycin agar to obtain single colonies and incubated overnight at 37° C. Three single colonies from each mutant are picked into 4 mL LB+spectinomycin and incubated at 37° C. shaker for approximately 6 hours. 1.5 mL of the LB culture is used to inoculate 15 mL FA2 (2 g/L NH₄Cl)+0.05% Triton X-100+ spectinomycin in a baffled 125 mL flask and incubated overnight at 32° C. 3 mL of the overnight culture is used to inoculate 15 mL FA2 (2 g/L NH₄Cl)+0.05% Triton X-100+1 mM IPTG, 2% Methanol+spectinomycin in a baffled 125 mL flask and incubated for approximately 24 hours at 32° C. At time of extraction 300 µL of culture is sampled from the flask into a 96 well plate and extracted using a standard extraction protocol (Protocol 5).

A standard extraction protocol was carried out as described above (Protocol 5). The results from screening saturation libraries are shown in Tables 8 through 11. Table 8 presents results for single beneficial mutations found using the wild type as template, i.e., SEQ ID NO: 51. Table 9 presents results for combination mutants found using the wild type as template. Table 10 presents results for single beneficial mutations found using SEQ ID NO: 53 (referred to as PROF 1) as template (SEQ ID NO: 53 already contained two beneficial mutations in its sequence including S46V and Y116H mutations). For PROF1, the S46V and Y116H mutations were "fixed" into the template, such that future rounds of saturation/combination would be starting from that variant rather than from the wild type sequence. Table 11 presents results for double mutations found as beneficial using PROF1 as template. Table 12 presents results for combination mutants found, using the wild type as template. Table 13 presents results for combination mutants found using P1B9v2.0 (SEQ ID NO: 58) as template in BD64.

Table 8 below shows a list of single beneficial mutations found using Ppro wild type (SEQ ID NO: 51) as template.

TABLE 8

Mutation	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
T3L	0.911	0.882	0.889	1.023
P29A	1.015	1.088	1.071	0.947
P29F	0.984	0.984	0.984	1.000
P29G	1.008	1.009	1.009	0.999
P29S	1.018	1.030	1.027	0.992
I38Y	0.765	0.592	0.632	1.211
I45A	0.420	0.304	0.320	1.317
I45G	0.311	0.109	0.137	2.242
I45K	1.205	1.145	1.154	1.046
I45R	1.313	1.257	1.265	1.038
I45S	0.398	0.242	0.264	1.509
I45V	1.162	1.096	1.105	1.050
I45W	0.281	0.202	0.213	1.322
S46K	0.387	0.224	0.247	1.556

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TABLE 8-continued

Mutation	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
S46V	1.166	0.899	0.937	1.250
S46W	0.346	0.211	0.230	1.517
D48A	1.297	1.417	1.396	0.929
D48E	1.267	1.278	1.276	0.993
D48F	0.892	0.773	0.794	1.123
D48G	0.921	0.835	0.851	1.083
D48H	1.081	0.955	0.977	1.106
D48L	1.774	1.816	1.808	0.980
D48M	1.880	2.159	2.109	0.891
D48V	0.626	0.535	0.551	1.136
D48W	1.263	1.252	1.254	1.010
T49D	0.556	0.378	0.408	1.364
T49G	0.273	0.263	0.265	1.024
T49L	0.595	0.425	0.453	1.311
T49M	0.461	0.334	0.356	1.292
T49R	0.402	0.214	0.244	1.651
T49S	0.447	0.360	0.374	1.194
T49V	0.503	0.336	0.362	1.389
T49W	0.678	0.542	0.565	1.203
G51A	1.079	1.068	1.070	1.009
G51E	1.025	0.938	0.952	1.076
G51F	0.921	0.830	0.844	1.091
G51H	1.083	1.087	1.086	0.997
G51L	0.956	0.871	0.887	1.076
G51M	1.013	0.952	0.962	1.053
G51P	0.460	0.308	0.332	1.385
G51Q	1.047	1.082	1.075	0.972
G51R	1.377	1.452	1.440	0.955
G51S	1.088	1.046	1.053	1.033
G51T	1.062	1.115	1.107	0.959
G51V	1.003	1.036	1.031	0.973
G51Y	0.852	0.833	0.837	1.016
N52A	0.970	0.856	0.883	1.099
N52C	0.825	0.676	0.710	1.161
N52F	0.797	0.668	0.697	1.141
N52G	1.026	1.093	1.078	0.951
N52H	0.959	1.035	1.017	0.943
N52L	0.974	0.848	0.877	1.110
N52M	1.040	1.045	1.044	0.996
N52R	1.405	1.641	1.587	0.885
N52S	0.903	0.810	0.832	1.084
N52T	0.874	0.785	0.806	1.084
N52V	0.812	0.699	0.725	1.118
N76I	0.518	0.285	0.326	1.584
N76L	0.366	0.160	0.200	1.834
N76V	0.642	0.336	0.395	1.624
S92A	1.001	1.054	1.042	0.960
S92H	1.026	1.106	1.087	0.943
S92S	1.002	0.972	0.979	1.022
S92Y	1.022	1.199	1.158	0.881
I95V	0.910	0.876	0.884	1.028
Y116A	0.923	0.889	0.895	1.032
Y116F	1.077	1.035	1.042	1.034
Y116I	1.392	1.303	1.318	1.055
Y116L	1.203	1.246	1.239	0.971
Y116N	1.430	1.424	1.425	1.003
Y116P	0.628	0.460	0.488	1.284
Y116R	1.442	1.701	1.657	0.872
Y116S	0.876	0.757	0.777	1.127
Y116T	1.079	0.939	0.963	1.121
Y120R	0.490	0.399	0.420	1.164
N156A	1.048	1.090	1.082	0.969
N156C	0.903	0.858	0.866	1.042
N156G	0.984	1.070	1.055	0.933
N156L	0.841	0.821	0.825	1.018
N156R	1.477	1.942	1.856	0.796
N156W	0.824	0.678	0.705	1.168
L159G	0.399	0.245	0.273	1.455
L159M	1.254	1.651	1.577	0.795
L159N	0.513	0.323	0.358	1.433
L159R	0.554	0.540	0.542	1.021
L159S	0.390	0.237	0.265	1.466
L159Y	0.476	0.318	0.347	1.368

Table 9 below shows a list of combination mutants found using Ppro wild type (SEQ ID NO: 51) as template.

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TABLE 9

Mutation	Mutation	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
E37D	I45R	1.189	1.255	1.246	0.953
P29A	Stop_182I	1.014	1.058	1.047	0.967
I38L	V41I	0.843	0.760	0.779	1.082
Y116I	Q146H	1.554	1.906	1.846	0.841
Y116P	E175D	0.341	0.231	0.252	1.350
Y116Q	G117R	0.438	0.350	0.365	1.202
G8V	T49S	0.405	0.277	0.299	1.358
A30T	G51C	0.566	0.483	0.499	1.134
T49Q	Y116H	0.383	0.247	0.269	1.417
T49R	Y116H	0.441	0.303	0.324	1.362
N156K	L159K	1.456	1.955	1.862	0.782
G158D	L159V	0.334	0.224	0.244	1.362

Table 10 below shows a list of single beneficial mutations found using PROF1 (SEQ ID NO: 53) as template.

TABLE 10

Mutation	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
V41D	0.394	0.310	0.330	1.195
V41E	0.891	0.855	0.863	1.033
V41K	0.999	1.016	1.012	0.989
V41Q	1.015	1.048	1.040	0.976
V41R	0.996	1.019	1.014	0.983
V41S	0.577	0.495	0.515	1.122
V41T	0.732	0.669	0.684	1.070
V41W	0.856	0.778	0.797	1.075
F82A	1.036	0.940	0.963	1.077
F82E	0.861	0.747	0.775	1.111
F82G	0.770	0.577	0.625	1.233
F82K	1.167	1.220	1.207	0.967
F82Q	0.996	0.927	0.944	1.054
F82R	1.426	1.552	1.521	0.938
F82V	0.975	0.883	0.906	1.076
F82W	1.089	1.141	1.128	0.966
Q84A	0.989	0.936	0.949	1.043
Q84E	0.990	1.050	1.036	0.957
Q84G	0.822	0.774	0.785	1.047

Table 11 shows a list of double mutations found as beneficial using PROF1 (SEQ ID NO: 53) as template.

TABLE 11

Mutation	Mutation	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
V41L	D48G	0.684	0.601	0.621	1.103
F82G	T86P	0.589	0.399	0.446	1.323

Discussed herein are the results of feeding ethanol to one of the Ppro libraries. One of the combination libraries was fed either methanol or ethanol, in order to test if genes that have improved ester synthase activity using methanol, also show improved ester synthase activity in the presence of ethanol. As can be seen in FIGS. 7A and 7B, the improved ester synthase activity was observed with either methanol or ethanol feeding, though higher percentage esters were obtained using methanol. Table 12 shows a list of combination mutants found using Ppro wild type (SEQ ID NO: 51) as template.

TABLE 12

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
D48E, R119C	1.119	1.116	1.117	1.001
D48E, R88H, Y116H	1.265	1.559	1.506	0.839
G51R, D102V	1.012	1.026	1.023	0.988
G51R, R88H, R119C, L159M	1.597	1.818	1.778	0.897
I45T, D48E, S92R, Y116H	1.060	0.991	1.004	1.055
I45T, D48E, Y116H	1.027	1.019	1.020	1.007
I45T, D48E, Y116H, R119C	0.812	0.766	0.774	1.046
I45T, R88H	0.643	0.490	0.518	1.237
I45T, S46V, R119C	0.635	0.565	0.578	1.099
I45T, S46V, Y116H	1.004	0.891	0.912	1.100
I45T, Y116H	1.158	1.223	1.211	0.957
K32M, L159M	1.400	1.717	1.659	0.843
K32M, N156Y, L159M	1.345	1.278	1.290	1.042
R18C, G51R, N52T, D102V, N115I, P129L, L159M	1.377	1.524	1.497	0.918
R88H, Y116H	1.131	1.258	1.235	0.915
S46V, D48E, Y116H	1.209	1.052	1.081	1.117
S46V, D48E, Y116H, R119C	1.118	0.993	1.016	1.101
S46V, R88H	1.086	0.820	0.868	1.249
S46V, T86S, Y116H	1.401	1.092	1.149	1.219
S46V, Y116H	1.321	1.203	1.224	1.079
T2_50_T4, R88H, R119C	0.833	0.806	0.811	1.026
V41I, I45T, Y116H	1.057	1.129	1.116	0.945
V41L, S46V, R119C	0.897	0.631	0.680	1.318
Y116H, R119C	1.107	1.277	1.246	0.888
D48E, G51R, R88H, R119C, L159M	1.403	1.389	1.391	1.010
D48E, Y116H, R119C	1.493	1.570	1.558	0.958
G51R, N156Y	1.097	1.133	1.127	0.972
G51R, R119C, N156Y, L159M	1.364	1.566	1.534	0.888
G51R, R88H, Y116H, S121R, L159M	1.562	1.631	1.621	0.962
G51R, V112A, N115I, N156Y	0.317	0.216	0.230	1.339
G51R, Y116H, R119C	1.609	1.856	1.818	0.884
H36Q, S46V, R88H, Y116H, R119C, L159M	0.843	0.652	0.682	1.236
I45N, S46V, D48E, G51R, Y116H, L159M	1.016	0.848	0.875	1.162
K32M, D48E, R88H, Y116H, N156Y	0.920	0.811	0.828	1.110
K32M, D48E, Y116H	1.992	1.929	1.938	1.027
K32M, G51R, Y116H	1.722	1.943	1.908	0.902
K32M, H36Q, G51R, Y116H, L159M	1.629	1.651	1.648	0.988
K32M, S46V, D48E, A55T, R119C	0.710	0.518	0.547	1.294
K32M, S46V, D48E, G51R, Y116H, L159M	2.067	1.562	1.634	1.264
K32M, S46V, D48E, Y116H, N156Y, L159M	1.211	1.007	1.039	1.165
K32M, S46V, G51R	1.305	1.144	1.169	1.115
K32M, S46V, G51R, F82I, Q84L, Y116H, L159M	2.570	2.098	2.165	1.187
K32M, S46V, G51R, L159M	1.646	1.512	1.533	1.073
K32M, S46V, G51R, R88H, L159M	1.730	1.552	1.579	1.097
K32M, S46V, G51R, R88H, Y116H, R119C	1.516	1.397	1.415	1.072
K32M, S46V, G51R, R88H, Y116H, R119C, S121R	0.601	0.532	0.543	1.116
K32M, S46V, G51R, Y116H	2.066	1.634	1.695	1.219
K32M, S46V, G51R, Y116H, N156Y, L159M	1.992	1.859	1.880	1.060
K32M, S46V, G51R, Y116H, R119C, N156Y, L159M	1.866	1.578	1.619	1.153
K32M, S46V, G51R, Y116H, S121R, N156Y, L159M	0.774	0.664	0.681	1.137
K32M, S46V, Y116H	1.408	1.193	1.223	1.152
K32M, S46V, Y116H, E152*, L159M	0.441	0.385	0.394	1.136
K32M, S46V, Y116H, L159M	1.845	1.695	1.718	1.073
K32M, S46V, Y116H, N156Y	1.261	1.061	1.093	1.156
K32M, Y116H, R119C, L159M	1.693	1.928	1.891	0.895
P24Q, K32M, S46V, G51R, R88H, L159M	0.615	0.491	0.511	1.189
R88H, L159M	1.414	1.525	1.507	0.939
R88H, N89S, Y116H	1.263	1.368	1.352	0.934
S46V, D48E	0.975	0.755	0.790	1.234
S46V, D48E, G51R, R119C	1.160	0.946	0.979	1.181
S46V, D48E, G51R, R119C, L159*	0.343	0.199	0.221	1.556
S46V, D48E, G51R, R88H, Y116H, L159M	1.527	1.329	1.360	1.123
S46V, D48E, G51R, R88H, Y116H, R119C, N156Y	0.818	0.655	0.681	1.201

TABLE 12-continued

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
S46V, D48E, G51R, Y116H	1.645	1.355	1.399	1.175
S46V, D48E, G51R, Y116H, L159M	1.669	1.453	1.487	1.122
S46V, D48E, G51R, Y116H, N156Y, L159M	1.517	1.195	1.244	1.220
S46V, D48E, G51R, Y116H, S121N	0.532	0.485	0.492	1.078
S46V, D48E, N156K	1.411	1.112	1.159	1.217
S46V, D48E, N90K, Y116H, L159M	0.366	0.248	0.266	1.373
S46V, D48E, R119C	0.946	0.725	0.759	1.245
S46V, D48E, R88H, Y116H	1.199	1.016	1.044	1.148
S46V, D48E, Y116H, L159M	1.534	1.232	1.279	1.198
S46V, D48E, Y116H, N156Y, L159M	1.265	1.007	1.048	1.206
S46V, G51R	1.315	1.219	1.234	1.065
S46V, G51R, L79I, R88H, Y116H, N156Y, L159M	1.013	0.845	0.871	1.161
S46V, G51R, N156Y	1.285	1.073	1.106	1.161
S46V, G51R, R88H, Y116H	1.561	1.389	1.416	1.103
S46V, G51R, R88H, Y116H, M154K	0.520	0.413	0.430	1.210
S46V, G51R, R88H, Y116H, N156Y, L159M	1.677	1.571	1.588	1.055
S46V, G51R, R88H, Y116H, R119C	1.454	1.277	1.305	1.114
S46V, G51R, R88H, Y116H, R119C, S163F	0.554	0.392	0.417	1.327
S46V, G51R, Y116H	1.656	1.486	1.513	1.095
S46V, G51R, Y116H, L159M	1.989	1.875	1.893	1.050
S46V, G51R, Y116H, N156Y	1.788	1.461	1.508	1.186
S46V, G51R, Y116H, R119C	1.869	1.480	1.535	1.216
S46V, G51R, Y116H, R119C, L159M	2.256	2.015	2.050	1.101
S46V, L159M	1.487	1.300	1.329	1.118
S46V, N156Y, L159M	1.360	1.113	1.148	1.184
S46V, R88H, L159M	1.684	1.434	1.469	1.146
S46V, R88H, Y116H	1.219	1.024	1.054	1.157
S46V, R88H, Y116H, L159M	1.647	1.439	1.471	1.119
S46V, R88H, Y116H, N156Y	1.213	0.951	0.988	1.228
S46V, R88H, Y116H, S125I	0.685	0.501	0.530	1.290
S46V, R88Y, Y116H, R119C, L159M	1.254	1.119	1.140	1.101
S46V, Y116H, L159M	1.671	1.510	1.534	1.089
S46V, Y116H, N156Y	1.320	1.070	1.105	1.193
S46V, Y116H, R119C	1.191	0.964	0.999	1.192
S46V, Y116H, R119C, L159M	1.588	1.397	1.427	1.113
S46V, Y116H, R119C, N156Y	1.045	0.851	0.881	1.186
S46V, Y116H, R119C, N156Y, L159M	1.295	1.086	1.118	1.158
V41E, S46V, D48E, R119C	0.570	0.396	0.423	1.347
V41L, S46V, R88H, L159M	1.138	0.858	0.898	1.268
Y116H, N156Y, L159M	1.455	1.644	1.615	0.901
S46V, D48F, G51R, Y116H, E152D, L159M	1.100	0.904	0.936	1.176
S46V, D48F, G51R, Y116H, G117D, L159M	0.695	0.439	0.482	1.444
S46V, D48F, G51R, Y116H, L159M	1.297	1.263	1.269	1.021
S46V, D48F, G51R, N52C, Y116N	0.304	0.163	0.187	1.632
S46V, D48F, G51R, N52C, N76V, Y116S, N156K	0.218	0.197	0.201	1.091
S46V, D48F, G51R, N76V, Y116N	0.656	0.291	0.350	1.882
S46V, D48F, G51R, N76V, Y116H, N156K, L159M	0.838	0.360	0.434	1.941
S46V, D48F, G51R, P114R, Y116H	0.636	0.518	0.536	1.194
S46V, D48L, G51R, Y116S, N156K	0.854	0.766	0.780	1.096
S46V, D48L, G51R, Y116H, M155I, N156S	0.606	0.461	0.483	1.254
S46V, D48L, G51R, N52C, N76V, Y116H	1.033	0.397	0.495	2.089
S46V, D48L, G51R, N52F, Y116H, N156K	1.438	1.285	1.310	1.099
S46V, D48L, G51R, N52F, N76I, Y116N	0.446	0.272	0.298	1.492
S46V, D48L, G51R, N52F, N76I, Y116N, L149V	0.536	0.279	0.321	1.667
S46V, D48L, G51R, N52F, N76V, Y116H, L159M	1.023	0.319	0.428	2.389
S46V, D48L, G51R, N76I, Y116P, L159M	0.545	0.243	0.293	1.857
S46V, D48L, G51R, N76I, Y116S	0.398	0.238	0.265	1.501
S46V, D48M, G51R, Y116P	1.353	1.209	1.233	1.098
S46V, D48M, G51R, N52F, Y116H	1.951	1.559	1.620	1.204
S46V, D48M, G51R, N52F, Y116P	0.864	0.659	0.692	1.247
S46V, D48M, G51R, N52F, Y116H, L159M	1.432	1.225	1.260	1.136

TABLE 12-continued

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
S46V, D48M, G51R, N76I, Y116N, N156K	0.913	0.362	0.447	2.042
S46V, D48M, G51R, N76I, Y116S, L159M	0.971	0.387	0.481	2.012
S46V, D48M, G51R, N76V, Y116P	1.353	0.418	0.562	2.405
S46V, D48M, G51R, N76V, Y116P, K162R	1.179	0.377	0.501	2.356
S46V, D48M, G51R, P114R, Y116H, Q151R	1.481	1.318	1.345	1.102
S46V, G47S, D48L, G51R, N52F, Y116H, N156K	0.558	0.351	0.385	1.448
S46V, G51R, Y116N, L159M	2.259	1.823	1.890	1.196
S46V, G51R, Y116N, N156K	1.798	1.818	1.814	0.992
S46V, G51R, Y116P, N156K	1.257	1.000	1.039	1.211
S46V, G51R, Y116S, N156K	1.833	1.363	1.435	1.277
S46V, G51R, N52C, Y116H, L159M	2.011	1.595	1.659	1.212
S46V, G51R, N52C, Y116H, N156K	2.141	1.869	1.913	1.119
S46V, G51R, N52C, N76I, Y116H	0.287	0.141	0.163	1.759
S46V, G51R, N52C, N76V, Y116P	0.535	0.187	0.241	2.151
S46V, G51R, N52F, Y116H	1.296	1.006	1.051	1.234
S46V, G51R, N52F, Y116P, N156K, L159M	0.263	0.216	0.224	1.145
S46V, G51R, N52F, Y116S, L159M	0.271	0.215	0.224	1.210
S46V, G51R, N76I, Y116H	0.210	0.188	0.191	1.098
S46V, G51R, N76V, Y116H	0.740	0.331	0.397	1.865
S46V, G51R, N76V, Y116S	0.653	0.252	0.317	2.065
S46V, G51R, V112M, Y116N, N156K	1.384	0.949	1.016	1.363
I38Y, S46V, D48F, G51R, Y116N, L159M	0.258	0.115	0.139	1.867
I38Y, S46V, D48L, G51R, Y116S	0.218	0.127	0.142	1.510
I38Y, S46V, D48L, G51R, N76I, Y116H, N156K, L159M	0.954	0.379	0.468	2.041
I38Y, S46V, D48L, G51R, N76I, R80G, Y116H, N156K, L159M	0.497	0.245	0.284	1.714
I38Y, S46V, D48M, G51R, N52C, Y116S	0.441	0.305	0.327	1.348
I38Y, S46V, D48M, G51R, N52F, N76V, Y116H	1.638	0.417	0.604	2.711
I38Y, S46V, D48M, G51R, N52F, N76V, Y116H, L181S, *182Y	1.836	0.485	0.692	2.653
I38Y, S46V, D48M, G51R, N76V, Y116N, N156K, L159M	1.627	0.468	0.648	2.513
I38Y, S46V, G51R, Y116N	0.764	0.551	0.585	1.305
I38Y, S46V, G51R, Y116H, L159M	2.024	1.801	1.835	1.103
I38Y, S46V, G51R, N52C, N76I, Y116H, N156K	0.170	0.138	0.144	1.190
S46V, D48L, G51R, F82I, Q84L, Y116H, L159M	1.663	1.293	1.373	1.211
S46V, D48L, G51R, N76V, Y116H	0.678	0.340	0.413	1.639
S46V, D48L, G51R, N76V, F82I, Q84L, Y116N, L159M	0.846	0.375	0.484	1.745
S46V, D48L, G51R, N76V, Y116N	0.899	0.395	0.513	1.754
S46V, D48L, G51R, N76V, Y116N, L159M	0.897	0.403	0.517	1.731
S46V, D48L, G51R, N76V, Y116N, L159M, K162E	0.721	0.342	0.430	1.675
S46V, D48L, G51R, N76V, Y116N, N156K	0.888	0.382	0.492	1.805
S46V, D48L, G51R, N76V, Y116N, N156K, L159M	0.809	0.369	0.471	1.714
S46V, D48L, G51R, N76V, Y116H, L159M	1.073	0.414	0.557	1.924
S46V, D48L, G51R, N76V, Y116H, N156K	0.874	0.401	0.511	1.712
S46V, D48L, G51R, N76V, Y116H, N156K, L159M	0.949	0.425	0.547	1.735
S46V, D48L, G51R, N76V, V112M, Y116N	0.696	0.347	0.429	1.623
S46V, D48M, G51R, N52Y, N76V, Y116N, L159M	1.153	0.424	0.594	1.941
S46V, D48M, G51R, N52Y, N76V, Y116H, L159M	1.157	0.437	0.605	1.914
S46V, D48M, G51R, N76V, Y116H	1.225	0.445	0.614	1.994
S46V, D48M, G51R, N76V, Y116N	1.178	0.446	0.605	1.947
S46V, D48M, G51R, N76V, Y116N, L159M	1.267	0.480	0.663	1.908

TABLE 12-continued

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
S46V, D48M, G51R, N76V, Y116N, N156K	1.058	0.406	0.547	1.931
S46V, D48M, G51R, N76V, Y116N, N156K, L159M	1.176	0.490	0.650	1.811
S46V, D48M, G51R, N76V, Y116H, L159M	1.312	0.529	0.711	1.845
S46V, D48M, G51R, N76V, Y116H, M155R	0.799	0.352	0.449	1.779
S46V, D48M, G51R, N76V, Y116H, N156K	1.254	0.532	0.700	1.791
S46V, D48M, G51R, N76V, Y116H, N156K, L159M	1.257	0.524	0.695	1.810
S46V, D48M, G51R, R56P, N76V, Y116H, L159M	0.406	0.165	0.221	1.836
S46V, G51R, N76V, Y116N, L159M	0.822	0.386	0.487	1.687
S46V, G51R, N76V, Y116N, N156K, L159M	0.942	0.386	0.515	1.826
S46V, G51R, N76V, Y116N, R119H, N156K, L159M	0.916	0.386	0.509	1.796
S46V, G51R, N76V, Y116H, L159M	0.935	0.373	0.504	1.854
S46V, G51R, N76V, Y116H, N156K	0.834	0.340	0.447	1.864
S46V, G51R, N76V, Y116H, N156K, L159M	0.965	0.381	0.508	1.900
S46V, G51R, N76V, Y116H, R119L, N156K	0.761	0.326	0.420	1.809
K32M, S46V, D48L, G51R, F82I, Q84L, Y116H, L159M	1.724	1.281	1.377	1.251
K32M, S46V, G51R, F82I, Y116H, L159M	1.956	2.094	2.064	0.947
K32M, S46V, G51R, F82I, Q84L, Y116H, L159M	2.066	2.023	2.032	1.016
K32M, S46V, G51R, F82I, Q84L, Y116H, N156K, L159M	2.050	2.348	2.284	0.897
S46V, T49S, G51R, N76V, Q84I, Y116H, N156K, L159M	0.674	0.298	0.379	1.774
A30E, S46V, D48M, G51R, N76V, F82R, Y116H, I148Y	0.446	0.198	0.235	1.846
E37A, I45R, S46V, D48M, T49W, G51R, N76V, F82G, Y116H, N156K	0.625	0.318	0.367	1.704
S46V, D48M, G51R, N76V, F82G, Y116H, A136D, I148Y	2.627	0.613	0.913	2.878
S46V, D48M, G51R, N76V, F82G, Y116H, I148Y	2.416	0.621	0.908	2.661
S46V, D48M, G51R, N76V, F82G, Y116H, I148Y, W168C	0.339	0.239	0.254	1.331
S46V, D48M, G51R, N76V, F82G, Y116H, N156R, L159M	1.381	0.438	0.579	2.387
S46V, D48M, G51R, N76V, F82G, Q84L, Y116H	1.602	0.483	0.650	2.467
S46V, D48M, G51R, N76V, F82I, Y116H, N156K, L159K	1.164	0.465	0.569	2.047
S46V, D48M, G51R, N76V, F82I, Q84L, Y116H, N156R	1.419	0.466	0.618	2.289
S46V, D48M, G51R, N76V, F82R, Y116H, L159K	2.760	0.734	1.035	2.661
S46V, D48M, G51R, N76V, F82R, N115A, Y116H, N156K, L159K	1.874	0.582	0.788	2.375
S46V, D48M, G51R, N76V, F82R, Y116H, N156K, L159K	1.907	0.647	0.848	2.229
S46V, D48M, G51R, N76V, Y116H, I148Y	2.793	0.796	1.114	2.505
I45R, S46V, D48M, G51R, N76V, F82I, Y116H, N156K, L159K	1.335	0.491	0.626	2.125
I45R, S46V, D48M, G51R, N76V, Q84L, Y116H, N156R, L159K	0.448	0.238	0.269	1.660
I45R, S46V, D48M, T49L, G51R, A67S, N76V, F82G, Y116H, L159K	0.476	0.319	0.345	1.382
I45R, S46V, D48M, T49L, G51R, N76V, F82G, Y116H, L159K	0.532	0.335	0.366	1.451
I45R, S46V, D48M, T49L, G51R, N76V, Q84L, Y116H, N156R, L159M	0.318	0.201	0.220	1.444
I45R, S46V, D48M, T49R, G51R, N52D, N76V, F82I, Q84L, Y116H, I148Y, N156Y	1.056	0.383	0.490	2.152
I45R, S46V, D48M, T49W, G51R, N76V, F82I, Q84L, Y116H, I148Y	0.204	0.174	0.179	1.147

TABLE 12-continued

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
K32M, S46V, D48M, G51R, N76V, F82G, Y116H, N156R, L159M	1.289	0.436	0.574	2.246
K32M, S46V, D48M, G51R, N76V, F82G, Q84L, Y116H	1.565	0.493	0.664	2.353
K32M, S46V, D48M, G51R, N76V, F82I, Y116H, L159M	1.743	0.534	0.729	2.382
K32M, S46V, D48M, G51R, N76V, F82I, Q84L, Y116H, L159M	1.926	0.538	0.745	2.583
K32M, I45R, S46V, D48M, T49D, G51R, N76V, F82G, Y116H, L159Q	0.927	0.405	0.489	1.896
K32M, I45R, S46V, D48M, T49D, G51R, N76V, F82I, Y116H, I148Y	1.884	0.599	0.806	2.335
K32M, S46V, D48M, G51R, N76V, Y116H, L159K	1.289	0.472	0.603	2.138
K32M, S46V, D48M, N76V, F82G, Q84L, Y116H	0.902	0.353	0.441	2.045
K32M, S46V, D48M, T49D, G51R, N76V, F82I, Q84L, Y116H	0.537	0.229	0.275	1.956
K32M, S46V, D48M, T49D, G51R, N76V, Y116H, I148Y	0.841	0.343	0.423	1.988
K32M, S46V, D48M, T49L, G51R, N76V, F82G, Q84L, Y116H, N156K, L159M	2.118	0.623	0.861	2.456
K32M, S46V, D48M, T49L, G51R, N76V, F82R, Y116H, L159K	0.099	0.089	0.091	1.056
K32M, S46V, D48M, T49W, G51R, P58L, L73F, N76V, Q84L, Y116H, N156K, L159K	0.323	0.236	0.250	1.299
K32M, S46V, D48M, T49W, G51R, N76V, Q84L, Y116H, N156K, L159K	1.310	0.494	0.624	2.096
S46V, D48M, G51R, N76V, Y116H, N156K, L159M	2.019	0.595	0.824	2.450
Q34R, I45R, S46V, D48M, T49D, G51R, N76V, F82R, Y116H	1.421	0.492	0.642	2.214
S46V, D48M, G51R, N76V, Q84L, Y116H, N156K	1.631	0.522	0.701	2.325
S46V, D48M, T49D, G51R, N76V, F82G, Y116H, L159K	0.302	0.220	0.233	1.284
S46V, D48M, T49D, G51R, N76V, F82G, Y116H, N156R	0.596	0.267	0.319	1.871
S46V, D48M, T49D, G51R, N76V, F82I, Y116H, L159M	0.554	0.284	0.327	1.691
S46V, D48M, T49D, G51R, N76V, F82R, Y116H, N156R	0.746	0.326	0.388	1.923
S46V, D48M, T49D, G51R, N76V, F82R, Y116H, N156R, L159I	0.679	0.305	0.361	1.882
S46V, D48M, T49D, G51R, N76V, F82R, Q84L, Y116H, N156K	0.822	0.358	0.427	1.927
S46V, D48M, T49D, G51R, N76V, G81C, F82R, Q84L, Y116H, N156K	0.694	0.313	0.369	1.878
S46V, D48M, T49D, G51R, N76V, Y116H, N156K, L159M	0.676	0.284	0.343	1.973
S46V, D48M, T49D, G51R, N76V, Y116H, N156R, L159K	0.563	0.284	0.329	1.711
S46V, D48M, T49D, G51R, N76V, Q84L, Y116H, L159M	0.545	0.289	0.330	1.653
S46V, D48M, T49L, G51R, N76V, F82I, N156K	1.486	0.505	0.662	2.242
S46V, D48M, T49L, G51R, N76V, F82I, Y116H, N156K	1.729	0.569	0.754	2.292
S46V, D48M, T49L, G51R, N76V, F82I, Y116H, N156R, L159K	0.719	0.301	0.363	1.978
S46V, D48M, T49L, G51R, N76V, F82R, Y116H	0.854	0.345	0.427	1.998
S46V, D48M, T49L, G51R, N76V, F82R, Q84L, Y116H, I148Y	0.357	0.209	0.231	1.528
S46V, D48M, T49L, G51R, N76V, Y116H, N156K, L159M	2.230	0.621	0.877	2.543
S46V, D48M, T49L, G51R, N76V, Y116H, N156R	1.554	0.493	0.664	2.343
S46V, D48M, T49L, G51R, N76V, Q84L, Y116H	2.723	0.748	1.063	2.563
S46V, D48M, T49L, G51R, N76V, Q84L, Y116H, K150I	2.411	0.699	0.972	2.480

TABLE 12-continued

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
S46V, D48M, T49W, G51R, N76V, F82G, Q84L, Y116H, I148Y	1.432	0.483	0.634	2.259
S46V, D48M, T49W, G51R, N76V, F82G, Q84L, Y116H, N156K, L159K	0.364	0.287	0.300	1.215
S46V, D48M, T49W, G51R, N76V, F82I, Y116H	0.495	0.288	0.321	1.537
S46V, D48M, T49W, G51R, N76V, F82I, Y116H, N156R, L159M	0.466	0.270	0.302	1.543
S46V, D48M, T49W, G51R, N76V, F82R, Y116H, L159M	0.282	0.261	0.265	1.063
S46V, D48M, T49W, G51R, N76V, Q84L, Y116H, N156R, L159K	1.323	0.491	0.624	2.118

Table 13 shows a list of combination mutants found using P1B9v2.0 (SEQ ID NO: 58) as Template in BD64.

TABLE 13

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
E37D, S46V, D48M, T49L, G51R, N52K, N76V, F82I, V112L, N115W, Y116H, I148Y, L159M	3.143	2.067	2.668	1.175
E37D, S46V, D48M, T49L, G51R, N52K, N76V, N115Y, Y116H, I148Y	3.859	1.827	2.962	1.305
S46V, D48M, T49L, G51R, N76V, F82I, Y116H, I148Y, L159M	1.088	1.211	1.143	0.953
S46V, D48M, T49L, G51R, N76V, F82I, Y116H, I148Y, L159M, D164T	1.535	1.202	1.370	1.119
S46V, D48M, T49L, G51R, N76V, F82I, M94I, N115C, Y116H, I148Y	1.422	1.177	1.303	1.091
S46V, D48M, T49L, G51R, N76V, F82I, N115F, Y116H, I148Y	2.010	1.099	1.568	1.281
S46V, D48M, T49L, G51R, N76V, F82I, V112L, N115F, Y116H, I148Y	1.216	1.030	1.128	1.079
S46V, D48M, T49L, G51R, N76V, G85S, N115W, Y116H, I148Y, L159M	1.137	1.173	1.154	0.986
S46V, D48M, T49L, G51R, H64N, N76V, N115F, Y116H, I148Y, D164T	1.116	0.907	1.013	1.102
S46V, D48M, T49L, G51R, N76V, I87V, V112L, N115W, Y116H, I148Y	1.776	1.022	1.405	1.266
K32I, S46V, D48M, T49L, G51R, N52K, N76V, N115F, Y116H, I148Y, L159M	1.591	1.126	1.366	1.166
S46V, D48M, T49L, G51R, K65M, N76V, F82I, N115W, Y116H, I148Y, N156H	1.119	0.967	1.047	1.067
S46V, D48M, T49L, G51R, K65M, N76V, F82I, V112L, N115F, Y116H, I148Y	1.552	1.072	1.316	1.181
S46V, D48M, T49L, G51R, K65M, N76V, G85S, V112L, N115W, Y116H, I148Y, M155K	0.755	0.722	0.739	1.017
S46V, D48M, T49L, G51R, K65M, N76V, Y116H, I148Y, L159M	1.025	1.035	1.030	0.995
S46V, D48M, T49L, G51R, K65M, N76V, N115F, Y116H, I148Y	1.831	1.103	1.487	1.229
S46V, D48M, T49L, G51R, K65M, N76V, N89K, V112L, N115Y, Y116H, I148Y	1.215	1.045	1.132	1.077
S46V, D48M, T49L, G51R, K65M, N76V, V112L, N115W, Y116H, I148Y	1.650	0.957	1.325	1.244
S46V, D48M, T49L, G51R, K65M, N76V, V112L, N115Y, Y116H, I148Y	1.113	0.890	1.003	1.112
S46V, D48M, T49L, G51R, N76V, Y116H, I148Y, L159M, D164T	1.498	1.249	1.375	1.088
L31M, Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Y116H, I148Y	1.184	1.370	1.276	0.927
S46V, D48M, T49M, G51R, N52K, N76V, F82I, V112L, N115Y, Y116H, I148Y, D164T	2.161	1.375	1.780	1.217
S46V, D48M, T49L, G51R, N76V, N115F, Y116H, I148Y, L159M	1.358	1.100	1.230	1.103
S46V, D48M, T49L, G51R, N76V, N115W, Y116H, I148Y, D164T	1.337	1.013	1.184	1.126
S46V, D48M, T49L, G51R, N76V, N115W, Y116H, I148Y, L159M	1.906	1.118	1.537	1.240
S46V, D48M, T49L, G51R, N76V, N115Y, Y116H, I148Y, L159M	1.267	1.013	1.155	1.096

TABLE 13-continued

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
S46V, D48M, T49L, G51R, N52K, N76V, Y116H, I148Y, D164T	1.790	1.385	1.599	1.122
S46V, D48M, T49L, G51R, N52K, N76V, F82I, Y116H, I148Y	4.051	2.136	3.154	1.287
S46V, D48M, T49L, G51R, N52K, N76V, F82I, V112L, N115W, Y116H, I148Y, L159M	4.148	1.636	3.039	1.359
S46V, D48M, T49L, G51R, N52K, N76V, F82I, V112L, N115Y, Y116H, I148Y, D164T	2.175	1.471	1.834	1.189
S46V, D48M, T49L, G51R, N52K, K65M, N76V, Y116H, I148Y	1.733	1.303	1.525	1.136
S46V, D48M, T49L, G51R, N52K, K65M, N76V, N115W, Y116H, I148Y	4.254	1.591	2.944	1.441
S46V, D48M, T49L, G51R, N52K, K65M, N76V, N89I, N115W, Y116H, I148Y	1.068	0.794	0.933	1.146
S46V, D48M, T49L, G51R, N52K, K65M, N76V, V112L, N115W, Y116H, I148Y, L159M, D164T	1.721	1.094	1.426	1.205
S46V, D48M, T49L, G51R, N52K, K65M, N76V, V112M, Y116H, I148Y, D164V	1.112	1.085	1.099	1.011
S46V, D48M, T49L, G51R, N52K, N76V, N115F, Y116H, I148Y, L159M	2.353	1.381	1.882	1.252
S46V, D48M, T49L, G51R, N52K, N76V, N115W, Y116H, I148Y	4.639	1.722	3.264	1.424
S46V, D48M, T49L, G51R, N52K, N76V, N115Y, Y116H, I148Y	4.403	2.051	3.365	1.311
S46V, D48M, T49L, G51R, N52K, N76V, N115Y, Y116H, I148Y, D164T	4.704	1.945	3.411	1.381
S46V, D48M, T49L, G51R, N52K, N76V, T59A, F82I, V112L, N115W, Y116H, I148Y, L159M	1.898	1.171	1.577	1.200
S46V, D48M, T49L, G51R, N52K, N76V, V112L, Y116H, I148Y	3.675	2.069	2.966	1.245
S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115F, Y116H, I148Y, A165V	1.366	1.092	1.230	1.109
S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115F, Y116H, I148Y, L159M	3.776	2.131	2.962	1.256
S46V, D48M, T49L, G51R, N52S, N76V, V112L, N115Y, Y116H, I148Y	1.737	1.039	1.394	1.250
P29R, Q33R, S46V, D48M, T49L, G51R, N52K, N76V, F82I, N115F, Y116H, I148Y, D164T	3.873	2.125	3.054	1.271
Q28K, Q33R, S46V, D48M, T49L, G51R, N52K, N76V, N115F, Y116H, I148Y, D164T	4.385	2.210	3.366	1.303
Q28K, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, F82I, V112L, N115W, Y116H, I148Y, L159M	4.201	2.708	3.497	1.207
Q33A, S46V, D48M, T49L, G51R, N76V, F82I, N115W, Y116H, I148Y, D164T	1.639	1.072	1.360	1.207
Q33A, S46V, D48M, T49L, G51R, K65M, N76V, Y116H, K118R, I148Y	0.987	0.985	0.986	1.003
Q33A, S46V, D48M, T49L, G51R, K65M, N76V, V112L, N115Y, Y116H, I148Y	1.168	0.979	1.084	1.077
Q33A, S46V, D48M, T49V, G51R, K65M, N76V, N115Y, Y116H, I148Y, L159M	1.046	0.900	0.976	1.071
Q33A, S46V, D48M, T49L, G51R, N76V, N115Y	1.565	0.994	1.313	1.191
Q33A, S46V, D48M, T49L, G51R, N76V, N115Y, Y116H, I148Y, L159M	1.423	1.010	1.219	1.166
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, D77E, F82I, N115F, Y116H, I148Y	0.885	0.791	0.844	1.047
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, F82I, N115F, Y116H, I148Y, D164Y	2.639	1.713	2.230	1.187
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, F82I, R88C, N115F, Y116H, I148Y	4.148	2.157	3.269	1.270
Q33A, S46V, D48M, T49L, G51R, N52K, K65M, N76V, F82I, Y116H, I148Y	1.807	1.632	1.730	1.047
Q33A, S46V, D48M, T49L, G51R, N52K, K65M, N76V, F82I, Y116H, K118Q, I148Y	4.314	2.663	3.585	1.204
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, N115F, Y116H, I148Y	4.458	1.534	3.079	1.447
Q33A, S46V, D48M, T49L, G51R, N52K, T59S, N76V, F82I, N115F, Y116H, I148Y, K162R, D164E	4.215	2.129	3.295	1.280
Q33A, S46V, D48M, T49L, G51R, T59A, N76V, V112L, Y116H, I148Y, D164T	1.108	1.129	1.118	0.992
Q33A, S46V, D48M, T49L, G51R, N76V, V112L, Y116H, I148Y, D164T	1.265	1.119	1.196	1.059
Q33A, S46V, D48M, T49L, G51R, N76V, V112L, N115F, Y116H, I148Y, D164T	1.823	1.108	1.486	1.229
Q33A, S46V, D48M, T49L, G51R, N76V, V112L, N115F, Y116H, I148Y, K162N, D164T	1.559	0.990	1.291	1.207

TABLE 13-continued

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
Q33A, S46V, D48M, T49L, G51R, N76V, V112L, N115W, Y116H, I148Y	1.802	1.052	1.430	1.258
Q33H, S46V, D48M, T49L, G51R, N52K, K65M, N76V, N115W, Y116H, I148Y	2.564	1.315	1.949	1.318
Q33R, S46V, D48M, T49L, G51R, N76V, F82I, N115W, Y116H, I148Y	1.587	1.198	1.404	1.131
Q33R, S46V, D48M, T49L, G51R, N76V, F82I, V112L, N115Y, Y116H, I148Y, L159M	1.432	1.351	1.396	1.017
Q33R, S46V, D48M, T49L, G51R, N76V, I110S, N115W, Y116H, I148Y, L159M	0.983	0.920	0.952	1.030
Q33R, S46V, D48M, T49L, G51R, K65M, N76V, Y116H, I148Y, L159M	0.986	1.054	1.019	0.965
Q33R, S46V, D48M, T49L, G51R, K65M, N76V, F82I, N115W, Y116H, I148Y	1.051	1.122	1.085	0.969
Q33R, S46V, D48M, T49L, G51R, K65M, N76V, G81C, N115F, Y116H, I148Y, L159M, D164T	1.027	0.746	0.894	1.142
Q33R, S46V, D48M, T49L, G51R, K65M, N76V, N115F, Y116H, I148Y, L159M, D164T	1.846	1.048	1.469	1.254
Q33R, S46V, D48M, T49L, G51R, N76V, L79M, V112L, N115Y, Y116H, I148Y	0.761	0.725	0.743	1.024
Q33R, S46V, D48M, T49L, G51R, N76V, N115W, Y116H, I148Y	1.960	1.238	1.603	1.222
Q33R, S46V, D48M, T49L, G51R, N76V, N115W, Y116H, I148Y, L159M	1.396	1.048	1.225	1.139
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Y116H, I148Y	3.939	2.752	3.352	1.175
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, F82I, N115F, Y116H, I148Y, D164T	3.131	2.540	2.854	1.099
Q33R, S46V, D48M, T49L, G51R, N52K, G53C, N76V, F82I, N115F, Y116H, I148Y, D164T	1.593	1.380	1.493	1.069
Q33R, S46V, D48M, T49L, G51R, N52K, K65M, N76V, V112L, Y116H, I148Y, L159M	3.795	2.176	2.998	1.270
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, N115F, Y116H, I148Y	5.164	2.018	3.639	1.421
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, N115Y, Y116H, I148Y	4.774	1.710	3.289	1.454
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, N115Y, Y116H, I148Y, L159M, D164T	2.326	1.258	1.809	1.285
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, V112L, Y116H, I148Y	4.278	2.041	3.230	1.330
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, V112L, Y116H, I148Y, D164Y	2.435	1.530	2.011	1.216
Q33R, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, Q63K, Y68C, N115F, Y116H, I148Y	5.192	1.873	3.583	1.451
Q33R, S46V, D48M, T49L, G51R, R51S, N52K, K65M, N76V, V112L, Y116H, I148Y, L159M	1.149	1.087	1.118	1.027
Q33R, S46V, D48M, T49L, G51R, N76V, V112L, N115F, Y116H, I148Y, D164T	2.316	1.030	1.714	1.351
Q33R, S46V, D48M, T49L, G51R, N76V, V112L, N115Y, Y116H, I148Y	1.243	0.982	1.114	1.115
Q33R, S46V, D48M, T49L, G51R, Y68S, N76V, F82I, N115W, Y116H, I148Y, D164A	0.938	0.912	0.925	1.011
Q34L, S46V, D48M, T49L, G51R, N52K, K65M, N76V, N115Y, Y116H, I148Y, D164T	2.948	1.885	2.425	1.206
Q34R, S46V, D48M, T49L, G51R, N76V, F82I, Y116H, I148Y, D164T	1.292	1.083	1.191	1.084
Q34R, S46V, D48M, T49L, G51R, N52K, K65M, Y116H, I148Y	4.561	1.912	3.277	1.391
S46V, D48M, T49L, G51R, N76V, V112L, Y116H, I148Y, L159M	1.384	1.257	1.328	1.043
S46V, D48M, T49L, G51R, N76V, V112L, N115W, Y116H, I148Y	2.333	1.105	1.729	1.351
A30T, S46V, D48M, T49L, G51R, N76V, T86P, V112L, N115F, Y116H, I148Y	1.158	0.762	0.983	1.178
K32I, S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115F, Y116H, I148Y	4.584	2.512	3.624	1.264
K32N, S46V, D48R, T49L, G51R, N52K, N76V, Q84V, N115W, Y116H, I148Y, L149F	3.361	2.327	2.888	1.164
L5V, S22R, Q33A, S46V, D48M, T49L, G51R, N52K, N76V, I87V, N115F, Y116H, I148Y, L149M	3.457	1.962	2.765	1.250
N2T, Q34R, S46V, D48M, T49L, G51R, N76V, Q84T, V112L, N115F, Y116H, K118Q, I148Y, L149M	1.784	1.178	1.503	1.186
S46V, D48M, T49L, G51R, N52K, N76V, G81C, N115W, Y116H, I148Y	3.148	1.848	2.546	1.236
S46V, D48M, T49L, G51R, N52K, N76V, N115F, Y116H, I148Y	4.305	3.298	3.839	1.125

TABLE 13-continued

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
S46V, D48M, T49L, G51R, N52K, N76V, N115F, Y116H, K118Q, I148Y	4.547	2.847	3.743	1.214
S46V, D48M, T49L, G51R, N52K, N76V, Q84A, V112L, N115F, Y116H, I148Y	4.872	2.567	3.767	1.291
S46V, D48M, T49L, G51R, N52K, N76V, Q84T, N115W, Y116H, I148Y, L149F	4.840	2.692	3.809	1.270
S46V, D48M, T49L, G51R, N52K, N76V, Q84T, V112L, N115F, Y116H, I148Y	3.710	2.244	3.054	1.213
S46V, D48M, T49L, G51R, N52K, N76V, Q84V, N115W, Y116H, I148Y, L149F	2.744	1.925	2.369	1.159
S46V, D48M, T49L, G51R, N52K, N76V, Q84V, V112L, N115F, Y116H, I148Y	3.042	1.847	2.518	1.208
S46V, D48M, T49L, G51R, N52K, N76V, Q84V, V112L, N115W, Y116H, K118Q, I148Y	1.944	1.219	1.608	1.208
S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115F, Y116H, I148Y	4.686	2.599	3.719	1.259
S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115F, Y116H, K118Q, I148Y	5.314	2.799	4.163	1.276
S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115W, Y116H, I148Y	4.977	2.155	3.714	1.344
S46V, D48M, T49L, G51R, N52R, N76V, Q84A, V112L, N115W, Y116H, I148Y, L149M	5.972	2.734	4.541	1.315
P29L, S46V, D48M, T49L, G51R, N52K, N76V, Q84T, V112L, N115F, Y116H, I148Y	2.850	1.792	2.377	1.200
P29L, Q33A, Q34R, I45M, S46V, D48M, T49L, G51R, N52K, N76V, F82L, R88H, V112L, N115W, Y116H, I148Y	1.803	1.458	1.652	1.091
Q28H, Q33R, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, Q84T, V112L, N115W, Y116H, I148Y	3.820	1.882	2.889	1.322
Q33A, S46V, D48M, T49L, G51R, L54M, N76V, Q84V, V112L, N115F	1.590	1.032	1.326	1.199
Q33A, S46V, D48L, T49L, G51R, N52K, N76V, N115W, Y116H, I148Y, L149M	1.458	1.126	1.306	1.116
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, N115F, Y116H, I148Y, L149M	4.373	2.843	3.664	1.193
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, N115F, Y116H, I148Y, L149T	4.515	3.220	3.947	1.144
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, N115W, Y116H, I148Y, L149M	4.499	2.757	3.702	1.215
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, Q84A, V112L, N115W, Y116H, K118Q, I148Y, L149T	2.859	1.522	2.226	1.284
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, Q84A, V112L, N115W, Y116H, I148Y, L149M	3.673	2.095	2.982	1.232
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, Q84T, N115W, Y116H, I148Y	3.231	1.937	2.610	1.235
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, Q84V, V112L, N115W, Y116H, I148Y, L149F	3.489	1.854	2.741	1.273
Q33A, S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115F, Y116H, I148Y	4.389	2.814	3.699	1.187
Q33A, S46V, D48M, T49L, G51R, N52R, N76V, Y116H, I148Y, L149T	3.791	3.302	3.557	1.069
Q33A, S46V, D48M, T49L, G51R, N52S, N76V, Q84V, V112L, N115F, Y116H, I148Y	1.751	0.996	1.393	1.256
Q33A, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, P83R, Q84A, V112L, N115F, Y116H, K118Q, I148Y	4.215	4.136	4.181	1.008
Q33A, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, P83R, Q84A, V112L, N115F, Y116H, K118Q, I148Y	3.306	2.121	2.782	1.189
Q33A, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, Q84A, V112L, N115F, Y116H, K118Q, I148Y	4.736	3.094	4.010	1.181
Q33A, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115W, Y116H, I148Y	5.554	3.159	4.504	1.234
Q33A, Q34R, S46V, D48M, T49L, G51R, N52R, N76V, Y116H, I148Y, L149F	3.728	3.273	3.510	1.062
Q33A, Q34R, S46V, D48M, T49L, G51R, N52R, N76V, Q84V, V112L, N115W, Y116H, I148Y, L149T	5.915	2.816	4.427	1.335
Q33A, Q34R, S46V, D48M, T49L, G51R, N52R, N76V, V112L, N115F, Y116H, I148Y, L149F	4.973	2.606	3.839	1.295
Q33A, Q34R, S46V, D48M, T49L, G51R, N76V, Q84A, N115F, Y116H, I148Y	1.535	0.921	1.254	1.224
Q33A, Q34R, S46V, D48M, T49L, G51R, N76V, Q84V, V112L, N115W, Y116H, I148Y, L149T	1.508	1.013	1.270	1.187
Q33A, S46V, D48M, T49L, G51R, N76V, Q84A, V112L, N115W, Y116H, I148Y	1.676	1.063	1.407	1.192

TABLE 13-continued

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
Q33A, S46V, D48M, T49L, G51R, N76V, Q84T, V112L, N115W, Y116H, I148Y	1.373	0.956	1.186	1.158
Q33A, S46V, D48M, T49L, G51R, N76V, V112L, N115F, Y116H, K118Q, I148Y, L149T	2.007	1.269	1.653	1.222
Q33P, S46V, D48M, T49L, G51R, N52R, N76V, V112L, N115F, Y116H, I148Y	2.207	1.365	1.804	1.238
Q33R, E37D, S46V, D48M, T49L, G51R, N52K, N76V, G81C, Q84A, V112L, N115W, Y116H, I148Y	3.331	1.850	2.645	1.259
Q33R, S46V, D48M, T49L, G51R, N76V, N115F, Y116H, I148Y	1.776	1.148	1.485	1.195
Q33R, S46V, D48M, T49L, G51R, N76V, N115F, Y116H, K118Q, I148Y, L149F	1.966	1.146	1.599	1.230
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, F82V, V112L, N115F, Y116H, I148Y	3.586	2.808	3.242	1.106
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, G81C, N115F, Y116H, I148Y	3.470	1.980	2.780	1.248
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, G81C, Q84A, V112L, N115W, Y116H, I148Y, L149M	3.143	1.816	2.535	1.240
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, L79M, V112L, N115W, Y116H, I148Y, L149T	2.892	1.774	2.392	1.209
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, N115W, Y116H, I148Y	3.279	2.324	2.852	1.149
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84A, V112L, N115F, Y116H, I148Y, L149F	3.751	2.345	3.134	1.197
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84A, V112L, N115F, Y116H, I148Y, L149F, M154I	1.426	1.098	1.282	1.113
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84A, V112L, N115W, Y116H, I148Y, L149M	6.010	2.759	4.521	1.329
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84A, V112L, N115W, Y116H, I148Y, L149M	1.968	1.221	1.622	1.212
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84K, V112L, N115F, Y116H, I148Y	2.521	1.712	2.164	1.166
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84T, N115F, Y116H, I148Y, L149M	3.052	2.022	2.591	1.178
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84T, V112L, N115F, Y116H, I148Y	3.220	1.947	2.630	1.224
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84T, V112L, N115F, Y116H, K118Q, I148Y, L149T	4.980	2.989	4.068	1.224
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84T, V112L, N115W, Y116H, I148Y	3.012	1.832	2.490	1.210
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84V, V112L, N115F, Y116H, K118Q, I148Y, L149F	4.921	2.656	3.834	1.283
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84V, V112L, N115W, Y116H, K118Q, I148Y	2.999	1.686	2.398	1.249
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84V, V112L, N115W, Y116H, I148Y, L149F	3.750	1.961	2.931	1.279
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, Q84V, V112L, N115W, Y116H, I148Y, L149F	4.702	2.750	3.839	1.225
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115F, Y116H, I148Y	5.424	2.806	4.253	1.276
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115W, Y116H, I148Y	6.026	2.682	4.477	1.345
Q33R, S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115W, Y116H, I148Y, L149M	4.563	2.572	3.672	1.242
Q33R, S46V, D48M, T49L, G51R, N52R, N76V, V112L, N115W, Y116H, I148Y, L149T	4.017	3.432	3.736	1.075
Q33R, S46V, D48M, T49L, G51R, N52R, N76V, Y116H, I148Y, L149F	5.463	2.558	4.068	1.343
Q33R, S46V, D48M, T49L, G51R, N52R, N76V, N115W, Y116H, I148Y	4.830	2.877	3.893	1.241
Q33R, S46V, D48M, T49L, G51R, N52R, N76V, Q84A, N115F, Y116H, I148Y, L149F	2.939	1.666	2.381	1.235
Q33R, S46V, D48M, T49L, G51R, N52R, N76V, Q84A, V112L, N115F, Y116H, K118Q, I148Y, L149M	3.105	1.992	2.607	1.191
Q33R, S46V, D48M, T49L, G51R, N52R, N76V, Q84T, V112L, N115F, Y116H, I148Y	5.372	2.835	4.251	1.265
Q33R, S46V, D48M, T49L, G51R, N52R, N76V, T86P, V112L, N115W, Y116H, I148Y, L149T	4.105	2.503	3.388	1.211
Q33R, S46V, D48M, T49L, G51R, N52R, N76V, V112L, N115F, Y116H, I148Y	3.385	2.212	2.871	1.175
Q33R, S46V, D48M, T49L, G51R, N52R, N76V, V112L, N115F, Y116H, K118Q, I148Y, L149M	6.042	2.468	4.462	1.354
Q33R, S46V, D48M, T49L, G51R, N52R, N76V, V112L, N115W, Y116H, I148Y, L149T				

TABLE 13-continued

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
Q33R, Q34K, S46V, D48M, T49L, G51R, N76V, Q84V, V112L, N115W, Y116H, I148Y, L149T	1.484	0.957	1.231	1.205
Q33R, Q34R, S46V, D48I, T49L, G51R, N52K, N76V, N115W	1.665	1.313	1.497	1.112
Q33R, Q34R, S46V, D48M, T49L, G51R, N52K, A67T, N76V, V112L, N115F, Y116H, I148Y, L149M, M155I	1.891	1.298	1.610	1.174
Q33R, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, N115W, Y116H, K118Q, I148Y	4.801	2.910	3.895	1.231
Q33R, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, Q84A, V112L, N115F, Y116H, I148Y, L149F	5.024	2.719	4.005	1.254
Q33R, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, Q84T, V112L, N115W, Y116H, I148Y	5.007	2.275	3.695	1.354
Q33R, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, Q84V, N115F, Y116H, I148Y, K133R, L149T	3.340	2.027	2.711	1.231
Q33R, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, Q84V, N115F, Y116H, I148Y, L149T	3.305	2.106	2.730	1.210
Q33R, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115F, Y116H, I148Y	4.894	2.294	3.645	1.343
Q33R, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115F, Y116H, I148Y, L149M	4.942	2.527	3.799	1.300
Q33R, Q34R, S46V, D48M, T49L, G51R, N52K, N76V, V112L, N115W, Y116H, I148Y, L149F	6.145	2.696	4.566	1.346
Q33R, Q34R, S46V, D48M, T49L, G51R, N52R, N76V, Q84V, N115W, Y116H, K118Q, I148Y, L149M	5.301	2.946	4.223	1.255
Q33R, Q34R, S46V, D48M, T49L, G51R, N52R, N76V, V112L, N115W, Y116H, I148Y	5.818	2.358	4.270	1.363
Q33R, Q34R, S46V, D48M, T49L, G51R, N76V, Q84A, V112L, N115F, Y116H, I148Y, L149F, G158C	1.854	1.113	1.527	1.215
Q33R, Q34R, S46V, D48M, T49L, G51H, N52K, N76V, V112L, N115F, Y116H, I148Y	2.840	1.559	2.225	1.275
Q33R, Q34R, S46V, D48M, T49L, G51R, N76V, V112L, N115W, Y116H, I148Y	1.685	0.926	1.346	1.253
Q33R, Q34R, S46V, D48M, T49L, G51R, N76V, V112L, N115W, Y116H, K118Q, I148Y	1.501	1.097	1.314	1.142
Q33R, S46V, D48M, T49L, G51R, N76V, Q84A, N115F, Y116H, I148Y, L149F	1.737	1.143	1.452	1.196
Q33R, S46V, D48M, T49L, G51R, N76V, Q84A, N115W, Y116H, K118Q, I148Y	1.967	1.441	1.726	1.140
Q33R, S46V, D48M, T49L, G51R, N76V, Q84A, N89H, N115W, Y116H, K118Q, I148Y	1.225	1.050	1.145	1.070
Q33R, S46V, D48M, T49L, G51R, N76V, Q84T, V112L, N115F, Y116H, I148Y	1.695	1.019	1.382	1.226
Q33R, S46V, D48M, T49L, G51R, N76V, Q84V, N115F, Y116H, I148Y, L149T	1.640	1.024	1.348	1.216
Q33R, S46V, D48M, T49L, G51R, N76V, Q84V, T86P, V112L, N115F, Y116H, I148Y, L149M	2.225	1.168	1.717	1.295
Q33R, S46V, D48M, T49L, G51R, R56H, N76V, N115F, Y116H, K118Q, I148Y, L149F	0.986	0.818	0.911	1.083
Q33R, S46V, D48M, T49L, G51R, N76V, V112L, N115F, Y116H, I148Y	4.155	2.220	3.289	1.262
Q33R, S46V, D48M, T49L, G51R, N76V, V112L, N115F, Y116H, I148Y, L149M	2.305	1.130	1.742	1.322
Q33R, S46V, D48M, T49L, G51R, N76V, V112L, N115F, Y116H, I148Y, L149T	1.756	1.057	1.447	1.213
Q33R, S46V, D48M, T49L, G51R, N76V, V112L, N115W, Y116H, I148Y	1.825	1.178	1.529	1.194
Q33R, S46V, D48M, T49L, G51R, N76V, V112L, N115W, Y116H, I148Y, L149M	1.821	1.164	1.530	1.190
Q34K, S46V, D48M, T49L, G51R, N52K, N76V, Q84V, V112L, N115F, Y116H, I148Y	1.571	1.126	1.376	1.142
Q34K, S46V, D48M, T49L, G51R, N52R, N76V, V112L, N115F, Y116H, I148Y	2.881	1.734	2.374	1.214
Q34L, S46V, D48M, T49L, G51R, N52K, N76V, N115F, Y116H, K118Q, I148Y	5.058	2.819	3.998	1.264
Q34R, G35A, S46V, D48M, T49L, G51R, N76V, Q84A, V112L, N115F, Y116H, I148Y, L149T	1.792	1.116	1.493	1.200
Q34R, S46V, D48M, T49L, G51R, N76V, N115W, Y116H, I148Y	2.409	1.476	1.961	1.205
Q34R, S46V, D48M, T49L, G51R, N52K, N76V, Q84A, V112L, N115F, Y116H, I148Y, L149T	5.576	2.923	4.304	1.294
Q34R, S46V, D48M, T49L, G51R, N52R, N76V, N115W, Y116H, I148Y	5.448	2.512	4.039	1.348
Q34R, S46V, D48M, T49L, G51R, N52R, N76V, V112L, N115W, Y116H, I148Y, L149F	3.948	1.967	3.080	1.282

TABLE 13-continued

Mutations	Norm FAME	Norm FFA	Norm Titer	Norm % FAME
Q34R, S46V, D48M, T49L, G51R, N76V, Q84A, V112L, N115F, Y116H, I147L, I148Y, L149T	1.061	0.669	0.888	1.195
Q34R, S46V, D48M, T49L, G51R, N76V, Q84A, V112L, N115F, Y116H, I148Y, L149T	1.961	1.090	1.576	1.244
Q34R, S46V, D48M, T49L, G51R, N76V, Q84T, V112L, N115W, Y116H, I148Y, L149F	1.823	1.156	1.531	1.191
Q34R, S46V, D48M, T49L, G51R, N76V, Q84V, V112L, N115W, Y116H, I148Y, L149M	1.684	1.102	1.429	1.179
S46V, D48M, T49L, G51R, N76V, Q84T, N115W, Y116H, I148Y, L149T	1.146	0.904	1.034	1.108
S46V, D48M, T49L, G51R, N76V, Q84V, V112L, N115F, Y116H, K118Q, I148Y	1.599	0.886	1.257	1.271
S46V, D48M, T49L, G51R, N76V, V112L, N115F, Y116H, I148Y	2.233	1.193	1.735	1.287
S46V, D48M, T49L, G51R, N76V, V112L, N115W, Y116H, K118Q, I148Y, L149F	1.649	1.026	1.370	1.204
S46V, D48M, T49L, G51R, N76V, V112L, N115W, Y116H, I148Y, L149F	2.150	1.111	1.652	1.301
S46V, D48M, T49L, G51R, N76V, V112L, N115W, Y116H, I148Y, L149M	1.810	1.166	1.525	1.187
S46V, D48M, T49L, G51R, N76V, V112L, N115W, Y116H, I148Y, L149T	1.657	1.059	1.395	1.188

As is apparent to one with skill in the art, various²⁵ modifications and variations of the above aspects and embodiments can be made without departing from the spirit and scope of this disclosure. Such modifications and variations are within the scope of this disclosure.

SEQUENCE LISTING

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 organism = Marinobacter hydrocarbonoclasticus

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PVNIRPADDE	GTGTQISFMI	ASLATDEADP	LNRLQQIKTS	TRRAKEHLQK	LPKSALTQYT	360
MLLMSPYILQ	LMSGLGGRMR	PVFNVNTISNV	PGPEGTLYYE	GARLEAMYPV	SLIAHGGALN	420
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 organism = Marinobacter hydrocarbonoclasticus

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 organism = Marinobacter hydrocarbonoclasticus

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KTDKEASVPA AVSQAMDALK LQADMAPRLW QAGNRLVHSV RHPEDGLTAP FTGPVSVLNH 240
RVTAQRRFAT QHYQLDRLKN LAHASGGSLN DIVLYLCGTA LRRFLAEQNN LPDTPPLTAGI 300
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cgggtgaata tccggccggc agacgacgag ggtacgggca cccagatcag ttttatgatt 960
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accgacgggg caaggagcga cctgcagaaa cttccaaaaa gtgccctgac ccagtacacc 1080
atgctgctga tgtcacccta cattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
ccattcttca acgtgaccat ttccaaactg cccggccccg aaggcacgct gtattatgaa 1200
ggagccccgg ttgagggcat gtatcgggta tcgctaactg ctcacggcgg cgccctgaac 1260
atcacctgcc tgaagctatgc gggatcgctg aatttcggtt ttaccggctg tcgggatacg 1320
ctgccgagca tgcagaaact ggcgggttat accggtgaag ctctggatga gctggaatcg 1380
ctgattctgc caccacaaga ggcgcgccga acccgcaagt aa 1422

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SEQ ID NO: 8      moltype = AA  length = 473
FEATURE          Location/Qualifiers
source           1..473
                 mol_type = protein
                 organism = Marinobacter hydrocarbonoclasticus

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SEQUENCE: 8
MKRLGTLNAS WLAVESDTP MHVGTLLQIFS LPEGAPETFL RDMVTRMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKVDKD IDLDYHVRHS ALPRPGGERE LGILVSRLLS NPLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRLVTTD PERCNMPPPW TVRPHQRRGV 180
KTDKEASVPA AVSQAMDALK LQADMAPRLW QAGNRLVHSV RHPEDGLTAP FTGPVSVLNH 240
RVTAQRRFAT QHYQLDRLKN LAHASGGSLN DIVLYLCGTA LRRFLAEQNN LPDTPPLTAGI 300
PVNIRPADDE GTGTQISFMI ASLATDEADP LNRLQQIKTS TRRAKEHLQK LPKSALTQYT 360
MLLMSPYILQ LMSGLGGRMR PFFNVTISNV PGPEGTLYYE GARLEAMYVP SLIAHGGALN 420
ITCLSYAGSL NFGFTGCRDT LPSMQKLAVY TGEALDELES LILPPKKRAR TRK 473

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SEQ ID NO: 9      moltype = DNA  length = 1422
FEATURE          Location/Qualifiers
source           1..1422
                 mol_type = unassigned DNA
                 organism = Marinobacter hydrocarbonoclasticus

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SEQUENCE: 9
atgaaacgtc tcggaaccct gaacgcctcc tggctggcgg ttgaatctga agacaccccg 60
atgcatgtgg gtacgcttca gattttctca ctgccggaag gcgcaccaga aaccttctcg 120
cgtgacatgg tcactcgaat gaaagaggcc ggcgatgtgg caccaccctg gggatacaaa 180
ctggcctggt ctggtttctc cgggcgcgtg atcgccccgg cctggaaagt cgataaggat 240
atcgatctgg attatcacgt cgggcactca gccctgcctc gcccgcggcg ggagcgcgaa 300
ctgggtattc tggatccccg actgcactct aacccctcgg atttttccc cctcttttgg 360
gaatgccacg ttattgaagg cctggagaat aaccgttttg ccttttacac caaaatgcac 420
cactcgatga ttacgagcat cagcggcgctg cgactgatgc agaggggtgct caccaccgat 480

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cccgaacgct gcaatatgcc accgccctgg acggtacgcc cacaccaacg ccgtggtgta 540
aaaaaccgaca aagaggccag cgtgcccgca gcggtttccc aggcaatgga cgccctgaag 600
ctccaggcag acatggcccc caggctgtgg caggcccgca atcgctcgtt gcattcgggt 660
cgacacccgg aagacggact gaccgcgccc ttcactggac cggtttcggg gctcaatcac 720
cgggttaccg cgcagcgacg ttttgccacc cagcattatc aactggaccg gctgaaaaac 780
ctggcccatg cttccggcgg ttccctgaac gacatcgctc tttacctgtg tggcaccgca 840
ttgcgggcgt ttcctggctga gcagaacaat ctgccagaca ccccgctgac ggctggtata 900
ccggtgaata tccggccggc agacgacgag ggtacgggca cccagatcag ttttatgatt 960
gcctcgctgg ccaccgacga agctgatccg ttgaaccgcc tgcaacagat caaaacctcg 1020
acccgacggg ccaaggagca cctgaggaaa cttccaaaaa gtgccctgac ccagtacacc 1080
atgctgctga tgtcacccta cattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
ccattcttca acgtgaccca ttccaacgtg cccggcccg aaggcacgct gtattatgaa 1200
ggagcccgcc ttgagggcat gtatccggta tcgctaactg ctcacggcgg cgccctgaac 1260
atcacctgcc tgagctatgc cggatcgctg aatttcgggt ttaccggctg tcgggatacg 1320
ctgccgagca tgcagaaact ggcggtttat accggtgaag ctctggatga gctggaatcg 1380
ctgattctgc caccacaaga gcgcgcccga acccgcaagt aa 1422

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SEQ ID NO: 10      moltype = AA length = 473
FEATURE           Location/Qualifiers
source            1..473
                  mol_type = protein
                  organism = Marinobacter hydrocarbonoclasticus

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SEQUENCE: 10
MKRLGTLNAS WLAVESDTP MHVGTLQIFS LPEGAPETFL RDMVTRMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKVDKD IDLDYHVRHS ALPRPGGERE LGILVSRlhs NPLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRLVLTDD PERCNMPPPW TVRPHQRRGV 180
KTDKEASVPA AVSQAMDALK LQADMAPRLW QAGNRLVHSV RHPEDGLTAP FTGPVSVLNH 240
RVTAQRRFAT QHYQLDLRKN LAHASGGSLN DIVLYLCGTA LRRFLAEQNN LPDTPLTAGI 300
PVNIRPADDE GTGTQISFMI ASLATDEADP LNRLQQIKTS TRRAKEHLRK LPKSALTQYT 360
MLLMSPYILQ LMSGLGGRMR PFFNVTISNV PGPEGTLYYE GARLEAMYVP SLIAHGALN 420
ITCLSYAGSL NFGFTGCRDT LPSMQKLAVY TGEALDELES LILPPKKRAR TRK 473

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SEQ ID NO: 11      moltype = DNA length = 1422
FEATURE           Location/Qualifiers
source            1..1422
                  mol_type = unassigned DNA
                  organism = Marinobacter hydrocarbonoclasticus

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SEQUENCE: 11
atgaaacgtc tcggaaccct gaacgcctcc tggctggcgg ttgaatctga agacaccccg 60
atgcatgtgg gtacgcttca gattttctca ctgccggaag gcgcaccaga aaccttctg 120
cgtgacatgg tcaactcgaat gaaagaggcc ggcgatgtgg caccacctg gggatacaaa 180
ctggcctggt ctggtttcct cggggcgctg atcgccccgg cctggaaaag cgataaggat 240
atcgatctgg attatcacgt ccgacactca gccctgcctc gccccggcgg ggagcgcgaa 300
ctgggtatct tgggtatccc actgcactct aacccctcgg atttttccc cctctcttgg 360
gaatgccacg ttattgaagg cctggagaat aaccggtttg ccttttacac caaaatgcac 420
cactcgatga ttgacggcat cagcggcgtg cgactgatgc agagggtgct caccaccgat 480
cccgaacgct gcaatatgcc accgccctgg acggtacgcc cacaccaacg ccgtggtgta 540
aaaaaccgaca aagaggccag caggcccgca gcggtttccc aggcaatgga cgccctgaag 600
ctccaggcag acatggcccc caggctgtgg caggcccgca atcgctcgtt gcattcgggt 660
cgacacccgg aagacggact gaccgcgccc ttcactggac cggtttcggg gctcaatcac 720
cgggttaccg cgcagcgacg ttttgccacc cagcattatc aactggaccg gctgaaaaac 780
ctggcccatg cttccggcgg ttccctgaac gacatcgctc tttacctgtg tggcaccgca 840
ttgcgggcgt ttcctggctga gcagaacaat ctgccagaca ccccgctgac ggctggtata 900
ccggtgaata tccggccggc agacgacgag ggtacgggca cccagatcag ttttatgatt 960
gcctcgctgg ccaccgacga agctgatccg ttgaaccgcc tgcaacagat caaaacctcg 1020
acccgacggg ccaaggagca cctgcagaaa cttccaaaaa gtgccctgac cgtgtacacc 1080
atgctgctga tgtcacccta cattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
ccattcttca acgtgaccca ttccaacgtg cccggcccg aaggcacgct gtattatgaa 1200
ggagcccgcc ttgagggcat gtatccggta tcgctaactg ctcacggcgg cgccctgaac 1260
atcacctgcc tgagctatgc cggatcgctg aatttcgggt ttaccggctg tcgggatacg 1320
ctgccgagcg gccagaaact ggcggtttat accggtgaag ctctggatga gctggaatcg 1380
ctgattctgc caccacaaga gcgcgcccga acccgcaagt aa 1422

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SEQ ID NO: 12      moltype = AA length = 473
FEATURE           Location/Qualifiers
source            1..473
                  mol_type = protein
                  organism = Marinobacter hydrocarbonoclasticus

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SEQUENCE: 12
MKRLGTLNAS WLAVESDTP MHVGTLQIFS LPEGAPETFL RDMVTRMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKVDKD IDLDYHVRHS ALPRPGGERE LGILVSRlhs NPLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRLVLTDD PERCNMPPPW TVRPHQRRGV 180
KTDKEASRPA AVSQAMDALK LQADMAPRLW QAGNRLVHSV RHPEDGLTAP FTGPVSVLNH 240
RVTAQRRFAT QHYQLDLRKN LAHASGGSLN DIVLYLCGTA LRRFLAEQNN LPDTPLTAGI 300
PVNIRPADDE GTGTQISFMI ASLATDEADP LNRLQQIKTS TRRAKEHLQK LPKSALTQYT 360
MLLMSPYILQ LMSGLGGRMR PFFNVTISNV PGPEGTLYYE GARLEAMYVP SLIAHGALN 420
ITCLSYAGSL NFGFTGCRDT LPSMQKLAVY TGEALDELES LILPPKKRAR TRK 473

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SEQ ID NO: 13      moltype = DNA length = 1422

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FEATURE
source Location/Qualifiers
1..1422
mol_type = unassigned DNA
organism = Marinobacter hydrocarbonoclasticus

SEQUENCE: 13

atgaaacgctc	tccggtaccc	ggacgcctcc	tggtctggcgg	ttgaaggtga	agacaccccc	60
atgcatgtgg	gtacgcttca	gattttctca	ctgccggaag	gcgcaccaga	aaccttctctg	120
cgtgacatgg	tcactcgaat	gaaagaggcc	ggcgtatgtg	caccaccctg	gggatacaaaa	180
ctggcctcgt	ctggtttctc	cgggcgcgtg	atcgcccccg	cctggaaaagt	cgataaggat	240
atcgatctgg	attatcacgt	ccggcactca	gccctgcctc	gccccggcgg	ggagcgcgaa	300
ctgggtattc	tggtatcccg	actgcactct	aacagtctgg	atttttcccc	ccctcttttg	360
gaatgccacg	ttattgaagg	cctggagaat	aaccgttttg	ccctttacac	caaaatgcac	420
cactcgatga	ttgacggcat	cagcggcgtg	cgactgatgc	agaggggtgt	caccaccgat	480
cccgaacgct	gcaatatgcc	accgccctgg	acgcgcgcgc	cacaccagcg	ccgttggtgca	540
aaaaccgaca	aagaggccag	cgtgcgggca	gcggtttccc	aggcaatgga	cgccctgaag	600
ctccaggcag	acatggcccc	caggctgtgg	caggccggca	atcgccctgt	gcattcggtt	660
cgacacccgg	aagacggact	gaccgcgcgc	ttcactggac	cggtttcggg	gctcaatcac	720
cgggttaccc	cgcagcgacg	ttttgccacc	cagcattatc	aactggaccg	gctgaaaaac	780
ctggcccatg	ttccggcggg	ttccttgaac	gacatcgttc	tttacctgtg	tggcaccgca	840
ttgcggcgct	ttctggctga	gcagaacaat	ctgccagaca	ccccgctgac	ggctggtata	900
ccggtgaata	tccggccggc	agacgacgag	ggtacgggca	cccagatcag	ttggatgatt	960
gcctcgctgg	ccaacgacga	agctgatccg	ttgaaccgcc	tgcaacagat	caaaacctcg	1020
acccgacggg	ccaaggagca	cctgcagaaa	cttcacaaaa	cgccctgac	ccagtacacc	1080
atgctgctga	tgtcacccca	cattctgcaa	ttgatgtcag	gtctcggggg	gaggatgcga	1140
ccagtcctta	acgtgacctat	ttccaacgtg	cccgcccgcg	aaggcacgct	gtattatgaa	1200
ggagcccgcc	ttgaggccat	gtatccgttg	tcgctaatac	ctcacggcgg	cgccctgaac	1260
atcacctgcc	tgagctatgc	cggatcgctg	aatttcgggt	ttaccgctg	tcgggatacg	1320
ctgccgggga	tgcagaaact	ggcggtttat	accggtgaag	ctctggatga	gctggaatcg	1380
ctgattctgc	cacccaagaa	gcgcgccccg	acccgcaagt	aa		1422

SEQ ID NO: 14 moltype = AA length = 473

FEATURE Location/Qualifiers
source 1..473
mol_type = protein
organism = Marinobacter hydrocarbonoclasticus

SEQUENCE: 14

MKRLGSLDAS	WLAVEGEDTP	MHVGTLLQIFS	LPEGAPETFL	RDMVTRMKEA	GDVAPPWGYK	60
LAWSGFLGRV	IAPAWKVDKD	IDLDYHVRHS	ALPRPGGERE	LGILVSRlhs	NSLDFSRPLW	120
ECHVIEGLN	NRFLALYTKM	HSMDIGISGV	RLMQRVLTDT	PERCNMPPPW	TRRPHQRRGA	180
KTDKEASVRA	AVSQAMDALK	LQADMAPRLW	QAGNRLVHSV	RHPEDGLTAP	FTGPVSVLNH	240
RVTQRRFAT	QHYQLDRlKN	LAHASGGSLN	DIVLYLCGTA	LRRFLAEQNN	LPDTPLTAGI	300
PVNIRPADDE	GTGTQISWMI	ASLATDEADP	LNRLQQIKTS	TRRAKEHLQK	LPKTALTQYT	360
MLLMSPIILQ	LMSGLGGRMR	PVFNVTISNV	PGPEGTLYYE	GARLEAMYPL	SLIAHGGALN	420
ITCLSYAGSL	NFGFTGCRDT	LPGMQKLAVY	TGEALDELES	LILPPKKRAR	TRK	473

SEQ ID NO: 15 moltype = DNA length = 1422

FEATURE Location/Qualifiers
source 1..1422
mol_type = unassigned DNA
organism = Marinobacter hydrocarbonoclasticus

SEQUENCE: 15

atgaaacgctc	tccggtaccc	ggacgcctcc	tggtctggcgg	ttgaaggtga	agacaccccc	60
atgcatgtgg	gtacgcttca	gattttctca	ctgccggaag	gcgcaccaga	aaccttctctg	120
cgtgacatgg	tcactcgaat	gaaagaggcc	ggcgtatgtg	caccaccctg	gggatacaaaa	180
ctggcctcgt	ctggtttctc	cgggcgcgtg	atcgcccccg	cctggaaaagt	cgatttcgat	240
atcgatctgg	attatcacgt	ccggcactca	gccctgcctc	gccccggcgg	ggagcgcgaa	300
ctgggtattc	tggtatcccg	actgcactct	aacagtctgg	atttttcccc	ccctcttttg	360
gaatgccacg	ttattgaagg	cctggagaat	aaccgttttg	ccctttacac	caaaatgcac	420
cactcgatga	ttgacggcat	cagcggcgtg	cgactgatgc	agaggggtgt	caccaccgat	480
cccgaacgct	gcaatatgcc	accgccctgg	acgcgcgcgc	cacaccagcg	ccgttggtgca	540
aaaaccgaca	aagaggccag	cgtgcgggca	gcggtttgtg	aggcaatgga	cgccctgaag	600
ctccaggcag	acatggcccc	caggctgtgg	caggccggca	atcgccctgt	gcattcggtt	660
cgacacccgg	aagacggact	gaccgcgcgc	ttcactggac	cggtttcggg	gctcaatcac	720
cgggttacca	ggcagcgacg	ttttgccacc	cagcattatc	aactggaccg	gctgaaaaac	780
ctggcccatg	ttccggcggg	ttccttgaac	gacatcgttc	tttacctgtg	tggcaccgca	840
ttgcggcgct	ttctggctga	gcagaacaat	ctgccagaca	ccccgctgac	ggctggtata	900
ccggtgaata	tccggccggc	agacgacgag	ggtacgggca	cccagatcag	ttggatgatt	960
gcctcgctgg	ccaacgacga	agctgatccg	ttgaaccgcc	tgcaacagat	caaaacctcg	1020
acccgacggg	ccaaggagca	cctgcagcac	cttcacaaaa	cgccctgac	ccagtacacc	1080
atgctgctga	tgtcacccca	cattctgcaa	ttgatgtcag	gtctcggggg	gaggatgcga	1140
ccagtcctta	acgtgacctat	ttccaacgtg	cccgcccgcg	aaggcacgct	gtattatgaa	1200
ggagcccgcc	ttgaggccat	gtatccgttg	tcgctaatac	ctcacggcgg	cgccctgaac	1260
atcacctgcc	tgagctatgc	cggatcgctg	aatttcgggt	ttaccgctg	tcgggatacg	1320
ctgccgggga	tgcagaaact	ggcggtttat	accggtgaag	ctctggatga	gctggaatcg	1380
ctgattctgc	cacccaagaa	gcgcgccccg	acccgcaagt	aa		1422

SEQ ID NO: 16 moltype = AA length = 473

FEATURE Location/Qualifiers
source 1..473

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mol_type = protein
organism = Marinobacter hydrocarbonoclasticus

SEQUENCE: 16
MKRLGSLDAS WLAVEGEDTP MHVGTLLQIFS LPEGAPETFL RDMVTRMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKVDFF IDLDYHVRHS ALPRPGGERE LGILVSRLLS NSLDFSRPLW 120
ECHVIEGLEN NRPALYTKMH HSMIDGISGV RLMQRLVLTDD PERCNMPPPW TRRPHQRRGA 180
KTDKEASVRA AVVQAMDALK LQADMAPRLW QAGNRLVHSV RHPEDGLTAP FTGPFVSVLNH 240
RVTRQRRFAT QHYQLDLRLN LAHASGGSLN DIVLYLCGTA LRRFLAEQNN LPDTPPLTAGI 300
PVNIRPADDE GTGTQISWMI ASLATDEADP LNRLQQIKTS TRRAKEHLQH LPKTALTQYT 360
MLLMSPIYLQ LMSGLGGRMR PVFNVITISNV PGPEGTLYYE GARLEAMYPL SLIAHGGALN 420
ITCLSYAGSL NFGFTGCRDT LPGMQKLAVY TGEALDELES LILPPKKRAR TRK 473

SEQ ID NO: 17      moltype = DNA length = 1422
FEATURE            Location/Qualifiers
source              1..1422
                    mol_type = unassigned DNA
                    organism = Marinobacter hydrocarbonoclasticus

SEQUENCE: 17
atgaaacgtc tcggatccct ggagcgcctcc tggctggcgg ttgaaggtga agacaccccg 60
atgcatgtgg gtacgcttca gattttctca ctgccggaag gcgcaccaga aaccttctcg 120
cgtgacatgg tcaactcgaat gaaagaggcc ggcatgtgg caccaccccg gggatacaaa 180
ctggcctggt ctggttttct cgggcgcgtg atcgccccgg cctggaaact ggataaggat 240
atcgatctgg attatcacgt ccggcactca gccctgcctc gccccggcgg ggagcgcgaa 300
ctgggtatcc tggatcccg actgcactct aacagtctgg atttttcccg cctctcttgg 360
gaatgccacg ttattgaagg cctggagaat aaccgttttg ccttttacac caaaatgcac 420
cactcgatga ttgacggcat cagcggcgtg cgactgatgc agaggggtgt caccaccgat 480
cccgaacgct gcaatatgcc accgcctcgg acgcgcgcgc cacaccagcg ccgtggtgca 540
aaaaccgaca aagaggccag cgtgcgggca gcggtttccc aggcaatgga cgccctgaag 600
ctccaggcag acatggcccc caggctgtgg caggccggca atcgctcgtt gcattcgggt 660
cgacacccgg aagacggaact gaccgcgcgc ttcactggac cggtttcggg gctcaatcac 720
cgggttacgg cgcagcgacg ttttgccacc cagcattatc aactggaccg gctgaggaac 780
ctggcccatg ctcccgcggg ttctctgaac gacatcgctt tttacctgtg tggcaccgca 840
ttgcggcgct ttctggctga gcagaacaat ctgccagaca ccccgctgac ggctggtata 900
ccggtgaata tccggccggc agacgacgag ggtacgggca cccagatcgg gtggatgatt 960
gctcgcgtgg ccacgcgaca agctgatccg ttgaaccgcc tgcaacagat caaaacctcg 1020
acccgacggg ccaaggagca cctgcagaaa cttccaaaaa cgccctgac ccagtacacc 1080
cgctgtctga tgtcacccca cattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
ccagtcctca acgtgacctt ttccaacgtg cccggcccgg aaggcacgct gtattatgaa 1200
ggagcccgcg ttgagcccat gtatccgttg tcgctaatac ctcacggcgg cgccctgaac 1260
atcacctgcc ttgactatgc cggatcgctg aatttcggtt ttaccgctg tcgggatacg 1320
ctgccgggga tgcagaaact ggcggtttat accggtgaag ctctggatga gctggaatcg 1380
ctgattctgc caccacaaga gcgcgccccg acccgcaagt aa 1422

SEQ ID NO: 18      moltype = AA length = 473
FEATURE            Location/Qualifiers
source              1..473
                    mol_type = protein
                    organism = Marinobacter hydrocarbonoclasticus

SEQUENCE: 18
MKRLGSLDAS WLAVEGEDTP MHVGTLLQIFS LPEGAPETFL RDMVTRMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKLDKD IDLDYHVRHS ALPRPGGERE LGILVSRLLS NSLDFSRPLW 120
ECHVIEGLEN NRPALYTKMH HSMIDGISGV RLMQRLVLTDD PERCNMPPPW TRRPHQRRGA 180
KTDKEASVRA AVSQAMDALK LQADMAPRLW QAGNRLVHSV RHPEDGLTAP FTGPFVSVLNH 240
RVTAQRRFAT QHYQLDLRLN LAHASGGSLN DIVLYLCGTA LRRFLAEQNN LPDTPPLTAGI 300
PVNIRPADDE GTGTQIGWMI ASLATDEADP LNRLQQIKTS TRRAKEHLQH LPKTALTQYT 360
RLLMSPYILQ LMSGLGGRMR PVFNVITISNV PGPEGTLYYE GARLEAMYPL SLIAHGGALN 420
ITCLSYAGSL NFGFTGCRDT LPGMQKLAVY TGEALDELES LILPPKKRAR TRK 473

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SEQ ID NO: 19      moltype = DNA length = 1422
FEATURE            Location/Qualifiers
source              1..1422
                    mol_type = unassigned DNA
                    organism = Marinobacter hydrocarbonoclasticus

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SEQUENCE: 19
atgaaacgtc tcggatccct ggagcgcctcc tggctggcgg ttgaaggtga agacaccccg 60
atgcatgtgg gtacgcttca gattttctca ctgccggaag gcgcaccaga aaccttctcg 120
cgtgacatgg tcaactcgaat gaaagaggcc ggcatgtgg caccaccccg gggatacaaa 180
ctggcctggt ctggttttct cgggcgcgtg atcgccccgg cctggaaact cgataaggat 240
atcgatctgg attatcacgt ccggcactca gccctgcctc gccccggcgg ggagcgcgaa 300
ctgggtatcc tggatcccg actgcactct aacagtctgg atttttcccg cctctcttgg 360
gaatgccacg ttattgaagg cctggagaat aaccgttttg ccttttacac caaaatgcac 420
cactcgatga ttgacggcat cagcggcgtg cgactgatgc agaggggtgt caccaccgat 480
cccgaacgct gcaatatgcc accgcctcgg acgcgcgcgc cacaccagcg ccgtggtgca 540
aaaaccgaca aagaggccag cgtgcgggca gcggtttccc aggcaatgga cgccctgaag 600
ctccaggcag acatggcccc caggctgtgg caggccggca atcgctcgtt gcattcgggt 660
cgacacccgg aagacggaact gaccgcgcgc ttcactggac cggtttcggg gctcaatcac 720
cgggttacgg cgggcgcagc ttttgccacc cagcattatc aactggaccg gctgaaaaac 780
ctggcccatg ctcccgcggg ttgggtgaac gacatcgctt tttacctgtg tggcaccgca 840
ttgcggcgct ttctggctga gcagaacaat ctgccagaca ccccgctgac ggctggtata 900

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ccggtgaata tccggccggc agacgacgag gtcacgggca cccagatcag ttggatgatt 960
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acccgacggg ccaaggagca cctgcagaaa cttccaaaaa cgccctgac ccagtacacc 1080
atgctgctga tgtcacctcg gattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
ccagtcttca acgtgacctt ttccaacgtg cccggcccgg aaggcacgct gtattatgaa 1200
ggagcccgcc ttgaggccat gtatccgttg tcgctaatac ctcacggcgg cgccctgaac 1260
atcacctgcc tgagctatgc cggatcgctg aatttcggtt ttaccggctg tcgggatacg 1320
ctgccgggga tgcagaaact ggcggtttat accggtgaag ctctggatga gctggaatcg 1380
ctgattctgc caccacaaga ggcgcgccga acccgcaagt aa 1422

```

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SEQ ID NO: 20      moltype = AA  length = 473
FEATURE           Location/Qualifiers
source            1..473
                  mol_type = protein
                  organism = Marinobacter hydrocarbonoclasticus

```

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SEQUENCE: 20
MKRLGSLDAS WLAVEGEDTP MHVGTLLQIFS LPEGAPETFL RDMVTRMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKVDKD IDLDYHVRHS ALPRPGGERE LGILVSRlhs NSLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRVLTTD PERCNMPPPW TRRPHQRRGA 180
KTDKEASVRA AVSQAMDALK LQADMAPRLW QAGNRLVHSV RHPEDGLTAP FTGPVSVLNH 240
RVTAGRRFAT QHYQLDRLKN LAHASGGGLN DIVLYLCGTA LRRFLAEQNN LPDTPPLTAGI 300
PVNIRPADDE VTGTQISWMI CSLATDEADP LNRLQQIKTS TRRAKEHLQK LPKTALTQYT 360
MLLMSPWILQ LMSGLGGRMR PVFNVITISNV PGPEGTLTYE GARLEAMYPL SLIAHGGALN 420
ITCLSYAGSL NFGFTGCRDT LPGMQKLAVY TGEALDELES LILPPKKRAR TRK 473

```

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SEQ ID NO: 21      moltype = DNA  length = 1422
FEATURE           Location/Qualifiers
source            1..1422
                  mol_type = unassigned DNA
                  organism = Marinobacter hydrocarbonoclasticus

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SEQUENCE: 21
atgaaacgtc tcggaacctc ggacgcctcc tggctggcgg ttgaaggtga agacaccccg 60
atgcatgtgg gtacgcttca gattttctca ctgccggaag gcgcaccaga aaccttctcg 120
cgtgacatgg tcactcgaaat gaaagaggcc ggcgatgtgg caccacctcg gggatacaaa 180
ctggcctggg ctgggttctc cgggcgcgtg atcgcccgcg cctggaaagt cgataaggat 240
atcgatctgg attatcacgt ccggcactca gccctgcctc gccccggcgg ggagcgcgaa 300
ctgggtattc tggatatccg actgcactct aacagtctgg atttttcccg cctcttttgg 360
gaatgccacg ttattgaagg cctggagaat aaccgttttg ccctttacac caaaatgcac 420
cactcgatga ttgacggcat cagcggcgctg cgaactgatgc agaggggcct caccaccgat 480
cccgaaacgt gcaatatgtc accgccctgg acgcgccgcc cacaccagcg ccgtgggtgca 540
aaaaccgaca aagaggccag cgtgcgggca gcggtttccc aggcgaatgga cgccctgaag 600
ctccaggcag acatggcccc caggctgtgg caggccggca atcgccctgg gcattcgggt 660
cgacacccgg aagacggact gaccgcgccc ttcaactggac cggtttcggt gctcaatcac 720
ggggttaccc gcgacgagc ttttgccacc cagcattatc aactggaccg gctgaaaaac 780
ctggcccatg cttccggcgg ttcttgtaac gacatcgctt tttacctgtg tggcaccgca 840
ttgcggcgct ttctggctga gcagaacaat ctgccagaca ccccgctgac ggctgggtata 900
ccggtgaata tccggccggc agacgacgag ggtacgggca cccagatcag ttggatgatt 960
gcctcgctgg ccacgcagca agctgatccg ttgaaccgcc tgcaacagat caaaacctcg 1020
acccgacggg ccaaggagca cctggcgaaa cttccaaaaa cgccctgac ccagtacacc 1080
atgctgctga tgtcacctca cattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
cattcttca acgtgacctt ttccaacgtg cccggcccgg aaggcacgct gtattatgaa 1200
ggagcccgcc ttgaggccat gtatccgttg tcgctaatac ctcacggcgg cgccctgaac 1260
atcacctgcc tgagctatgc cggatcgctg aatttcggtt ttaccggctg tcgggatacg 1320
ctgccgggga tgcagaaact ggcggtttat accggtgaag ctctggatga gctggaatcg 1380
ctgattctgc caccacaaga ggcgcgccga acccgcaagt aa 1422

```

```

SEQ ID NO: 22      moltype = AA  length = 473
FEATURE           Location/Qualifiers
source            1..473
                  mol_type = protein
                  organism = Marinobacter hydrocarbonoclasticus

```

```

SEQUENCE: 22
MKRLGTLTLDAS WLAVEGEDTP MHVGTLLQIFS LPEGAPETFL RDMVTRMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKVDKD IDLDYHVRHS ALPRPGGERE LGILVSRlhs NSLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRGLTTD PERCNMSPPW TRRPHQRRGA 180
KTDKEASVRA AVSQAMDALK LQADMAPRLW QAGNRLVHSV RHPEDGLTAP FTGPVSVLNH 240
RVTAQRFFAT QHYQLDRLKN LAHASGGSLN DIVLYLCGTA LRRFLAEQNN LPDTPPLTAGI 300
PVNIRPADDE GTGTQISWMI ASLATDEADP LNRLQQIKTS TRRAKEHLAK LPKTALTQYT 360
MLLMSPYILQ LMSGLGGRMR PFFNVITISNV PGPEGTLTYE GARLEAMYPL SLIAHGGALN 420
ITCLSYAGSL NFGFTGCRDT LPGMQKLAVY TGEALDELES LILPPKKRAR TRK 473

```

```

SEQ ID NO: 23      moltype = DNA  length = 1422
FEATURE           Location/Qualifiers
source            1..1422
                  mol_type = unassigned DNA
                  organism = Marinobacter hydrocarbonoclasticus

```

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SEQUENCE: 23
atgaaacgtc tcggaacctc ggacgcctcc tggctggcgg ttgaaggtga agacaccccg 60
atgcatgtgg gtacgcttca gattttctca ctgccggaag gcgcaccaga aaccttctcg 120

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cgtgacatgg tcaactcgaat gaaagaggcc ggcgatgtgg caccaccctg gggatacaaa 180
ctggcctggg ctggttttct cgggcgcgtg atcgcccccg cctggaaagt cgcgaaggat 240
atcgatctgg attatcacgt ccggcactca gccctgcctc gccccggcgg ggagcgcgaa 300
ctgggtatcc tggatccccg actgcactct aacagtctgg atttttcccg cctcttttgg 360
gaatgccacg ttattgaagg cctggagaat aaccgttttg ccttttacac caaaatgcac 420
cactcgatga ttgacggcat cagcggcgtg cgaactgatgc agagggtgct caccaccgat 480
cccgaacgct gcaatatgcc accgcctcgg acgcgccgcc cacaccagcg cctgggtgca 540
aaaaccgaca aagaggccag cgtgcgggca gcggtttccc aggcaatgga cgccctgaag 600
ctccaggcag acatggcccc caggctgtgg caggccggca atcgccctgg gattcgggtt 660
cgacaccggg aagacggact gaccgcgcc ttcactggac cggtttcggg gctcaatcac 720
cgggttaccg cgcagcgacg ttttgccacc cagcattatc aactggaccg gctgaaaaac 780
ctggcccatg cttccggcgg ttccctgaac gacatcgctt tttacctgtg tggcaccgca 840
ttgcggcgct ttctggctga gcagacaat ctgccagaca ccccgctgac ggctggtata 900
ccgggtgaata tccggccggc agacgacgag ggtacgggca gtcagatcag ttggatgatt 960
gcctcgctgg ccaccgacga agctgatccg ttgaaccgcc tgcaacagat caaaacctcg 1020
acccgacggg ccaaggagca cctgcagaaa cttccaaaaa cggccctgac ccagtacacc 1080
atgctgctga tgtcacccta cattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
ccattcttca acgtgaccat ttccaacgtg cccggcccgg aaggcacgct gtattatgaa 1200
ggagcccgcc ttgaggccat gtatccgttg tcgctaactc ctcacggcgg cgccctgaac 1260
gtgacctgcc tgaactatgc cggatcgctg aatttcgggt ttaccggctg tcgggatacg 1320
ctgccgggga tgcagaaact ggcggtttat accggtgaag ctctggatga gctggaatcg 1380
ctgattctgc caccacaaga gcgcgccgca acccgcaagt aa 1422

```

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SEQ ID NO: 24      moltype = AA length = 473
FEATURE
source            Location/Qualifiers
                  1..473
                  mol_type = protein
                  organism = Marinobacter hydrocarbonoclasticus

```

```

SEQUENCE: 24
MKRLGTLTLDAS WLAVEGEDTP MHVGTLLQIFS LPEGAPETFS RDMVTRMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKVAKD IDLDYHVRHS ALPRPGGERE LGILVSRlhs NSLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRVLTTD PERCNMPPPW TRRPHQRRGA 180
KTDKEASVRA AVSQAMDALK LQADMAPRLW QAGNRLVHSV RHPEDGLTAP FTGPVSVLNH 240
RVTAQRFRFAT QHYQLDLRLN LAHASGGS LN DIVLYLCGTA LRRFLAEQNN LPDPTPLTAGI 300
PVNIRPADDE GTGSQISWMI ASLATDEADP LNRLQQIKTS TRRAKEHLAK LPKTALTQYT 360
MLLMSPYLLQ LMSGLGGRMR PPFNVTISNV PGPEGTLYYE GARLEAMYPL SLIAHGALN 420
VTCLSYAGSL NPGFTGCRDT LPGMQKLAVY TGEALDELES LILPPKKRAR TRK 473

```

```

SEQ ID NO: 25      moltype = DNA length = 1422
FEATURE
source            Location/Qualifiers
                  1..1422
                  mol_type = unassigned DNA
                  organism = Marinobacter hydrocarbonoclasticus

```

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SEQUENCE: 25
atgaaacgtc tcggatccct ggacgcctcc tggctggcgg ttgaaggatga agacaccccg 60
atgcatgtgg gttggtctca gattttctca ctgccggaag gcgcaccaga aaccttctcg 120
cgtgacatgg tcttccgaat gaaagaggcc ggcgatgtgg caccaccctg gggatacaaa 180
ctggcctggg ctggttttct cgggcgcgtg atcgcccccg cctggaaagt cgataaggat 240
atcgatctgg attatcacgt ccggcactca gccctgcctc gccccggcgg ggagcgcgaa 300
ctgggtatcc tggatccccg actgcactct aacagtctgg atttttcccg cctcttttgg 360
gaatgccacg ttattgaagg cctggagaat aaccgttttg ccttttacac caaaatgcac 420
cactcgatga ttgacggcct gacgcggcgt cgaactgatgc agagggtgct caccaccgat 480
cccgaacgct gcaatatgcc accgcctcgg acgcgccgcc cacaccagcg cctgggtgca 540
aaaaccgaca aagaggccag cgtgcgggca gcggtttccc aggcaatgga cgccctgaag 600
ctccaggcag acatggcccc caggctgtgg caggccggca atcgccctgg gattcgggtt 660
cgacaccggg aagacggact gaccgcgcc ttcactggac cggtttcggg gctcaatcac 720
cgggttaccg cgcagcgacg ttttgccacc cagcattatc aactggaccg gctgaaaaac 780
ctggcccatg cttccggcgg ttccctgaac gacatcgctt tttacctgtg tggcaccgca 840
ttgcggcgct ttctggctga gcagacaat ctgccagaca ccccgctgac ggctggtata 900
ccgggtgaata tccggccggc aaacgacgag ggtacgggca cccagatcag ttggatgatt 960
gcctcgctgg ccaccgacga agctgatccg ttgaaccgcc tgcaacagat caaaacctcg 1020
acccgacggg ccaaggagca cctgcagaaa cttccaaaaa cggccctgac ccagtacacc 1080
atgctgctga tgtcacccta cattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
ccagtcttca nrfalytkmh ttccaacgtg cccggcccgg aaggcacgct gtattatgaa 1200
ggagcccgcc ttgaggccat gtatccgttg tcgctaactc ctcacggcgg cgccctgaac 1260
atcacctgcc tgaactatgc cggatcgctg aatttcgggt ttaccggctg tcgggatacg 1320
ctgccgggga tgcagaaact ggcggtttat accggtgaag ctctggatga gctggaatcg 1380
ctgattctgc caccacaaga gcgcgccgca acccgcaagt aa 1422

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SEQ ID NO: 26      moltype = AA length = 473
FEATURE
source            Location/Qualifiers
                  1..473
                  mol_type = protein
                  organism = Marinobacter hydrocarbonoclasticus

```

```

SEQUENCE: 26
MKRLGSLDAS WLAVEGEDTP MHVGLWLQIFS LPEGAPETFL RDMVFRMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKVDKD IDLDYHVRHS ALPRPGGERE LGILVSRlhs NSLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGLSGV RLMQRVLTTD PERCNMPPPW TRRPHQRRGA 180
KTDKEASVRA AVSQAMDALK LQADMAPRLW QAGNRLVHSV RHPEDGLTAP FTGPVSVLNH 240

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RVTAQRFFAT QHYQLDRLKN LAHASGGSLN DIVLYLCGTA LRRFLAEQNN LPDTPLTAGI 300
 PVNIRPANDE GTGTQISWMI ASLATDEADP LNRLQQIKTS TRRAKEHLQK LPKTALTQYT 360
 MLLMSPYILQ LMSGLGGRMR PVFNTISNV PGPEGTLYYE GARLEAMYPL SLIAHGGALN 420
 ITCLSYAGSL NFGFTGCRDT LPGMQKLAVY TGEALDELES LILPPKKRAR TRK 473

SEQ ID NO: 27 moltype = DNA length = 24
 FEATURE Location/Qualifiers
 misc_feature 1..24
 note = Description of Artificial Sequence: Synthetic
 oligonucleotide
 source 1..24
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 27
 atgaaacgctc tcggaaccct ggac 24

SEQ ID NO: 28 moltype = DNA length = 1446
 FEATURE Location/Qualifiers
 misc_feature 1..1446
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..1446
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 28
 atgaaacgctc tcggaaccct ggacatgaaa cgtctcggaa cctcggacgc ctccctggctg 60
 gcggttgaat ctgaagacac cccgatgcat gtgggtacgc ttcagatttt ctcaactgccg 120
 gaaggcgcac cagaaacctt cctgcgtgac atggtcactc gaatgaaaga ggcggcgcat 180
 gtggcaccac cctggggata caaactggcc tggctctggt tcctcgggcg cgtgatcgcc 240
 ccggcctgga aagtcgataa ggatctgatc ctggattatc acgtccggca ctccagccctg 300
 cctcgccccc gcggggagcg cgaactgggt attctgggat cccgactgca ctctaaccct 360
 ctggattttt cccgcctctc ttgggaatgc cagcttattg aaggcctgga gaataaccgt 420
 ttgccccttt acacaaaaat gcaccactcg atgattgacg gcacacagcg cgtgacgactg 480
 atgcagagggg tgctcaccac cgatcccgaa cgctgcaata tgccaccgcc ctggacggta 540
 cgcccacacc agcgcctgtg tgcaaaaaac gacaaagagg ccagcgtgcc cgcagcggtt 600
 tcccaggcaa tggacgcctt gaagctccag gcagacatgg ccccagggtt gtggcaggcc 660
 ggcaatcgcc tgggtgattc ggttcgacac ccggaagacg gactgaccgc gcccttcaact 720
 ggaccgggtt cgggtgctcaa tcaccgggtt accgcgcagc gacgttttgc caccacagcat 780
 tatcaactgg accggctgaa aaacctggcc catgcttccg cgggttccct gaacgacatc 840
 gttctttacc tctgtggcac cgattgcgg cgctttctgg ctgagcagaa caatctgcc 900
 gacacccccc tgacggctgg tataccgggt aatatccggc cggcagacga cgagggtacg 960
 ggcacccaga tcagttttat gattgcctcg ctggccaccg acgaagtga tccgttgaac 1020
 cgctgcaac agatcaaac cgcacccga cgggccaagg agcacctgca gaaacttcca 1080
 aaaaagtgcct tgaccagta caccatgctg ctgatgtcac cctacattct gcaattgatg 1140
 tcaggtctcg gggggaggat gcgaccagtc ttcaacgtga ccatttccaa cgtgcccggc 1200
 ccggaaggca cgcgtgatta tgaaggagcc cggcttgagg ccatgatatc ggtatcgcta 1260
 atcgctcacg cggcgccctt gaacatcacc tgcctgagct atgcccggatc gctgaatttc 1320
 ggttttaccg gctgtcgga tacgtgccc agcatgcaga aactggcggg ttataccggt 1380
 gaagctctgg atgagctgga atcgctgatt ctgccacca agaagcgccg ccgaaccgcg 1440
 aagtaa 1446

SEQ ID NO: 29 moltype = AA length = 481
 FEATURE Location/Qualifiers
 REGION 1..481
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..481
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 29
 MKRLGTLDMK RLGLDASWL AVESEDTPMH VGTLQIFSLP EGAPETFLRD MVTRMKEAGD 60
 VAPPWGYKLA WSGFLGRVIA PAWKVDKID LDYHVRHSAL PRPGERELG ILVSRHLSNP 120
 LDFSRLWEC HVIENLENNR FALYTKMHHS MIDGISGVRL MQRVLTTDPE RCNMPPPWTV 180
 RPHQRRGAKT DKEASVPAAV SQAMDALKLQ ADMAPRLWQA GNRLVHVSRL PEDGLTAPFT 240
 GPVSVLNRHV TAQRFFATQH YQLDRLKNLA HASGGSLNDI VLYLCGTALR FLAEQNNLP 300
 DTPLTAGIPV NIRPADDEGT GTQISFMIA LATDEADPLN RLQIKTSTR RAKEHLQKLP 360
 KSALTQYTML LMSPYILQLM SGLGGRMRPV FNVITISNVP PEGTLYYEGA RLEAMYPSVL 420
 IAHGGALNIT CLSYAGSLNF GFTGCRDTLP SMQKLAVYTG EALDELES LI PPKKRARTR 480
 K 481

SEQ ID NO: 30 moltype = DNA length = 549
 FEATURE Location/Qualifiers
 misc_feature 1..549
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..549
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 30

-continued

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atgggcaaca ccttctctgt cgtcgccgat tcactctccg caggetacca aatgctgtca 60
gaacaaagct ggcgggtatt gctgcaacct gccctgaac agcagggtca tgagatcact 120
gtggtaaatg caagcattgt gggatgctgt acgcgtccgg gtttggtctg tctgcccaacc 180
ctgttacaac aacacaagcc ggcgtatgtt atcatcgaat tgggtgcggt cgatggcttg 240
cgcggttttc cggcgggcac gattcgtaat aacctgagcc agatgattac cgagattcag 300
aacgctgacg cgaagccgat gctgggttcag atcaaaactcc cgccgtggca cggtaaacgc 360
tacagcgaca tgttcagcag catttaccg cagctgtcta aggaactggc gacgccactg 420
ctgccgtttc ttctggagca gatttatatg aagcaagaat ggatgatgaa tgacggtctg 480
catccgaaaa gcgatgcgca gccgtggatc gcagagtata tggccgagaa tatcgcgccg 540
tatttgttaa                                     549

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```

SEQ ID NO: 31      moltype = AA length = 182
FEATURE           Location/Qualifiers
REGION            1..182
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..182
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 31
MGNTLLVVGD SLSAGYQMR EQSWPVLLQP ALKQQGHEIT VVNASIVGML TRRGLARLPT 60
LLQQHKPAYV IIELGAVDGL RGFPAGTIRN NLSQMITEIQ NADAKPMLVQ IKLPPWHGKR 120
YSDMFSSIYP QLSKELATPL LPFFLEQIYM KQEWMMNDGL HPKSDAQFPI AEYMAENIAP 180
YL                                     182

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SEQ ID NO: 32      moltype = DNA length = 1422
FEATURE           Location/Qualifiers
source            1..1422
                  mol_type = unassigned DNA
                  organism = Marinobacter hydrocarbonoclasticus

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SEQUENCE: 32
atgaacgctc tcggatccct ggacgcctcc tggctggcgg ttgaaggtga agacaccccg 60
atgcatgtgg gtacgcttca gattttctca ctgccggaag gcgcaccaga aaccttctcg 120
cgtgacatgg tcaactcgaat gaaagaggcc ggcgatgtgg caccaccctg gggatacaaa 180
ctggcctggt ctggtttctc cgggcgcgtg atcgccccgg cctggaaagt cgatttcgat 240
atcgatctgg attatcacgt ccggcactca gccctgcctc gccccggcgg ggagcgcgaa 300
ctgggtattc ttgtatcccg actgcactct aacagtctgg atttttcccg cctcttttgg 360
gaatgccacg ttattgaagg cctggagaat aaccgttttg ccctttacac caaaatgcac 420
cactcgatga ttgacggcat cagcggcggtg cgaactgatgc agaggggtgct caccaccgat 480
cccgaaacgt gcaatatgcc accgccctgg acgcgccgcc cacaccagcg ccgtgggtgca 540
aaaaccgaca aagaggccag cgtgcgggca gcggttgtgc aggcaatgga cgcctcgaag 600
ctccaggcag acatggcccc cgcgctgtgg caggccggca atcgccctggt gcattcgggt 660
cgacaccggg aagacggact gaccgcgccc ttcaactggac cggtttcggt gctcaatcac 720
cgggttacca gcagcgcgct ttttgcaacc caacactatc agctggagcg catcaagcag 780
gtggcgacgg cgagcaatgg ctccctgaat gacatcgtct tgatctctgt cggcaccgca 840
ttgctgcgct ttctggttga acaagacggt ttgccggata cccactgac cgcaggcatc 900
cgggtgaata tccggccggc agacgacgag ggtacgggca ccagatcag ttggatgatt 960
gcctcgctgg ccaccgacga agctgatccg ttgaaccgcc tgcaacagat caaacctcgt 1020
acccgacggg ccaagagaca cctgaggaaa cttccaaaaa cgccctgac ccagtacacc 1080
atgctgctga tgtcacccca cattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
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ggagccccgg ttgagggcat gatacggttg tcgctaactg ctcacggcgg cgccctgaac 1260
atcacctgcc tgagctatgc cggatcgctg aatttcggtt ttaccggctg tcgggatacg 1320
ctgccgggga tgcagaaact ggcggtttat accggtgaag ctctggatga gctggaatcg 1380
ctgattctgc cacccaagaa gcgcgccccg acccgcaagt aa 1422

```

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SEQ ID NO: 33      moltype = AA length = 473
FEATURE           Location/Qualifiers
source            1..473
                  mol_type = protein
                  organism = Marinobacter hydrocarbonoclasticus

```

```

SEQUENCE: 33
MKRLGSLDAS WLAVEGEDTP MHVGTLLQIFS LPEGAPETFL RDMVTRMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKVDVF IDLDYHVRHS ALPRPGGERE LGILVSRlhs NSLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRLVLTG PERCNMPPPW TRRPQRRGA 180
KTDKEASVRA AVVQAMDALK LQADMAPALW QAGNRLVHSV RHPEDGLTAP FTGPVSVLNH 240
RVTRQRRFAT QHYQLERIKQ VAQASNGSLN DIVLYLCGTA LRRFLVEQDG LPDTPPLTAGI 300
PVNIRPADDE GTGTQISWMI ASLATDEADP LNRLQQIKTS TRRAKEHLRK LPKTALTQYT 360
MLLMSPYILQ LMSGLGGRMR PFFNVTISNV PGPEGTLYYE GARLEAMYPL SLIAHGALN 420
ITCLSYAGSL NFGFTGCRDT LPGMQKLAVY TGEALDELES LILPPKKRAR TRK 473

```

```

SEQ ID NO: 34      moltype = DNA length = 1422
FEATURE           Location/Qualifiers
source            1..1422
                  mol_type = unassigned DNA
                  organism = Marinobacter hydrocarbonoclasticus

```

```

SEQUENCE: 34
atgaacgctc tcggatccct ggacgcctcc tggctggcgg ttgaaggtga agacaccccg 60
atgcatgtgg gtacgcttca gattttctca ctgccggaag gcgcaccaga aaccttctcg 120

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cgtgacatgg tcaactcgaat gaaagaggcc ggcgatgtgg caccaccctg gggatacaaa 180
ctggcctggg ctggttttct cgggcgcgtg atcgcccccg cctggaaagt cgatttcgat 240
atcgatctgg attatcacgt ccggcactca gccctgcctc gccccggcgg ggagcgcgaa 300
ctgggtatcc tggatccccg actgcactct aacagtctgg atttttcccg cctcttttgg 360
gaatgccacg ttattgaagg cctggagaat aaccgttttg ccttttacac caaaatgcac 420
cactcgatga ttgacggcat cagcggcgtg cgaactgatgc agagggtgct caccaccgat 480
cccgaacgct gcaatatgcc accgcctcgg acgcgccgcc cacaccagcg cctgggtgca 540
aaaaccgaca aagaggccag cgtgcgggca gcggttctgc aggcaatgga cgccctgaag 600
ctccaggcag acatggcccc cgctcgttgg caggccggca atcgccctgg gattcgggtt 660
cgacaccggg aagacggact gaccgcgcc ttcactggac cgttctcgtt gctcaatcac 720
cgggttacca ggcagcgccg ttttgcaacc caacactatc agctggagcg catcaagcag 780
gtggcgacag cgagcaatgg ctccctgaat gacatcgtct tgtatctgtg cggcaccgca 840
ttgctgcgct ttctgggtga acaagacggg ttgcccggata cccactgac cgcaggcact 900
ccgggtgaata tccggccggc agacgcagag ggtacgggca cccagatcag ttggatgatt 960
gcctcgctgg ccaccgcaga agctgatccg ttgaaccgcc tgcaacagat caaaacctcg 1020
acccgacggg ccaaggagca cctgcagcac cttccaaaaa cggccctgac ccagtacacc 1080
atgctgctga tgtcacccta cattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
ccattcttca acgtgacctt ttccaacgtg cccggcccgg aaggcacgct gtattatgaa 1200
ggagcccgcc ttgaggccat gtatccgttg tcgctaactc ctcacggcgg cgccctgaac 1260
atcacctgcc tgagctatgc cggatcgctg aatttcggtt ttaccggctg tcgggatacg 1320
ctgccgggga tgcagaaact ggcggtttat accggtgaag ctctggatga gctggaatcg 1380
ctgattctgc cacccaagaa gcgcgccgca acccgcaagt aa 1422

```

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SEQ ID NO: 35      moltype = AA length = 473
FEATURE
source            Location/Qualifiers
                  1..473
                  mol_type = protein
                  organism = Marinobacter hydrocarbonoclasticus

```

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SEQUENCE: 35
MKRLGSLDAS WLAVEGEDTP MHVGTLLQIFS LPEGAPETFL RDMVTRMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKVDFF IDLDYHVRHS ALPRPGGERE LGILVSRLLS NSLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRLVLTDD PERCNMPPPW TRRPHQRRGA 180
KTDKEASVRA AVVQAMDALK LQADMAPALW QAGNRLVHSV RHPEDGLTAP FTGPVSVLNH 240
RVTRQRRFAT QHYQLERIKQ VAQASNGSLN DIVLYLCGTA LRRFLVEQDG LPDPTPLTAGI 300
PVNIRPADDE GTGTQISWMI ASLATDEADP LNRLQQIKTS TRRAKEHLQH LPKTALTQYT 360
MLLMSPYLLQ LMSGLGGRMR PFFNVTISNV PGPEGTLYYE GARLEAMYPL SLIAHGALN 420
ITCLSYAGSL NFGFTGCRDT LPGMQKLAVY TGEALDELES LILPPKKRAR TRK 473

```

```

SEQ ID NO: 36      moltype = DNA length = 1422
FEATURE
source            Location/Qualifiers
                  1..1422
                  mol_type = unassigned DNA
                  organism = Marinobacter hydrocarbonoclasticus

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```

SEQUENCE: 36
atgaaacgtc tcggaccctc ggacgcctcc tggctggcgg ttgaaggatga agacaccccg 60
atgcatgtgg gtacgcttcc gattttctca ctgccggaag acgcaccaga aaccttctcg 120
cgtgacatgg tcaagcgaat gaaagaggcc ggcgatgtgg caccaccctg gggattaaaa 180
ctggcctggg ctggttttct cgggcgcgtg atcgcccccg cctggaaagt cgataaggat 240
atcgatctgg attatcacgt ccggcactca gccctgcctc gccccggcgg ggagcgcgaa 300
ctgggtatcc tggatccccg actgcactct aacagtctgg atttttcccg cctcttttgg 360
gaatgccacg ttattgaagg cctggagaat aaccgttttg ccttttacac caaaatgcac 420
cactcgatga ttgacggcat cagcggcgtg cgaactgatgc agagggtgct caccaccgat 480
cccgaagagt gcaatatgcc accgcctcgg acgcgccgcc cacaccagcg cctgggtgca 540
aaaaccgaca aagaggccag cgtgcgggca gcggttctcc aggcaatgga cgccctgaag 600
ctccaggcag acatggcccc cgctcgttgg caggccggca atcgccctgg gattcgggtt 660
cgacaccggg aagacggact gaccgcgcc ttcactggac cgtgctcgtt gctcaatcac 720
cgggttaccg cgggcgcagc ttttgccacc cagcattatc aactggagcg gctgaaaaac 780
ctggcccatg cttccggcgg tgggttgaac gacatcgttc tttacctgtg tggcaccgca 840
ttgcccgcgt ttctggctga gcagaacaat ctgccagaca ccccgctgac ggctgggtata 900
ccgggtgaata tccggccggc aaacgcagag gtcacgggca cccagatcag ttggatgatt 960
tggttcgctg ccaccgcaga agctgatccg ttgaaccgcc tgcaacagat caaaacctcg 1020
acccgacggg ccaaggagca cctgcagaaa cttccaaaaa cggccctgac ccagtacacc 1080
atgctgctga tgtcaccctg gattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
ccagtcttca nrfalytkmh ttccaacgtg cccggcccgg aagagacgct gtattatgaa 1200
ggagcccgcc ttgaggccat gtatccgttg tcgctaactc ctcacggcgg cgccctgaac 1260
atcacctgcc tgagctatgc cggatcgctg aatttcggtt ttaccggctg tcgggatacg 1320
ctgccgggga tgcagaaact ggcggtttat accggtgaag ctctggatga gctggaatcg 1380
ctggttctgc cacccaagaa gcgcgccgca acccgcaagt aa 1422

```

```

SEQ ID NO: 37      moltype = AA length = 473
FEATURE
source            Location/Qualifiers
                  1..473
                  mol_type = protein
                  organism = Marinobacter hydrocarbonoclasticus

```

```

SEQUENCE: 37
MKRLGSLDAS WLAVEGEDTP MHVGTLLQIFS LPEDAPETFL RDMVTRMKEA GDVAPPWGLK 60
LAWSGFLGRV IAPAWKVDKD IDLDYHVRHS ALPRPGGERE LGILVSRLLS NSLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRLVLTDD PEECNMPPPW TRRPHQRRGA 180
KTDKEASVRA AVSQAMDALK LQADMAPALW QAGNRLVHSV RHPEDGLTAP FTGPVSVLNH 240

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RVTAGRRFAT	QHYQLERLKN	LAHASGGGLN	DIVLYLCGTA	LRRFLAEQNN	LPDTPLTAGI	300
PVNIRPANDE	VTGTQISWMI	CSLATDEADP	LNRLQQIKTS	TRRAKEHLQK	LPKTALTQYT	360
MLLMSPWILQ	LMSGLGGRMR	PVFNVITISNV	PGPEETLYYE	GARLEAMYPL	SLIAHGGALN	420
ITCLSYAGSL	NFGFTGCRDT	LPGMQKLAVY	TGEALDELES	LVLPPKKRAR	TRK	473

SEQ ID NO: 38 moltype = DNA length = 1422
 FEATURE Location/Qualifiers
 source 1..1422
 mol_type = unassigned DNA
 organism = Marinobacter hydrocarbonoclasticus

SEQUENCE: 38

atgaaacgctc	tccgaccctc	ggacgcctcc	tggtggtg	ttgaaggtga	agacaccccg	60
atgcatgtgg	gtacgcttca	gattttctca	ctgccggaag	acgcaccaga	aaccttctcg	120
cgtgacatgg	tcaagcgaa	gaaagaggcc	ggcgatgtgg	caccaccctg	gggatgcaaa	180
ctggcctggt	ctggtttcct	cgggcgcgtg	atcgccccgg	cctggaaagt	cgataaggat	240
atcgatctgg	attatcacgt	cggcactca	gccctgcctc	gccccggcgg	ggagcgcgaa	300
ctgggtattc	tggtatcccc	actgcactct	aacagtctgg	atttttcccc	ccctctttgg	360
gaatgccacg	ttattgaagg	cctggagaat	aaccgttttg	ccctttacac	caaaatgcac	420
cactcgatga	ttgacggcat	cagcggcgtg	cgactgatgc	agaggggtgt	caccaccgat	480
cccgaacgct	gcaatatgcc	accgccctgg	acgcgccgcc	cacaccagcg	ccgttggtgca	540
aaaaccgaca	aagaggccag	cgtgcgggca	gcggtttccc	aggcaatgga	cgccctgaag	600
ctccaggcag	acatggcccc	cgcgctgtgg	caggccggca	atcgctggtg	gcattcggtt	660
cgacacccgg	aagacggact	gaccgcgccc	ttcactggac	cggtctcggt	gctcaatcac	720
cgggttaccg	cgggcgcacg	ttttgccacc	cagcattatc	aactggaccg	gatcaaaaac	780
ctggccccatg	cttcggcggt	tggtttgaac	gacatcgttc	tttacctgtg	tggaaccgca	840
ttgcggcgct	ttctggctga	gcagaacaat	ctgccagaca	ccccgctgac	ggctggtata	900
ccggtgaata	tccggccggc	aaacgacgag	gtcacgggca	cccagatcag	ttggatgatt	960
tgttcgctgg	ccaccgacga	agctgatccg	ttgaaccgcc	tgcaacagat	caaaacctcg	1020
acccgacggg	ccaaggagca	cctgcagaaa	cttccaaaaa	cggccctgac	ccagtacacc	1080
atgctgctga	tgtcacctcg	gattctgcaa	ttgatgtcag	gtctcggggg	gaggatgcga	1140
ccagtcctta	acgtgacctg	ttccaacgtg	cccgcccgcg	aagagacgct	gtattatgaa	1200
ggagcccgcc	ttgaggccat	gtatccgttg	tcgctaatac	ctcacggcgg	cgccctgaac	1260
atcacctgcc	tgagctatgc	cggatcgctg	aatttcggtt	ttaccgctg	tcgggatacg	1320
ctgccggggg	tgcaaaaact	ggcgggttat	accgggtaag	ctctggatga	gctggaatcg	1380
ctggttctgc	cacccaagaa	gcgcgccccg	acccgcaagt	aa		1422

SEQ ID NO: 39 moltype = AA length = 473
 FEATURE Location/Qualifiers
 source 1..473
 mol_type = protein
 organism = Marinobacter hydrocarbonoclasticus

SEQUENCE: 39

MKRLGFLDAS	WLAVEGEDTP	MHVGTLQIFS	LPEDAPETFL	RDMVKRMKEA	GDVAPPWGCK	60
LAWSGFLGRV	IAPAWKVDKD	IDLDYHVRHS	ALPRPGGERE	LGILVSRlhs	NSLDFSRPLW	120
ECHVIEGLEN	NRFALYTKMH	HSMDIGISGV	RLMQRVLTDT	PERCNMPPPW	TRRPHQRRGA	180
KTDKEASVRA	AVSQAMDALK	LQADMAPALW	QAGNRLVHSV	RHPEDGLTAP	FTGPCSVLNH	240
RVTAGRRFAT	QHYQLDRIKN	LAHASGGGLN	DIVLYLCGTA	LRRFLAEQNN	LPDTPLTAGI	300
PVNIRPANDE	VTGTQISWMI	CSLATDEADP	LNRLQQIKTS	TRRAKEHLQK	LPKTALTQYT	360
MLLMSPWILQ	LMSGLGGRMR	PVFNVITISNV	PGPEETLYYE	GARLEAMYPL	SLITHGGALN	420
ITCLSYAGSL	NFGFTGCRDT	LPGMQKLAVY	TGEALDELES	LVLPPKKRAR	TRK	473

SEQ ID NO: 40 moltype = DNA length = 1422
 FEATURE Location/Qualifiers
 source 1..1422
 mol_type = unassigned DNA
 organism = Marinobacter hydrocarbonoclasticus

SEQUENCE: 40

atgaaacgctc	tccgaccctc	ggacgcctcc	tggtggtg	ttgaaggtga	agacaccccg	60
atgcatgtgg	gtacgcttca	gattttctca	ctgccggaag	gcgcaccaga	aaccttctcg	120
cgtgacatgg	tcgctcgaa	gaaagaggcc	ggcgatgtgg	caccaccctg	gggatgcaaa	180
ctggcctggt	ctggtttcct	cgggcgcgtg	atcgccccgg	cctggaaagt	cgataaggat	240
atcgatctgg	attatcacgt	cggcactca	gccctgcctc	gccccggcgg	ggagcgcgaa	300
ctgggtattc	tggtatcccc	actgcactct	aacagtctgg	atttttcccc	ccctctttgg	360
gaatgccacg	ttattgaagg	cctggagaat	aaccgttttg	ccctttacac	caaaatgcac	420
cactcgatga	ttgacggcat	cagcggcgtg	cgactgatgc	agaggggtgt	ctccaccgat	480
cccgaacgct	gcatatgcc	accgccctgg	tcgcgccgcc	cacaccagcg	ccgttggtgca	540
aaaaccgaca	aagaggccag	cgtgcgggca	gcggtttccc	aggcaatgga	cgccctgaag	600
ctccaggcag	acatggcccc	cgcgctgtgg	caggccggca	atcgctggtg	gcattcggtt	660
cgacacccgg	aagacggact	gaccgcgccc	ttcactggac	cggtctcggt	gctcaatcac	720
cgggttaccg	cgggcgcacg	ttttgccacc	cagcattatc	aactggaccg	gatcaaaaac	780
ctggccccatg	cttcggcggt	tggtttgaac	gacatcgttc	tttacctgtg	tggaaccgca	840
ttgcggcgct	ttctggctga	gcagaacaat	ctgccagaca	ccccgctgac	ggctggtata	900
ccggtgaata	tccggccggc	aaacgacgag	gtcacgggca	cccagatcag	ttggatgatt	960
tgttcgctgg	ccaccgacga	agctgatccg	ttgaaccgcc	tgcaacagat	caaaacctcg	1020
acccgacggg	ccaaggagca	cctgcagaaa	cttccaaaaa	cggccctgac	ccagtacacc	1080
atgctgctga	tgtcacctcg	gattctgcaa	ttgatgtcag	gtctcggggg	gaggatgcga	1140
ccagtcctta	acgtgacctg	ttccaacgtg	cccgcccgcg	aagagacgct	gtattatgaa	1200
ggagcccgcc	ttgaggccat	gtatccgttg	tcgctaatac	ctcacggcgg	cgccctgaac	1260
atcacctgcc	tgagctatgc	cggatcgctg	aatttcggtt	ttaccgctg	tcgggatacg	1320

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ctgccgggga tgcagaaact ggcgggttat accggtgaag ctctggatga gctggaatcg 1380
ctggttctgc cacccaagaa gcgcgcccga acccgcaagt aa 1422

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```

SEQ ID NO: 41      moltype = AA length = 473
FEATURE           Location/Qualifiers
source            1..473
                  mol_type = protein
                  organism = Marinobacter hydrocarbonoclasticus

```

```

SEQUENCE: 41
MKRLGPLDAS WLAVEGEDTP MHVGTLLQIFS LPEGAPETFL RDMVARMKEA GDVAPPWGYK 60
LAWSGFLGRV IAPAWKVDKD IDLDYHVRHS ALPRPGGERE LGILVSRLLS NSLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRLVSTD PERCDMPPPW SRRPHQRRGA 180
KTDKEASVRA AVSQAMDALK LQADMAPALW QAGNRLIHSV RHPEDGLTAP FTGPSCVLNH 240
RVTAGRRFAT QHYQLDRIKN LAHASGGGLN DIVLYLCGTA LRRFLAEQNN LPDTPPLTAGI 300
PVNIRPANDE VTGTQISWMI CSLATDEADP LNRLKQIKTS TRRAKEHLQK LPKTALTQYT 360
MLLMSPWILQ LMSGLGGRMR PVFNVTISNV PGPEETLYYE GARLEAMYPL SLIAHGGALN 420
ITCLSYAGSL NFGFTGCRDT LPGMQKLAVY TGEALDELES LVLPPKKRAR TRK 473

```

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SEQ ID NO: 42      moltype = DNA length = 1422
FEATURE           Location/Qualifiers
source            1..1422
                  mol_type = unassigned DNA
                  organism = Marinobacter hydrocarbonoclasticus

```

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SEQUENCE: 42
atgaaacgtc tcggaccctc ggcgcctccc ttgctggcgg ttgaagggtga agacaccccg 60
atgcatgtgg gtacgcttca gattttctca ctgccggaag gcgcaccaga aaccttctcg 120
cgtgacatgg tcactcgaaat gaagagggcc ggcatgtgg caccacctcg gggattaaaa 180
ctggcctggt ctggtttctc cgggcgcgtg atcgccccgg cctggaaagt cgataaggat 240
atcgatctgg attatcacgt cgggcaactc gccctgcctc gcccgcgagg ggagcgcgaa 300
ctgggtatct tggatcccg actgcactct aacagtctgg atttttcccc ccctctttgg 360
gaatgccacg ttattgaagg cctggagaat aaccgttttg ccttttacac caaatgcac 420
cactcgatga ttgacggcat cagcggcgtg cgactgatgc agaggggtgct caccaccgat 480
ccgaagagtg gcaatatgcc accgcctcgg tcgcgcgcgc cacaccagcg ccgtggttca 540
aaaaccgaca aagaggccag cgtgcgggca gcggtttccc aggcaatgga cgccctgaag 600
ctccaggcag acatggcccc cgcgctgtgg caggccggca atcgctggtt gcattcggtt 660
cgacaccccg aagacggact gaccgcgcgc ttcactggac cgtgctcggt gctcaatcac 720
cgggtttacc cgggcgcgac ttttgccacc cagcattatc aactggaccg gatcaaaaac 780
ctggccccatg cttcccgagg tgggttgaac gacatcgctt tttacctgtg tggcaccgca 840
ttgcggcgct tctcggtgta gcagaacaat ctgcagaca ccccgctgac ggctggtata 900
ccggtgaata tcgggcggcg aaacgacgag gtcacgggca ccagatcag ttggatgatt 960
tgttcgctgg ccacgcgaca agctgatccg ttgaaccgcc tgcaacagat caaaacctcg 1020
acccgacggg ccaaggagca cctgcagaaa cttccaaaaa aggcctcgac ccagtacacc 1080
atgctgctga tgtcacctcg gattctgcaa ttgatgtcag gtctcggggg gaggatgcga 1140
ccagtcctca acgtgacctt ttccaacgtg cccggcccgg aagagacgct gtattatgaa 1200
ggagcccgcc ttgaggccat gtatccgttg tcgctaactc ctcacggcgg cgccctgaac 1260
atcacctgcc tcagctatgc cggatcgctg aatttcggtt ttaccgctg tcgggatacg 1320
ctgccgggga tgcagaaact ggcgggttat accggtgaag ctctggatga gctggaatcg 1380
ctggttctgc cacccaagaa gcgcgcccga acccgcaagt aa 1422

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```

SEQ ID NO: 43      moltype = AA length = 473
FEATURE           Location/Qualifiers
source            1..473
                  mol_type = protein
                  organism = Marinobacter hydrocarbonoclasticus

```

```

SEQUENCE: 43
MKRLGPLDAS WLAVEGEDTP MHVGTLLQIFS LPEGAPETFL RDMVTRMKEA GDVAPPWGLK 60
LAWSGFLGRV IAPAWKVDKD IDLDYHVRHS ALPRPGGERE LGILVSRLLS NSLDFSRPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRLVTTD PEECNMPPPW SRRPHQRRGS 180
KTDKEASVRA AVSQAMDALK LQADMAPALW QAGNRLVHSV RHPEDGLTAP FTGPSCVLNH 240
RVTAGRRFAT QHYQLDRIKN LAHASGGGLN DIVLYLCGTA LRRFLAEQNN LPDTPPLTAGI 300
PVNIRPANDE VTGTQISWMI CSLATDEADP LNRLQQIKTS TRRAKEHLQK LPKKALTQYT 360
MLLMSPWILQ LMSGLGGRMR PVFNVTISNV PGPEETLYYE GARLEAMYPL SLIAHGGALN 420
ITCLSYAGSL NFGFTGCRDT LPGMQKLAVY TGEALDELES LVLPPKKRAR TRK 473

```

```

SEQ ID NO: 44      moltype = DNA length = 1470
FEATURE           Location/Qualifiers
source            1..1470
                  mol_type = unassigned DNA
                  organism = Marinobacter adhaerens

```

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SEQUENCE: 44
atgaaacgcc tgggaacgct ggatgcctct ttgctggcgg tagaatctga agacaccccc 60
atgcatgtgg gcaacctgca gatctttctg ctaccggagg gcgccccctga aaccttctctc 120
cgggacatgg tgaccgcgat gaaggaaacc ggcatgtcg cgccccctcg gggtttaaaag 180
cttgctggtt ccggaactgt gggtcgggta ctggcgccgg gctggaaaagt cgacaagaaa 240
atcgatctgg attaccacgt gcgtcactcg gcactgccgc gcccgcgagg tgaacgggaa 300
ctgggcataat tggatcgcg cctgcactcc aacccccctg actttgcccc ccaactctgg 360
gaatgccacg ttatcgaggg cctcgagaat aaccgttttg cgctctacac caagatgcat 420
cattccatga ttgatggcat cagtggtgta cgccctgatgc aacgggtact gacaaccgat 480
cccgacaaac gggacatgcc gccaccctgg tcggttcggc cggagcgccg tcgcggcagc 540

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aaatccgatt cccagggccag cgtgcccggc gcgggtctcac agggccatgga ggcactgaaa 600
ctgcaagcgg atatggcgcc acgtctgttg caagcgggca acaggttggt gcattcggtc 660
cgtaaccccg aggatgtgtc caccggcaccg tttaacggggc ccgtgtccaa gatcaatcac 720
cgggttaccg gccagcgccg gtttgccacc cagcattacc agctcgacag gatcaaggag 780
cttgcccacg tttccggcgc ttccctcaac gatatcggtc tgtacctgtg cggaaactgcc 840
ctcaggcggtt ttctgctgga acagaatgaa ctgcccggacg ctccctccac cgcgcgcata 900
ccggttaaca tccggccgctc cgatgacgaa gggacaggca cccagatcag ttctatgatc 960
tcttcgcttg cgacgcagca agccgatccg ctgacccgcc tgcaaaacat caaggcggtca 1020
acgcgaaggc ccaagagcga ttgacagaag ctgcccacaaa gcgcgctcac ccagtacacc 1080
atgctctcga tgtcgcctta tattctgcag ctgatgtccg gccttgccgg tcgcatgcgg 1140
ccggtattta acgtcacgat ttccaatgtg ccgggggcccc agagaacgct ctactacgag 1200
ggcgcgaaac tgggaagccat gtaccctgtc tcgctgatta ctacgggtgg cgctttgaac 1260
atcacctgcc tgagctatga cggttcactg aacttcggct atacgggctg cagggatacc 1320
ctgccgagca tgcagagact tgcggtttac acgggtgagg cactggacga actggaaagc 1380
ctgattcttc cgcacaaaggc caaacccgaaa gctgcccga agcccagtcg gccgcgcaaa 1440
cagccgacga aaaagagtaa agcggactga 1470

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```

SEQ ID NO: 45      moltype = AA length = 489
FEATURE
source             Location/Qualifiers
                   1..489
                   mol_type = protein
                   organism = Marinobacter adhaerens

```

```

SEQUENCE: 45
MKRLGTLTLDAS WLAVESEDTP MHVGNLQIFS LPEGAPETFL RDMVTRMKET GDVAPPWGLK 60
LAWSGLLGRV LAPGWKVDKK IDLDYHVRHS ALPRPGGERE LGILVSRHS NPLDFARPLW 120
ECHVIEGLEN NRFALYTKMH HSMIDGISGV RLMQRVLTTD PDKRDMPPPW SVRPERRRGS 180
KSDSEASVPG AVSQAMEALK LQADMAPRL LQAGNRLVHSV RHPEDGLTAP FTGPVSKINH 240
RVTGQRRFAT QHYQLDRKE LAHVSGASLN DIVLYLCGTA LRRFLLEQNE LPDAPLTAGI 300
PVNIRPSDDE GTGTQISFMI SSLATDEADP LTRLQNIKAS TRRAKEHLQK LPKSALTQYT 360
MLLMSPYLLQ LMSGLGGRMR PVFNVTISNV PGPQRTLYYE GAKLEAMYVP SLITHGGALN 420
ITCLSYDGLS NFGYTGRDRT LPSMQRLAVY TGEALDELES LILPPKAKPK AAAPKAPRK 480
QPTKSKAD 489

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SEQ ID NO: 46      moltype = DNA length = 1458
FEATURE
source             Location/Qualifiers
                   1..1458
                   mol_type = unassigned DNA
                   organism = Marinobacter algicola

```

```

SEQUENCE: 46
atgaaacgcc tgggcacact ggacgcttcc tgggtggcgg tgggaatcgga agacaccccc 60
atgcacgtcg ggaacctgca gattttttca ctaccggaag atgctccgga gacgtttcta 120
agagacatgc tcgcccgcct gaaagccgat gccgatgtag cgcgcgcctg gtgctacaag 180
ctcgccctttt cgggggttctt gggccgcctg gtccgcccgt cctggaaggt cgacaagaag 240
ctggatctcg actaacacgt tcgacactcg gcgttgccgc gccccggcag tgaacgggaa 300
cttggtatcc tggatccag gctgcattcc aaccgcgtgg atttttcccg cccgctttgg 360
gaatgccaca ttatcgaggg cctggagaac aaccgttttg cctctacac caagatgcat 420
cattccatga ttgatggcat aagcgggtgt cggctgatgc agcgggtgct cagcagggac 480
cccggtgaga ttaatatgct gccgccatgg tcggtacgcc cggagcggac acggggcagc 540
aagacagatt ccgaagccag catctcagcc gccctgtccc agcccatgga agccctgagg 600
attcaggccg acatggcgcc gaggtctctg aatgcgatga accgcctgat ccagtccgca 660
cggcaccgcc aagaggggct gaccgcgccc tttgcggccc cggtttccgc cctcaatcac 720
cgggtcaccc gtcagcggcg gtttgccacc cagcactacc agctcgaaac gatcaaacag 780
gtcggccagg gtcaccaagg ctccctgaac gacattgtgc tctacctctg tggcactgcc 840
ctgcgcgcgt ttcttgttga acaggatggt ttgcgggata cgccactcac cgcgcggaat 900
ccggtgaata tccgcccctc cgatgaccag ggacacggga cccagatcag ctctcatgatt 960
gctcactcgg cgaccgcaga agccgatccc ctcaagcgcc tgaagagcat caaacactct 1020
acccgcaggg ccaacaacaa ccttcagaaa ctgcccgcta aagccctgac ccaatacacc 1080
atgctgtcga tgtcgcccta catcctgcag ttgatgtcag gcctggcgg gcgaatgcgc 1140
ccggtgttca acgtgacat ctccaacgtg ccaggggccag gggaaaccc ttactatgaa 1200
ggagcacgac tggaggcgat gtacccggtt tcgcttattg cccacggcgg tcgctcaac 1260
attacttgcc tgagctacgc cggctcgctc aacttcggtt tcacgggctg ccgcgatacg 1320
ttgcccagta tgcagaagct ggcgggtctac accggtgagg ctttggatga actggaaagc 1380
ctggtttcgc caccacaaa tcagacaaa accaacgctc gaaaggcgcc tcgcaaaaag 1440
actgcgaaaa agagctaa 1458

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SEQ ID NO: 47      moltype = AA length = 485
FEATURE
source             Location/Qualifiers
                   1..485
                   mol_type = protein
                   organism = Marinobacter algicola

```

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SEQUENCE: 47
MKRLGTLTLDAS WLAVESEDTP MHVGNLQIFS LPEDAPETFL RDMILARMKAD ADVAPPWCYK 60
LAFSGFLGRL VAPSWKVDKK LDLDYHVRHS ALPRPGSERE LGILVSRHS NPLDFSRPLW 120
ECHIEGLEN NRFALYTKMH HSMIDGISGV RLMQRVLSED PGEINMLPPW SVRPERTRGS 180
KTDSEASISA ALSQAMEALR IQADMAPRLW NAMNRLIQSA RHPDEGLTAP FAGPVSAALNH 240
RVTGQRRFAT QHYQLERIKQ VAQASNGSLN DIVLYLCGTA LRRFLVEQDG LPDTPLTAGI 300
PVNIRPSDDQ GTGTQISFMI ASLATDEADP LKRLKSIKHS TRRAKQHLQK LPRKALTQYT 360
MLLMSPYLLQ LMSGLGGRMR PVFNVTISNV PGPGETLYYE GARLEAMYVP SLIAHGGALN 420
ITCLSYAGSL NFGFTGRDRT LPSMQKLAVY TGEALDELES LVSPPPNQTK TNARKAPRKK 480

```

-continued

TAEKS 485
 SEQ ID NO: 48 moltype = DNA length = 1470
 FEATURE Location/Qualifiers
 source 1..1470
 mol_type = unassigned DNA
 organism = *Marinobacter* sp.
 SEQUENCE: 48
 atgaaacgcc tggcaacatt ggacgcgtct tggctagcgg tcgagctctga cgatacaccc 60
 atgcacgtgg gcaacttgca gattttcagc ctgcccgcaca acgccccatc tacattcgcg 120
 ggcgacttgg tcaaaagcat gaagcaagcc ggtaatgttg agcttccctg gggctgcaag 180
 ctgggtatgg caggttttct gggccgcgtt ctggcgccca cctggaagca cgacaagcat 240
 attgatctgg attatcacgt gcgccactcg gccctacca aacccgggtg tgaacgcgaa 300
 ctggggggaac tggatcgcg cctgcactcc aaccgcgtgg atctgtcgcg gccgctgtgg 360
 gagtgcacaca tgatcgagg gctggaacac aacggttttg ccctgtacac gaagatgcat 420
 cactgcatga ttgatggcat cagtgggtga cgctgtatgc aaagggtgct gagcaaatcc 480
 cccgacgagc ggcacatgct gccaccctgg tcagtacgcc cggaagcac gcgcggcaaa 540
 aagaccgaca gcgaggccag cgtgccgggt gctatatccc aagctatgga agctctcaaa 600
 ctgcagttgg gcttggcacc acggctgtgg caagccagca atcgctgat tcactcggtg 660
 cgccatccgg aagacggtct gaccgcgccc ttacacggcc cggtttccaa gatcaatcat 720
 cgggttactg gccagcgccg cttcgccacc cagcagatgc agttagaaga tatgaaagcc 780
 atggcccgcg cctcgggcacc ctcgatgaac gacattgtgc tgtatttgtg cggtactgcg 840
 ttgcccgtgt ttcgtctgga acaggacgat ttgacctgaa tatcattaac agcaggcata 900
 ccggtcaaca ttccgcccgg ggaatgacgaa ggcacaggaa cccagatcag cttcatgatt 960
 gccgccctgg ccaaccaacca acctgatccg ctaacgcgcc tgaatgcat caaggaaatct 1020
 tcgtgcgaaag ccaagagca cttgcaaaaa ttgcccgaag aagcgttgac ccaatacacc 1080
 atgatgctga tgtcgcccta catattgcag ctgatgtctg gcttggcgcg gcgcattgca 1140
 ccggtattta acgtaaacat ctccaacggt ccggggccca ccgaagatct ttattacgaa 1200
 ggccgcaaac tcgaagccat gtatccgggt tcgctgatca cccacggcgg agcgttgaaac 1260
 attacttgcc tgagctatgc cggatcattg aactttggtt tcaactggtg ccgcgcaccc 1320
 ttaccagca tcgagaagct ggcgtgtgat accgggggaa cattggaaga actcagaacc 1380
 ctgctgttac cgcaaaagaa aaaaccagc ccacgcaaac ctagaacggc cgcgaaaaag 1440
 aagcccgcg tgaacagcaa cgtagctga 1470
 SEQ ID NO: 49 moltype = AA length = 489
 FEATURE Location/Qualifiers
 source 1..489
 mol_type = protein
 organism = *Marinobacter* sp.
 SEQUENCE: 49
 MKRLATLDAS WLAVESDDTP MHVGNLQIFS LPDNAPSTFA GDLVSKMQA GNVELPWGCK 60
 LVWPGFLGRV LAPTWKHKH IDLDYHVRHS ALPKPGGERE LGELVSRHLS NPLDLSRPLW 120
 ECHMIEGLEH NRFALYTKMH HCMIDGISGV RLMQVRVLSKS PDERDMLPPW SVRPESTRGK 180
 KTDSEASVPG AISQAMEALK LQLGLAPRLW QASNRLIHSV RHPEDGLTAP FTGPVSKINH 240
 RVTGQRRFAT QQYQLEDMA MARASGSSMN DIVLYLCGTA LRRFLLEQDD LPEISLTAGI 300
 PYNIRPADDE GTGTQISFMI AALATNQDPD LTRLKCIKES SCKAKEHLQK LPKKALTQYT 360
 MMLMSPYILQ LMSGGLGGRMR PVFNVTISNV PGPTEDLYE GAKLEAMYPV SLITHGGALN 420
 ITCLSYAGSL NFGFTGCRDT LPSMQKLAVY TGEALEELRT LLLPPKKKPS PRKPRTAAKK 480
 KPAVNSNAS 489
 SEQ ID NO: 50 moltype = DNA length = 549
 FEATURE Location/Qualifiers
 source 1..549
 mol_type = unassigned DNA
 organism = *Photobacterium profundum*
 SEQUENCE: 50
 atgggtaaca cttactgggt tgttgggtgat agcttgagcg cgggctatca aatgcgtgca 60
 gaacaaagct ggcgggtgtt actgcaaccc gcattaaagc aacaaggtea cgaaatcacc 120
 gttgtaaatg ccagtatttc aggtgataca acaggaaagc gcttggctcg cttgcctaca 180
 ttactacaac aacataaacc agcttacgtc atcatagaac tcggggcgaa tgatggctta 240
 cgtggtttcc ctcaaggtac tattcgtaac aatctcagcc aatgatcac tgaattcaa 300
 aatgctgatg ccaagccaat gctcgtgcag atcaaagtgc cgcccaatta cggcaaacgc 360
 tacagtgata tgttcagttc tatttaccct caactcagta aagagttagc cacaccattg 420
 ttacctttct ttttagagca gatcatttta aaacaagaat ggatgatgaa tgacggttta 480
 catcctaata ctgatgctca gccgtggatt gccgaatata tggctgagaa tatcgcgct 540
 tatttataa 549
 SEQ ID NO: 51 moltype = AA length = 182
 FEATURE Location/Qualifiers
 source 1..182
 mol_type = protein
 organism = *Photobacterium profundum*
 SEQUENCE: 51
 MGNLLVVG D SLSAGYQMR EQSWPVLLQP ALKQQGHEIT VVNASISGDT TGNGLARLPT 60
 LLQQHKPAYV IIELGANDGL RGFPQGTIRN NLSQMITEIQ NADAKPMLVQ IKVPPNYGKR 120
 YSDMFSSIYP QLSKELATPL LPFFLEQIIL KQEWMMNDGL HPKSDAQFWI AEYMAENIAP 180
 YL 182
 SEQ ID NO: 52 moltype = DNA length = 549

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FEATURE          Location/Qualifiers
source            1..549
                  mol_type = unassigned DNA
                  organism = Photobacterium profundum

SEQUENCE: 52
atgggtaaca cattactggt tgttggtgat agcttgagcg cgggctatca aatgcgtgca 60
gaacaaagct ggccggtggt actgcaaccc gcattaaagc aacaagggtc cgaatcacc 120
gttgtaaatg ccagtattgt cggtgacaca acaggaacgc gcttggtctg cttgcctacg 180
ttactacaac aacataaacc agcttacgtc atcatagaac tcggggcgaa tgatggcctg 240
cgtgggtttc cgcaagggtac tattcgtaac aatctgagcc aaatgatcac tgaattcaa 300
aatgctgatg ccaagccaat gctcgtgcaa atcaaagtgc cgcccaatca cggcaaacgc 360
tacagtataa tggtcagttc tattaccct caactcagta aagagttagc cacaccattg 420
ttacctttct ttttagagca gatcatttta aaacaagaat ggatgatgaa tgacgggtta 480
catcctaaat ccgatgctca gccgtggatt gccgaatata tggctgagaa tatcgcgctt 540
tatttataa                                     549

SEQ ID NO: 53      moltype = AA length = 182
FEATURE            Location/Qualifiers
source              1..182
                  mol_type = protein
                  organism = Photobacterium profundum

SEQUENCE: 53
MGNTLLVVG D SLSAGYQMR EQSWPVLLQ ALKQQGHEIT VVNASIVGDT TGNGLARLPT 60
LLQQHKKPAYV IIELGANDGL RGFPQGTIRN NLSQMITEIQ NADAKPMLVQ IKVPPNHGKR 120
YSDMFSSIIYP QLSKELATPL LPFFLEQIIL KQEWMMNDGL HPKSDAQFPI AEYMAENIAP 180
YL                                     182

SEQ ID NO: 54      moltype = DNA length = 549
FEATURE            Location/Qualifiers
source              1..549
                  mol_type = unassigned DNA
                  organism = Photobacterium profundum

SEQUENCE: 54
atgggtaaca cattactggt tgttggtgat agcttgagcg cgggctatca aatgcgtgca 60
gaacaaagct ggccggtggt actgcaaccc gcattaaagc aacaagggtc cgaatcacc 120
gttgtaaatg ccagtattgt cggtatgaca acaagaagcg gcttggtctg cttgcctacg 180
ttactacaac aacataaacc agcttacgtc atcatagaac tcggggcggt ggatggcctg 240
cgtgggtttc cgcaagggtac tattcgtaac aatctgagcc aaatgatcac tgaattcaa 300
aatgctgatg ccaagccaat gctcgtgcaa atcaaagtgc cgcccaatca cggcaaacgc 360
tacagtataa tggtcagttc tattaccct caactcagta aagagttagc cacaccattg 420
ttacctttct ttttagagca gatcatttta aaacaagaat ggatgatgaa tgacgggtta 480
catcctaaat ccgatgctca gccgtggatt gccgaatata tggctgagaa tatcgcgctt 540
tatttataa                                     549

SEQ ID NO: 55      moltype = AA length = 182
FEATURE            Location/Qualifiers
source              1..182
                  mol_type = protein
                  organism = Photobacterium profundum

SEQUENCE: 55
MGNTLLVVG D SLSAGYQMR EQSWPVLLQ ALKQQGHEIT VVNASIVGMT TRNGLARLPT 60
LLQQHKKPAYV IIELGAVDGL RGFPQGTIRN NLSQMITEIQ NADAKPMLVQ IKVPPNHGKR 120
YSDMFSSIIYP QLSKELATPL LPFFLEQIIL KQEWMMNDGL HPKSDAQFPI AEYMAENIAP 180
YL                                     182

SEQ ID NO: 56      moltype = DNA length = 549
FEATURE            Location/Qualifiers
source              1..549
                  mol_type = unassigned DNA
                  organism = Photobacterium profundum

SEQUENCE: 56
atgggcaaca cccttctcgt cgtcggcgat tcactctccg caggctacca aatgcgtgca 60
gaacaaagct ggccggtatt gctgcaacct gccctgaaac agcagggtca tgagatcact 120
gtggttaaatg caagcattgt gggtatgacc acgcgtaacg gtttggtctg tctgccaacc 180
ctgttacaac aacacaagcc ggcgatggtt atcatcgaat tgggtgcggt cgatggcctg 240
cgcggttttc cgcaaggcac gattcgtaat aacctgagcc agatgattac cgagattcag 300
aacgctgacg cgaagccgat gctgggttcag atcaaagtgc cgccgaacca cggtaaacgc 360
tacagcgaca tggtcagcag catttaccg cagctgtcta aggaactggc gacgccactg 420
ctgccgtttc ttctggagca gattatcctg aagcaagaat ggatgatgaa tgacgggtctg 480
catccgaaaaa gcgatgcgca gccgtggatc gcagagtata tggccgagaa tatcgcgccg 540
tatttgtaa                                     549

SEQ ID NO: 57      moltype = DNA length = 549
FEATURE            Location/Qualifiers
source              1..549
                  mol_type = unassigned DNA
                  organism = Photobacterium profundum

SEQUENCE: 57
atgggtaaca cattactggt tgttggtgat agcttgagcg cgggctatca aatgcgtgca 60

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gaacaaagct ggccggtgtt actgcaaccc gcattaaagc aacaagggtca cgaaatcacc 120
gttgtaaatg ccagtattgt cggatatgtg acaagaaacg gcttggtctg cttgcctacg 180
ttactacaac aacataaacc agcttacgtc atcatagaac tcggggcggtt ggatggcctg 240
cgtgggtttcc cgcaaggtac tattcgtaac aatctgagcc aaatgatcac tgaaattcaa 300
aatgctgatg ccaagccaat gctcgtgcaa atcaaagtgc cgcccaatca cggcaaacgc 360
tacagtgata tggtcagttc tatttaccct caactcagta aagagttagc cacaccattg 420
ttacctttct ttttagagca gatctattta aaacaagaat ggatgatgaa tgacgggtta 480
catcctaaat ccgatgctca gccgtggatt gccgaatata tggctgagaa tatcgcgcc 540
tatttataa 549

```

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SEQ ID NO: 58      moltype = AA  length = 182
FEATURE           Location/Qualifiers
source            1..182
                  mol_type = protein
                  organism = Photobacterium profundum

```

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SEQUENCE: 58
MGNTLLVVG D SLSAGYQMRA EQSWPVLLQP ALKQQGHEIT VVNASIVGML TRNGLARLPT 60
LLQQHKPAYV I IELGAVDGL RGFPQGTIRN NLSQMITEIQ NADAKPMLVQ IKVPPNHGKR 120
YSDMFSSIYP QLSKELATPL LPFFLEQIYL KQEWMMNDGL HPKSDAQFPI AEYMAENIAP 180
YL 182

```

```

SEQ ID NO: 59      moltype = DNA  length = 549
FEATURE           Location/Qualifiers
source            1..549
                  mol_type = unassigned DNA
                  organism = Photobacterium profundum

```

```

SEQUENCE: 59
atgggcaaca cccttctcgt cgtcggcgat tcaactctcg caggctacca aatgcgtgca 60
gaacaaagct ggccggtatt gctgcaacct gccctgaaac agcagggtca tgagatcact 120
gtgggttaatg caagcattgt gggatatgtg acgcgtaacg gtttggtctg tctgcccaacc 180
ctgttacaac aacacaagcc ggcgtatgtt atcatcgaat tgggtgcggt cgatggcttg 240
cgcggttttc cgcaaggcac gattcgtaac aacctgagcc agatgattac cgagattcag 300
aacgctgacg cgaagccgat gctggttcag atcaaagtgc cgccgaacca cggtaaacgc 360
tacagcgaca tggtcagcag catttaccg cagctgtcta aggaactggc gacgccactg 420
ctgcggttct ttctggagca gatttatctg aagcaagaat ggatgatgaa tgacgggtctg 480
catccgaaaa gcgatgcgca gccgtggatc gcagagtata tggccgagaa tatcgcgccg 540
tatttgtaa 549

```

```

SEQ ID NO: 60      moltype = DNA  length = 549
FEATURE           Location/Qualifiers
misc_feature      1..549
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..549
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 60
atgggcaaca cccttctcgt cgtcggcgat tcaactctcg caggctacca aatgcgtgca 60
gaacaaagct ggccggtatt gctgcaacct gccctgaaac agcagggtca tgagatcact 120
gtgggttaatg caagcattgt gggatatgtg acgcgtaacg gtttggtctg tctgcccaacc 180
ctgttacaac aacacaagcc ggcgtatgtt atcatcgaat tgggtgcggt cgatggcttg 240
cgcggttttc cgcaaggcac gattcgtaac aacctgagcc agatgattac cgagattcag 300
aacgctgacg cgaagccgat gctggttcag atcaaagtgc cgccgtttca cggtaaacgc 360
tacagcgaca tggtcagcag catttaccg cagctgtcta aggaactggc gacgccactg 420
ctgcggttct ttctggagca gatttatctg aagcaagaat ggatgatgaa tgacgggtctg 480
catccgaaaa gcgatgcgca gccgtggatc gcagagtata tggccgagaa tatcgcgccg 540
tatttgtaa 549

```

```

SEQ ID NO: 61      moltype = AA  length = 182
FEATURE           Location/Qualifiers
REGION           1..182
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..182
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 61
MGNTLLVVG D SLSAGYQMRA EQSWPVLLQP ALKQQGHEIT VVNASIVGML TRNGLARLPT 60
LLQQHKPAYV I IELGAVDGL RGFPQGTIRN NLSQMITEIQ NADAKPMLVQ IKVPPFHGKR 120
YSDMFSSIYP QLSKELATPL LPFFLEQIYL KQEWMMNDGL HPKSDAQFPI AEYMAENIAP 180
YL 182

```

```

SEQ ID NO: 62      moltype = DNA  length = 549
FEATURE           Location/Qualifiers
misc_feature      1..549
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..549
                  mol_type = other DNA

```

-continued

```

                                organism = synthetic construct
SEQUENCE: 62
atgggcaaca cccttctcgt cgtcggcgat tcaactctccg caggctacca aatgcgtgca 60
gaacaaagct ggcgggtatt gctgcaacct gccctgaaac gccgggggtca tgagatcact 120
gtgggttaatg caagcattgt gggatatgctg acgcgtcggg gtttggctcg tctgcccaacc 180
ctgttacaac aacacaagcc ggcgtatggt atcatcgaat tgggtgcggt cgatggcttg 240
cgcggttttc cgcaaggcac gattcgtaat aacctgagcc agatgattac cgagattcag 300
aacgctgacg cgaagccgat gctggttcag atcaaaactcc cgcggtggca cggtaaacgc 360
tacagcgaca tggtcagcag catttaccgc cagctgtcta aggaactggc gacgccactg 420
ctgccgttct ttctggagca gatttatctg aagcaagaat ggatgatgaa tgacgggtctg 480
catccgaaaa gcgatgcgca gccgtggatc gcagagtata tggccgagaa tatcgcgccg 540
tatttgtaa                                     549

SEQ ID NO: 63                moltype = AA length = 182
FEATURE                      Location/Qualifiers
REGION                      1..182
                                note = Description of Artificial Sequence: Synthetic
                                polypeptide
source                      1..182
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 63
MGNTLLVVG D SLSAGYQMR EQSWPVLLQ ALKRRGHEIT VVNASIVGML TRGLARLPT 60
LLQQHKKPAYV IIELGAVDGL RGFPQGTIRN NLSQMITEIQ NADAKPMLVQ IKLPPWHGKR 120
YSDMFSSIIYP QLSKELATPL LPFFLEQIYL KQEWMMNDGL HPKSDAQFPI AEYMAENIAP 180
YL                                     182

SEQ ID NO: 64                moltype = DNA length = 549
FEATURE                      Location/Qualifiers
misc_feature                1..549
                                note = Description of Artificial Sequence: Synthetic
                                polynucleotide
source                      1..549
                                mol_type = other DNA
                                organism = synthetic construct

SEQUENCE: 64
atgggcaaca cccttctcgt cgtcggcgat tcaactctccg caggctacca aatgcgtgca 60
gaacaaagct ggcgggtatt gctgcaacct gccctgaaac gccaggggtca tgagatcact 120
gtgggttaatg caagcattgt gggatatgctg acgcgtcaagg gtttggctcg tctgcccaacc 180
ctgttacaac aacacaagcc ggcgtatggt atcatcgaat tgggtgcggt cgatggcttg 240
cgcggttttc cgcaaggcac gattcgtaat aacctgagcc agatgattac cgagattcag 300
aacgctgacg cgaagccgat gctggttcag atcaaaactcc cgcggtggca cggtaaacgc 360
tacagcgaca tggtcagcag catttaccgc cagctgtcta aggaactggc gacgccactg 420
ctgccgttct ttctggagca gatttatatg aagcaagaat ggatgatgaa tgacgggtctg 480
catccgaaaa gcgatgcgca gccgtggatc gcagagtata tggccgagaa tatcgcgccg 540
tatttgtaa                                     549

SEQ ID NO: 65                moltype = AA length = 182
FEATURE                      Location/Qualifiers
REGION                      1..182
                                note = Description of Artificial Sequence: Synthetic
                                polypeptide
source                      1..182
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 65
MGNTLLVVG D SLSAGYQMR EQSWPVLLQ ALKRQGHEIT VVNASIVGML TRGLARLPT 60
LLQQHKKPAYV IIELGAVDGL RGFPQGTIRN NLSQMITEIQ NADAKPMLVQ IKLPPWHGKR 120
YSDMFSSIIYP QLSKELATPL LPFFLEQIYM KQEWMMNDGL HPKSDAQFPI AEYMAENIAP 180
YL                                     182

SEQ ID NO: 66                moltype = AA length = 10
FEATURE                      Location/Qualifiers
REGION                      1..10
                                note = Description of Artificial Sequence: Synthetic peptide
VARIANT                    3
                                note = Ala or Gly
VARIANT                    4
                                note = Val, Ile or Leu
VARIANT                    7
                                note = Ala or Gly
source                      1..10
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 66
LGXXDAXLRG                                     10

SEQ ID NO: 67                moltype = AA length = 472
FEATURE                      Location/Qualifiers

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-continued

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source          1..472
mol_type       = protein
organism       = Marinobacter hydrocarbonoclasticus

SEQUENCE: 67
KRLGTLDAWV LAVESEDTPM HVGTLQIFSL PEGAPETFLR DMVTRMKEAG DVAPPWGYKL 60
AWSGFLGRVI APANKVDKDI DLDYHVRHSA LPRPGERREL GILVSRHLSN PLDFSRPLWE 120
CHVIEGLENN RFALYTKMHH SMIDGISGVR LMQRVLTTDP ERCNMPPPT VRPHQRRGAK 180
TDKEASVPAA VSQAMDALKL QADMAPRLWQ AGNRLVHVSF HPEDGLTAPF TGPVSVLNHR 240
VTAQRRFATQ HYQLDRNLNL AHASGGSLND IVLYLCGTAL RRFLAEQNNL PDTPLTAGIP 300
VNIRPADDEG TGTQISFMIA SLATDEADPL NRLQQIKTST RRAKEHLQKL PKSALTQYTM 360
LLMSPYILQL MSGLGGRMRP VFNVTTISNP GPEGTLYYEG ARLEAMYFVS LIAHGGALNI 420
TCLSYAGSLN FGFTGCRDTL PSMQKLAVYT GEALDELESF ILPPKKRART RK 472

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We claim:

1. A variant ester synthase polypeptide comprising an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 67, wherein said variant ester synthase polypeptide is genetically engineered to have at least one mutation selected from the group consisting of to K1N, K1T, R2S, G4R, T5P, T5S, L6Q, D7N, S15G, T24M, T24N, T24W, Q26P, Q26T, L30H, G33D, G33N, G33S, L39A, L39M, L39S, R40S, D41A, D41G, D41H, D41Y, V43K, V43S, T44A, T44F, T44K, E48A, E48Q, A49D, A49T, G50S, Y58C, Y58L, V69L, V69W, I70L, I70V, A73G, A73Q, A73S, A73T, V76L, D77A, K78F, K78W, D79H, D79K, I80V, R87W, R93T, R98D, E99Q, G101L, I102L, I102R, S105G, N110R, P111D, P111G, P111S, D113A, D113V, S115R, C121S, H122Q, H122S, V123I, V123M, G126D, L127G, R131M, L134T, T136G, I146K, I146L, I146R, S147A, S147T, V149L, R150P, V155G, T157S, T158E, T158K, T158R, E161D, E161G, R162E, R162K, C163I, C163R, N164D, N164R, M165K, P166L, P166S, T170M, T170R, T170S, V171E, V171F, V171H, V171L, V171R, V171W, R172S, R172W, P173R, P173S, P173W, H174E, Q175E, Q175R, Q175S, R176T, R177V, A179K, A179S, A179V, D182G, K183S, E184F, E184G, E184L, E184R, E184S, A185L, A185M, S186T, V187G, V187R, P188R, A189G, A190P, A190R, A190W, V191I, V191L, S192A, S192L, S192V, Q193R, Q193S, M195G, D196E, A197T, A197V, L200R, Q201A, Q201R, Q201V, Q201W, A202L, D203E, D203R, P206F, R207A, G212A, G212L, G212M, G212S, V216I, V219L, A228G, T231S, V234C, V236A, V236K, L237I, H239G, T242K, T242R, A243G, A243R, Q244G, R246A, R246G, R246L, R246Q, R246P, R246V, R246W, Q250W, Q253G, L254R, L254T, R255E, L257I, L257M, K258R, N259A, N259E, N259Q, L260M, L260V, H262Q, H262R, A263V, S264D, S264V, S264W, G265D, G265N, G266A, G266S, S267G, L268F, L268M, V272A, Y274G, A279G, A279V, A285L, A285R, A285V, E286H, Q287S, N288D, N289E, N289G, D292F, T293A, T293I, P294G, V301A, N302G, I303G, I303R, I303W, R304W, A306G, A306S, D307F, D307G, D307L, D307N, D307R, D307V, E309A, E309G, E309S, G310H, G310R, G310V, T311S, T313S, Q314G, Q314R, I315F, S316G, F317W, I319G, A320C, A323G, A327I, D328F, N331G, N331K, N331T, Q334K, Q334S, Q335C, Q335N, Q335S, T338A, T338E, T338H, Q348A, Q348R, K349A, K349C, K349H, K349Q, P351G, K352I, K352N, S353K, S353T, T356G, T356W, Q357V, M360Q, M360R, M360S, M360W, M363W, P365G, Y366G, Y366W, I367M, M371R, G373R, G375A, G375S, G375V, M378A, V381F, T385G, E393G, E393Q, E393R, E393T, E393W, G394E, G394R, T395E, T395V, R402K, E404K, E404R, V409L, S410G, L411A, L411P, L411V, A413T, I420V, C422G, S424G, S424Q, A426V, N430G, S442E, S442G, M443G, A447C, A447I, A447L, L454V, D455E, L457Y, E458G, E458W, I461G,

15 I461L, I461V, K466N, R467L, R467T, A468G, A468S, A468T, R469L, T470A, T470P, T470S, K472T, K472*, *474Y, and a combination thereof, and wherein said variant ester synthase polypeptide has improved ester synthase activity compared to a corresponding wild type polypeptide.

20 2. The variant ester synthase polypeptide of claim 1, wherein expression of the variant ester synthase polypeptide in a recombinant host cell results in a higher titer of fatty ester compositions compared to a recombinant host cell expressing the corresponding wild type polypeptide.

25 3. The variant ester synthase of claim 2, wherein said fatty ester compositions comprise fatty acid methyl esters, fatty acid ethyl esters, fatty acid propyl esters, fatty acid isopropyl esters, fatty acid butyl esters, monoglycerides, fatty acid isobutyl esters, fatty acid 2-butyl esters, fatty acid tert-butyl esters, or a combination thereof.

30 4. The variant ester synthase of claim 1, wherein said improved ester synthase activity results in an increased percentage of beta-hydroxy esters, a decreased percentage of beta-hydroxy esters, an increased chain lengths of fatty acid esters, a decreased chain lengths of fatty acid esters, or a combination thereof.

35 5. A recombinant host cell, genetically engineered to express the variant ester synthase polypeptide of claim 1.

40 6. The recombinant host cell of claim 5, wherein: the recombinant host cell produces a fatty acid ester composition with higher titer, higher yield and/or higher productivity of fatty acid esters than a host cell expressing the corresponding wild type polypeptide, when cultured in medium containing a carbon source under conditions effective to express the variant ester synthase polypeptide; or the recombinant host cell produces a fatty ester composition with an altered percentage of beta-hydroxy esters as compared to a host cell that expresses the corresponding wild type ester synthase polypeptide.

45 7. The recombinant host cell of claim 6, wherein the altered percentage of beta-hydroxy esters is an increased percentage of beta-hydroxy esters.

50 8. The recombinant host cell of claim 6, wherein the altered percentage of beta-hydroxy esters is a decreased percentage of beta-hydroxy esters.

55 9. A cell culture, comprising the recombinant host cell of claim 5 and a fatty ester composition.

60 10. The cell culture of claim 9, wherein the fatty ester composition comprises one or more of a saturated or unsaturated C6, C8, C10, C12, C13, C14, C15, C16, C17, or C18 fatty ester.

65 11. The cell culture of claim 9, wherein the fatty ester composition comprises a fatty ester having a double bond at position 7 in the carbon chain between C7 and C8 from the reduced end of the fatty ester.

12. The cell culture of claim 9, wherein the fatty ester composition comprises branched chain fatty esters.

13. A method of producing a fatty ester composition, the method comprising culturing the recombinant host cell of claim 5 in a medium comprising a carbon source. 5

14. The method of claim 13, further comprising isolating the fatty ester composition.

15. The method of claim 14, wherein the fatty ester composition comprises:

an increased or decreased percentage of beta-hydroxy 10
esters; and/or

one or more of a saturated or unsaturated C6, C8, C10,
C12, C13, C14, C15, C16, C17, or C18 fatty ester.

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